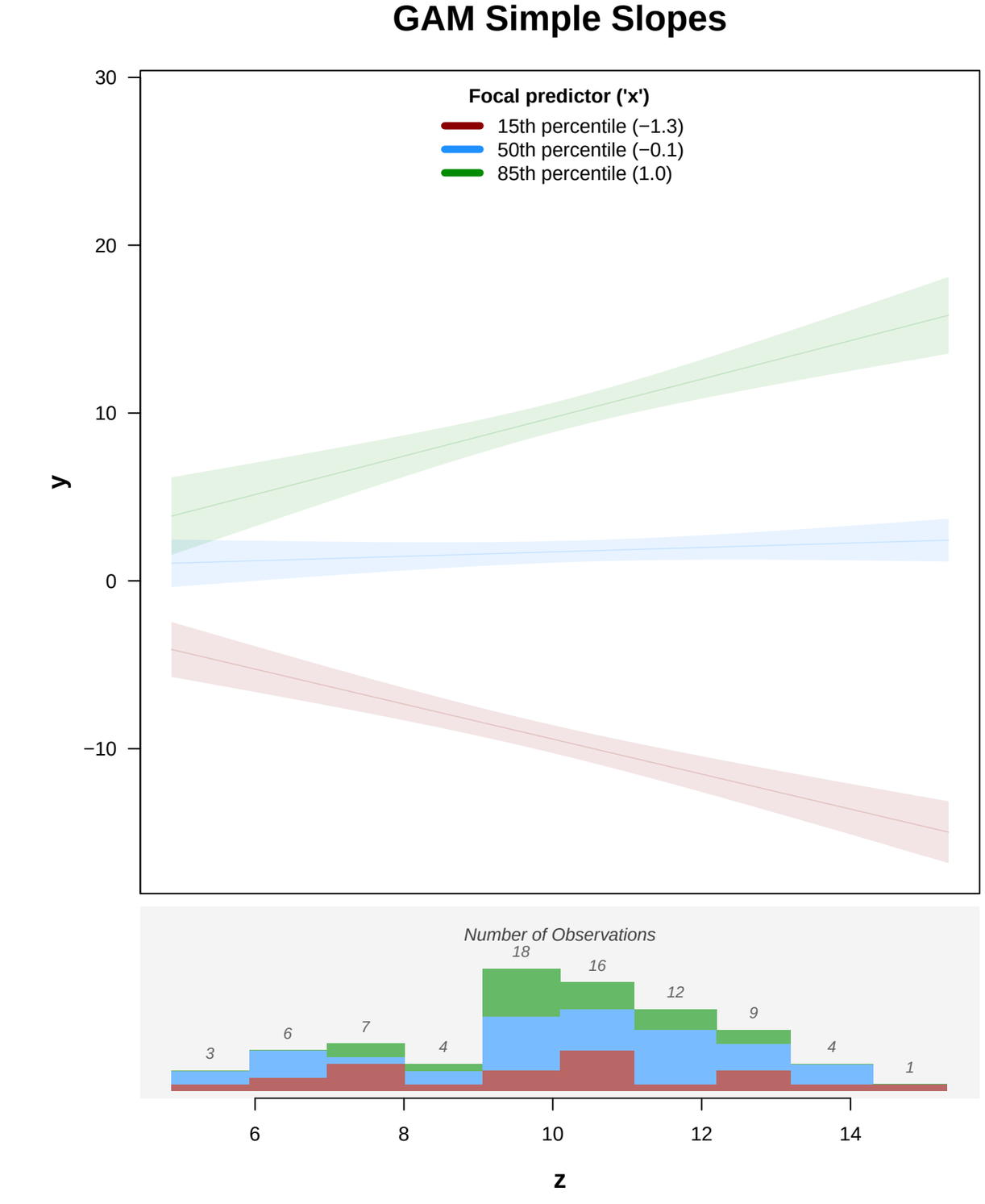
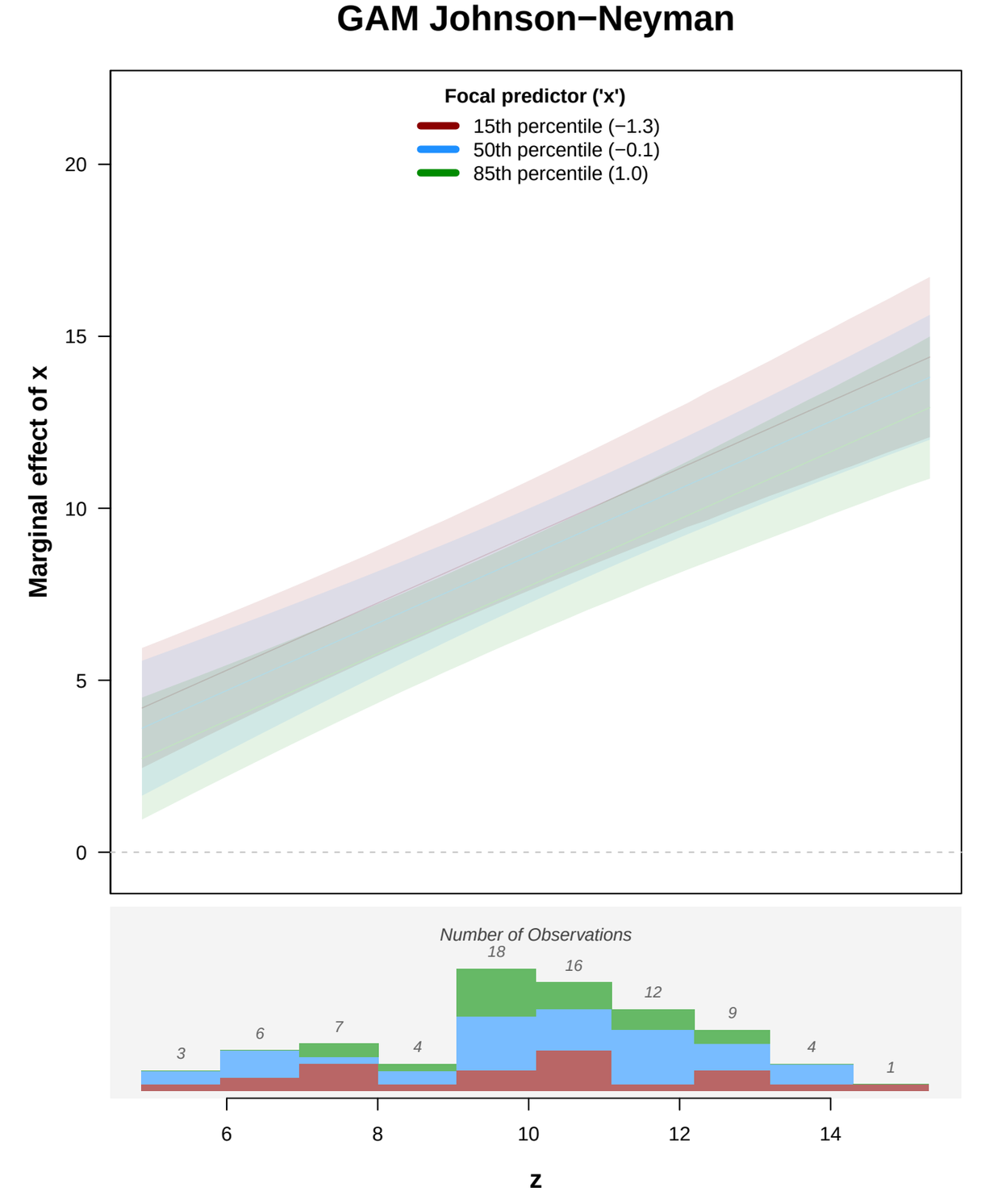
Part A (sample sizes) | Page 1 | continuous_both_hist | n=80 (small)

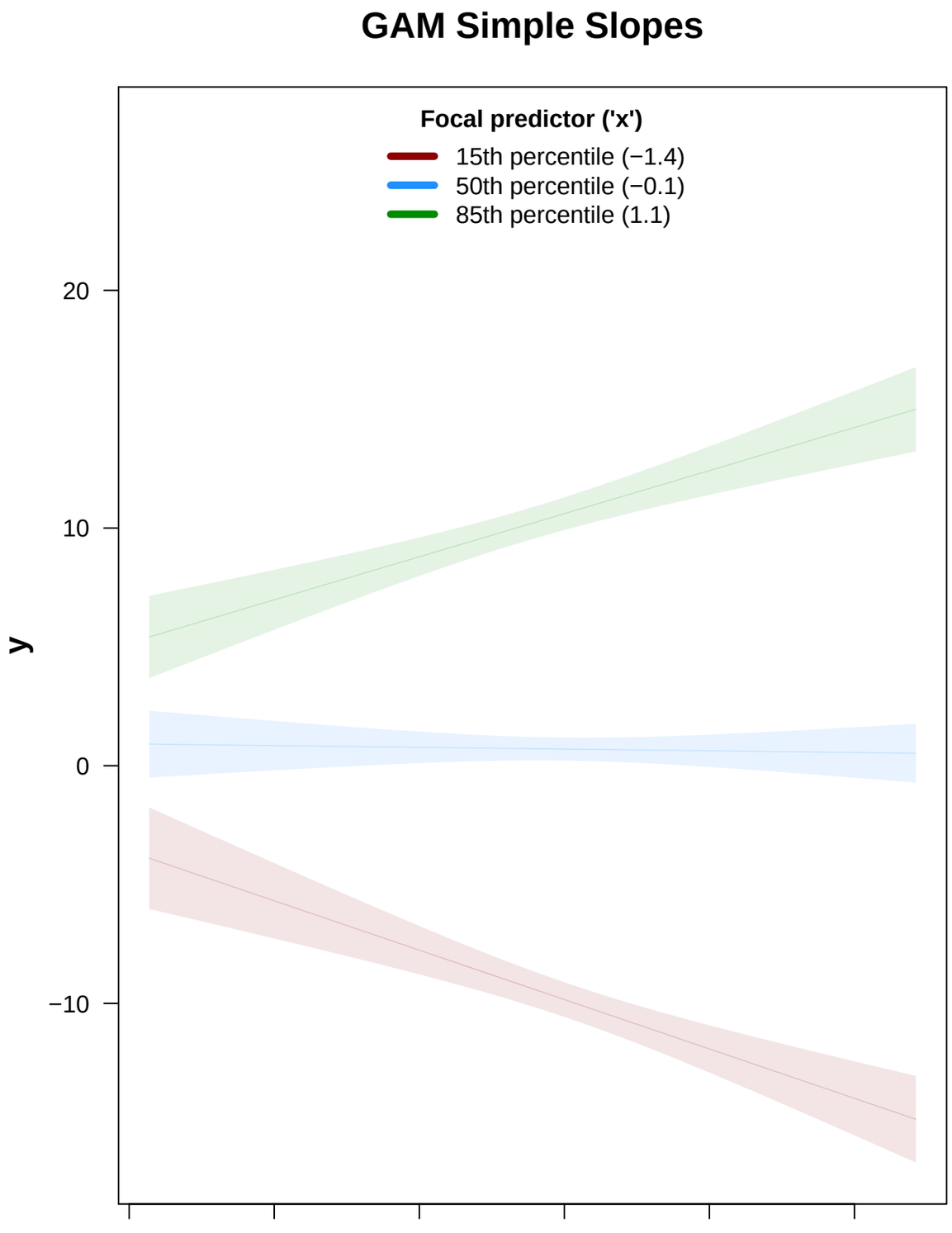
Continuous x & z; default GAM; draw='both'; frequency panels ON. Check stacked hist under both curves, count labels inside gray band, no collisions.

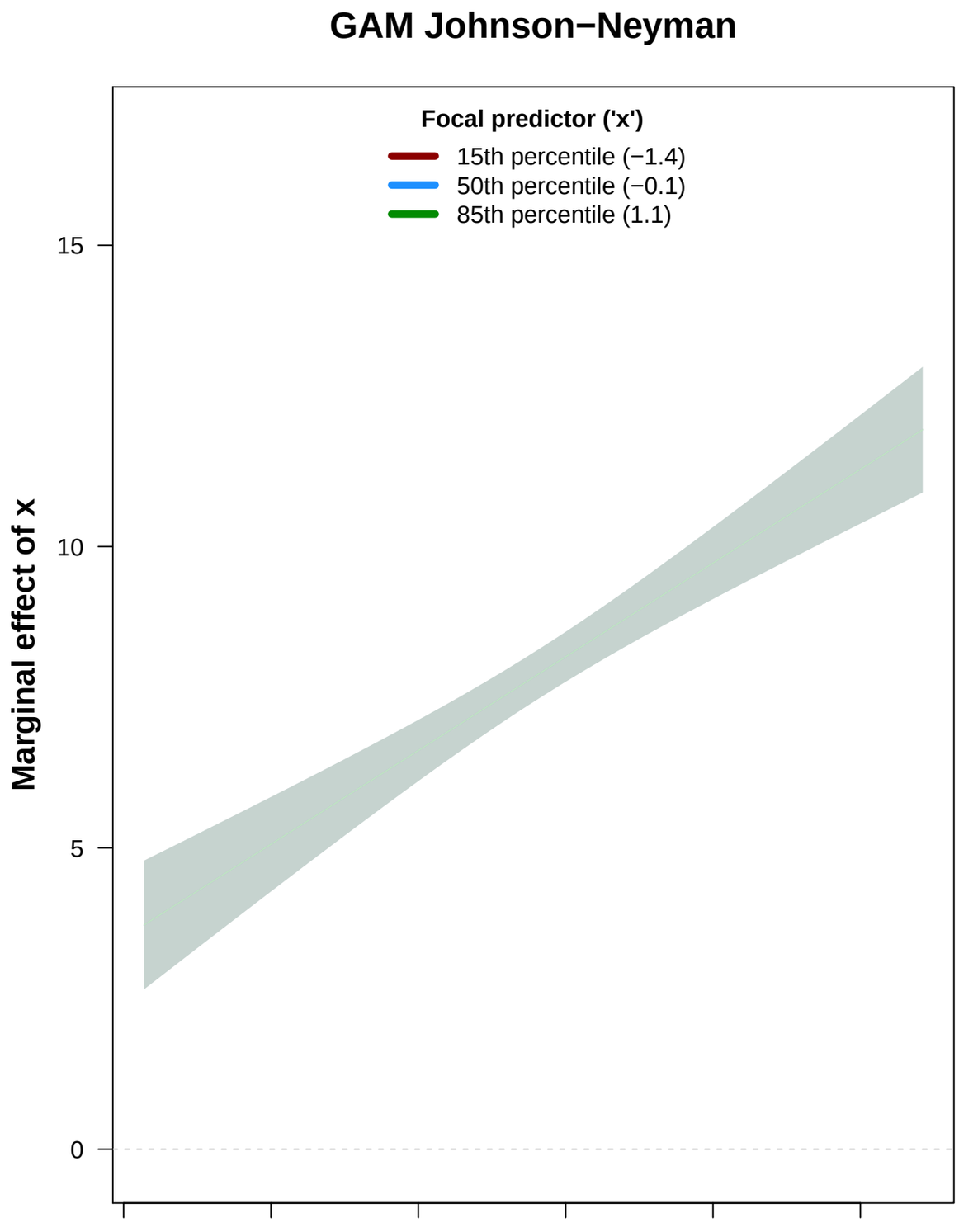

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.3)
50th percentile (−0.1)
85th percentile (1.0)
y
30
20
10
0
−10
Number of Observations
3
6
7
4
18
16
12
9
4
1
6
8
10
12
14
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.3)
50th percentile (−0.1)
85th percentile (1.0)
Marginal effect of x
20
15
10
5
0
Number of Observations
3
6
7
4
18
16
12
9
4
1
6
8
10
12
14
z

Part A (sample sizes) | Page 2 | continuous_both_nohist | n=80 (small)

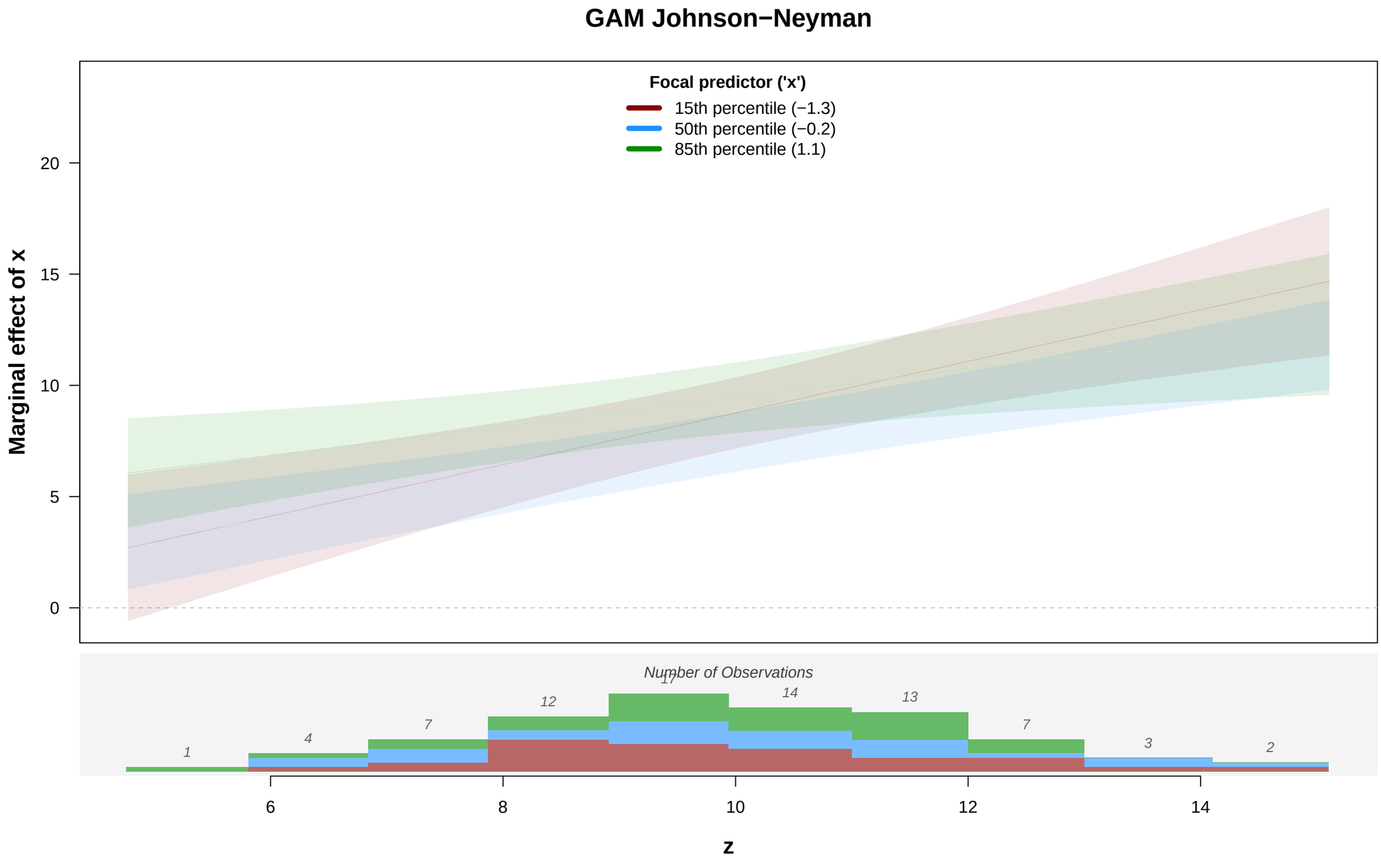
Continuous x & z; default GAM; draw='both'; frequency panels OFF. Baseline layout without the bottom frequency row.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.4)
50th percentile (−0.1)
85th percentile (1.1)
20
10
0
−10
y


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.4)
50th percentile (−0.1)
85th percentile (1.1)
15
10
5
0
Marginal effect of x

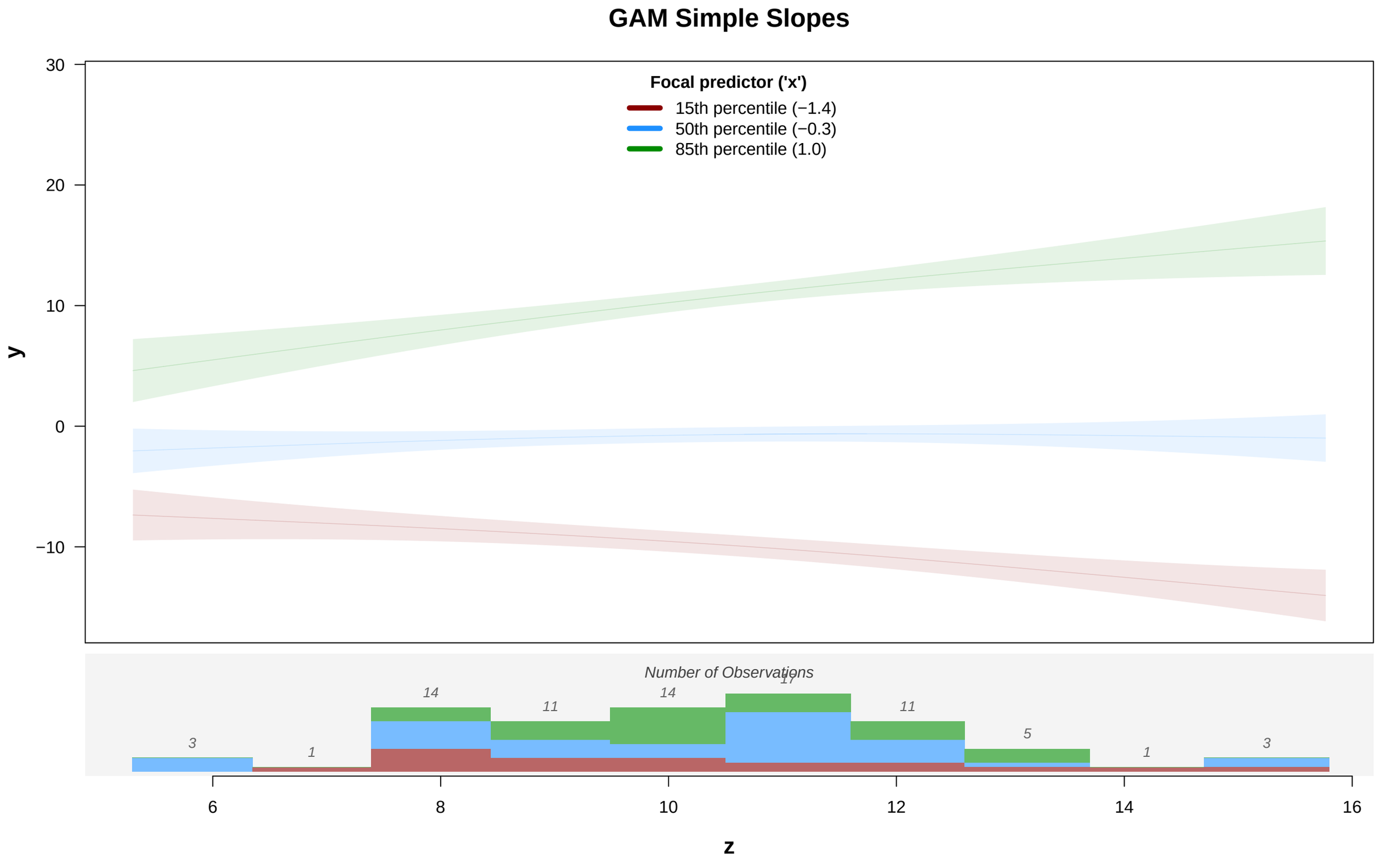
Part A (sample sizes) | Page 3 | continuous_jn_hist | n=80 (small)

Continuous x & z; single Johnson−Neyman panel with frequency subplot underneath.


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.3)
50th percentile (−0.2)
85th percentile (1.1)
Marginal effect of x
20
15
10
5
0
Number of Observations
1
4
7
12
17
14
13
7
3
2
6
8
10
12
14
z

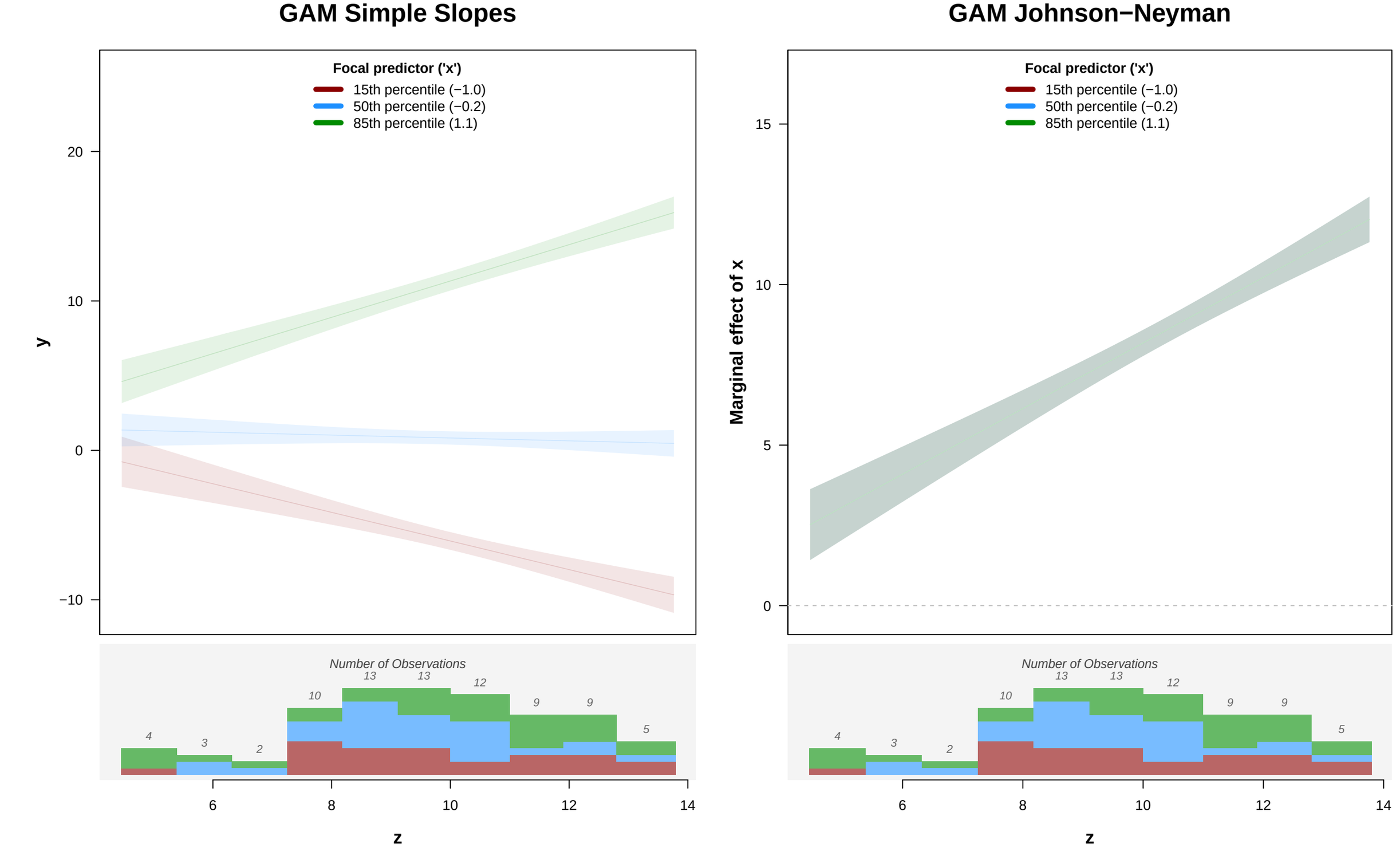
Part A (sample sizes) | Page 4 | continuous_slopes_hist | n=80 (small)

Continuous x & z; single simple−slopes panel with frequency subplot underneath.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.4)
50th percentile (−0.3)
85th percentile (1.0)
30
20
10
0
−10
y
Number of Observations
3
1
14
11
14
17
11
5
1
3
6
8
10
12
14
16
z

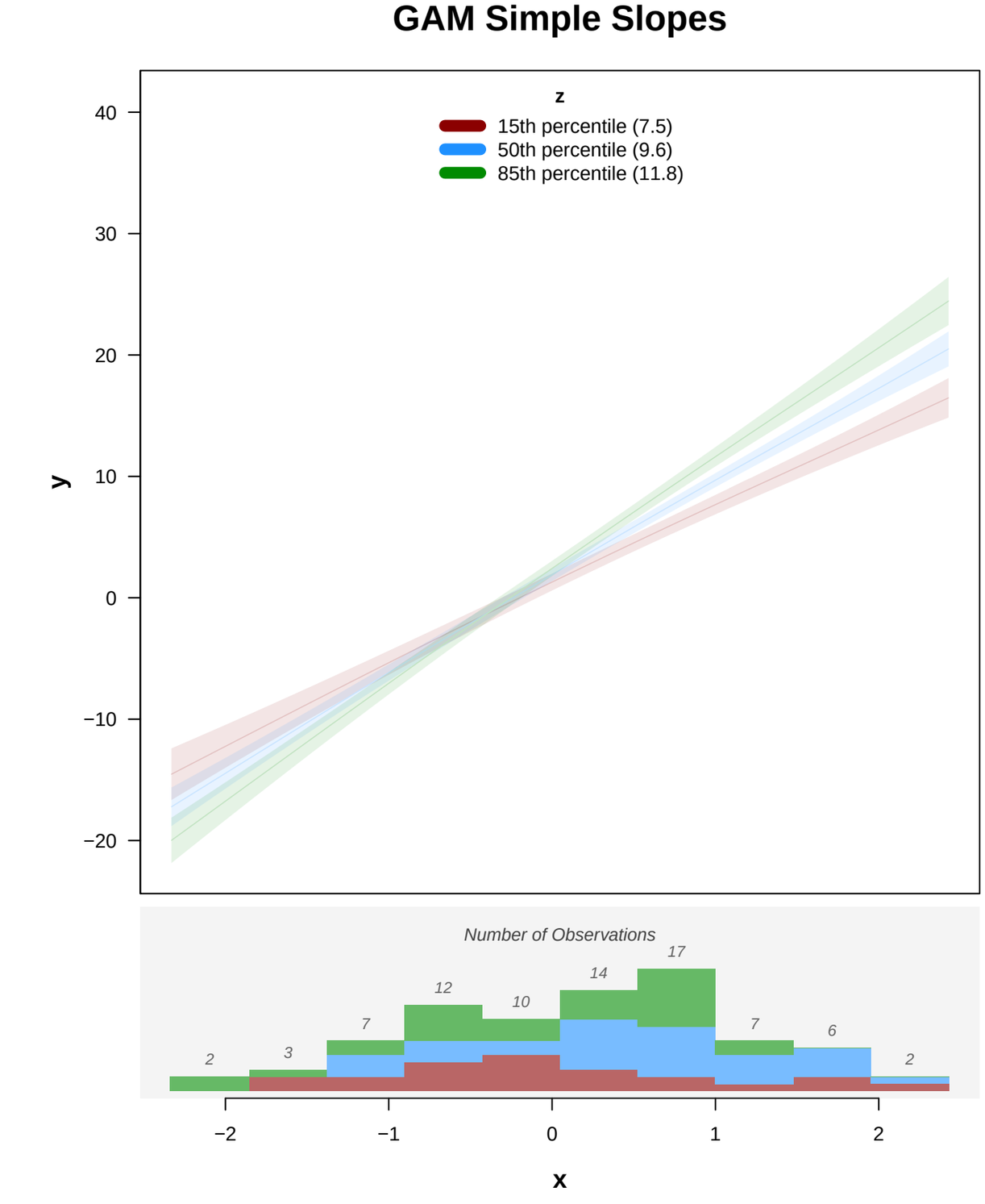
Part A (sample sizes) | Page 5 | continuous_linear_hist | n=80 (small)

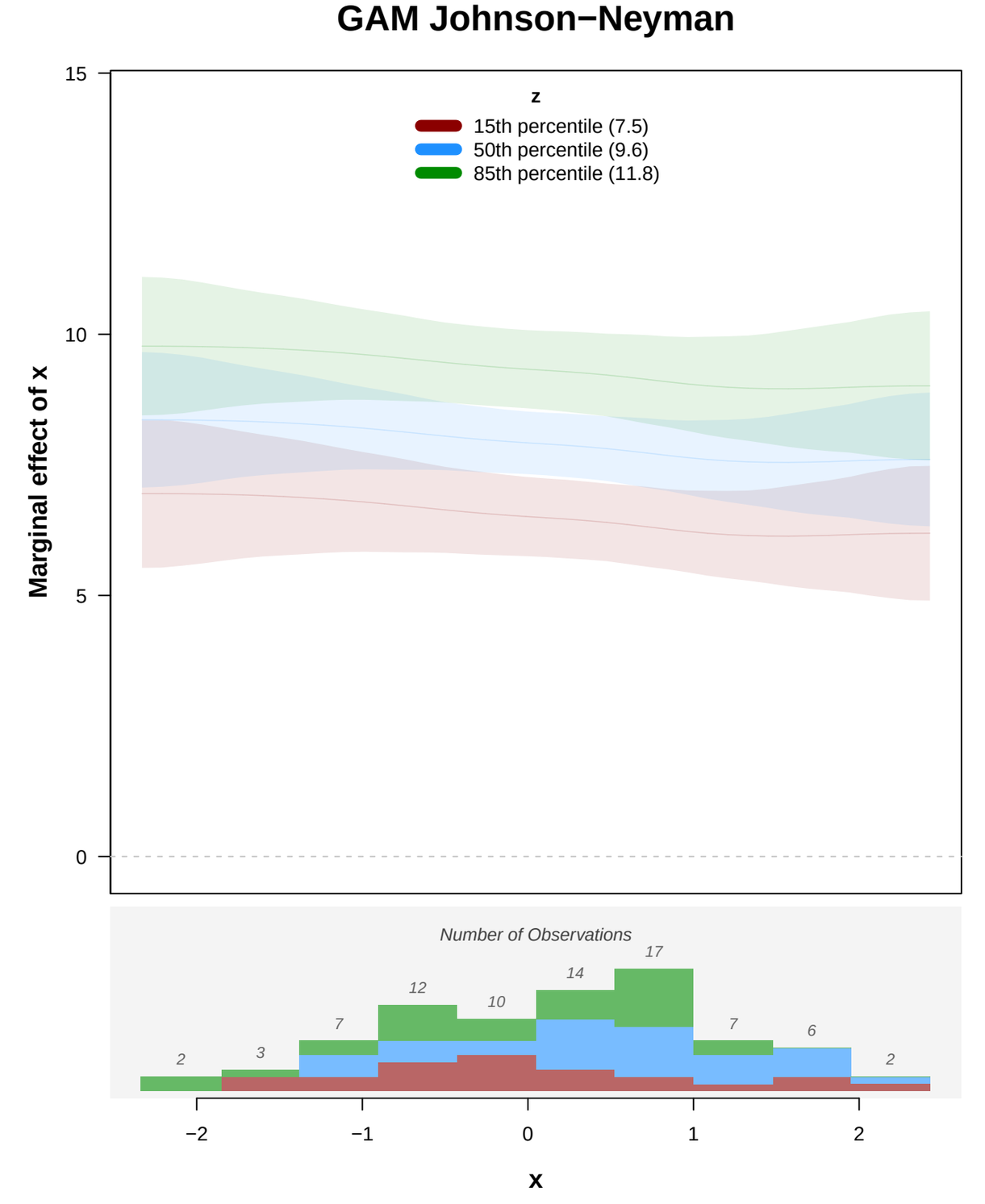
Continuous x & z; model='linear' (lm2) instead of GAM; both panels with frequency subplots.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.2)
85th percentile (1.1)
y
20
10
0
−10
Number of Observations
4
3
2
10
13
13
12
9
9
5
6
8
10
12
14
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.2)
85th percentile (1.1)
Marginal effect of x
15
10
5
0
Number of Observations
4
3
2
10
13
13
12
9
9
5
6
8
10
12
14
z

Part A (sample sizes) | Page 6 | continuous_x_on_axis_hist | n=80 (small)

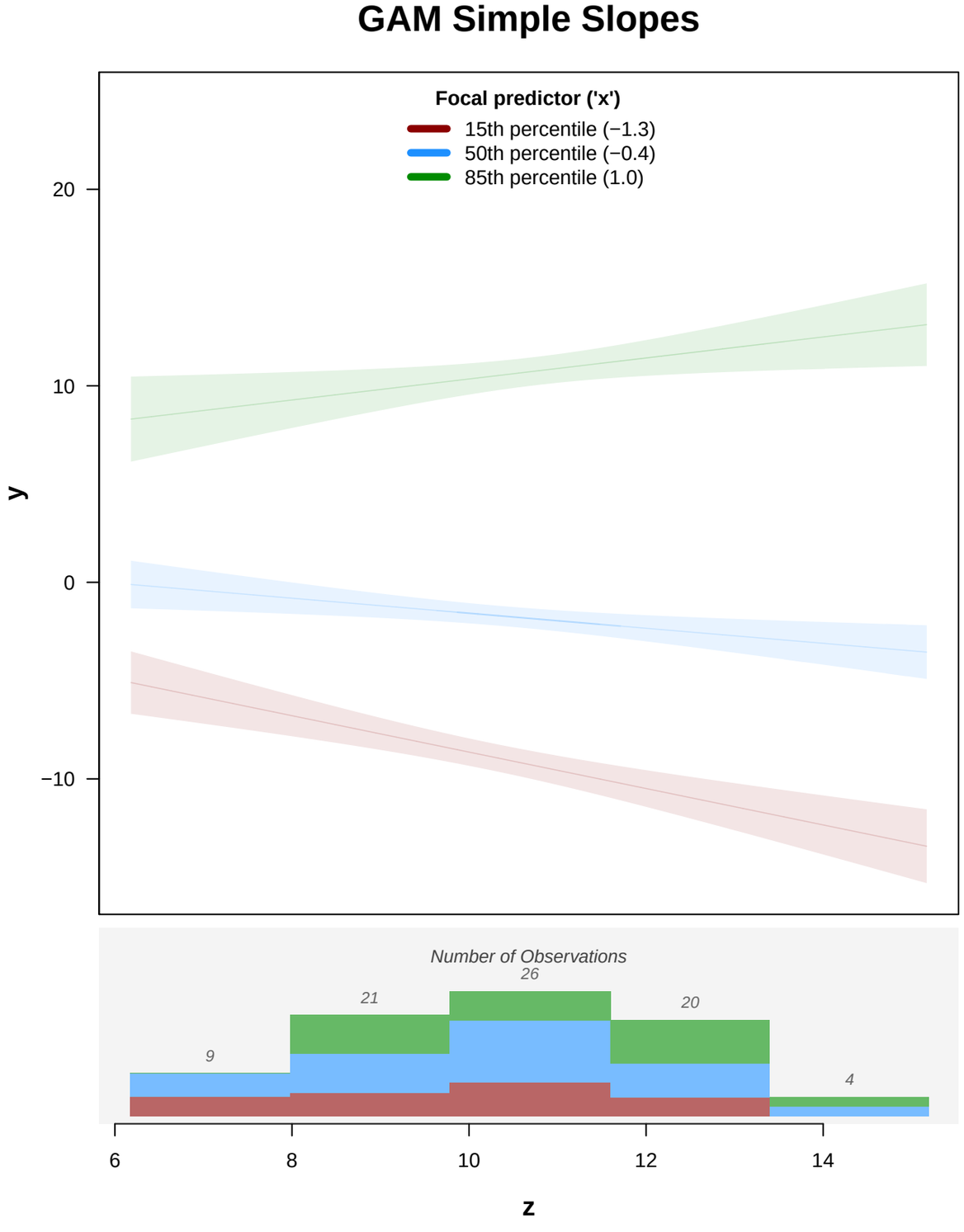
Continuous x & z; moderator.on.x.axis=FALSE so focal x is on the x−axis; frequency panels still aligned to that axis.

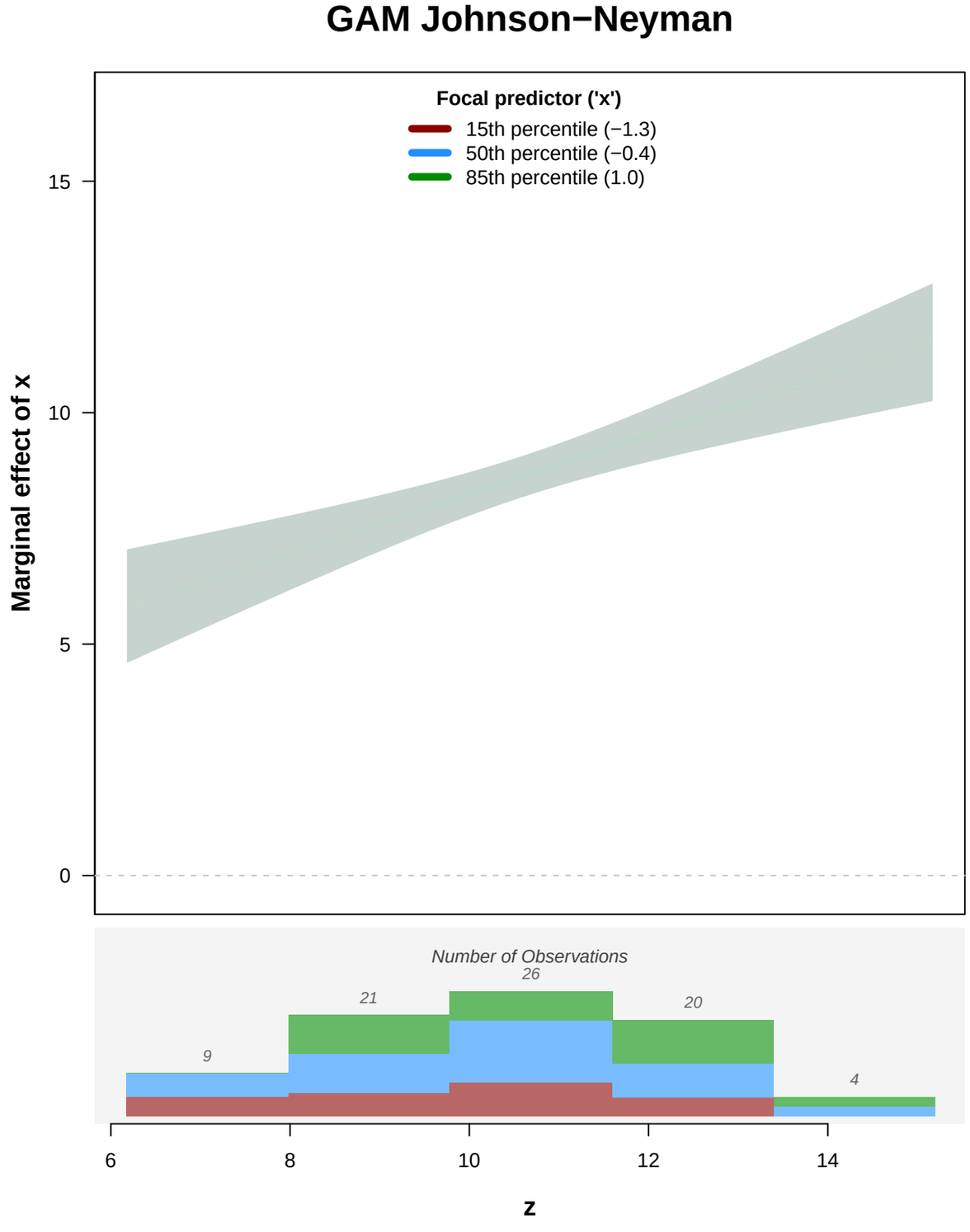

GAM Simple Slopes
z
15th percentile (7.5)
50th percentile (9.6)
85th percentile (11.8)
y
40
30
20
10
0
−10
−20
Number of Observations
2
3
7
12
10
14
17
7
6
2
−2
−1
0
1
2
x


GAM Johnson−Neyman
z
15th percentile (7.5)
50th percentile (9.6)
85th percentile (11.8)
Marginal effect of x
15
10
5
0
Number of Observations
2
3
7
12
10
14
17
7
6
2
−2
−1
0
1
2
x

Part A (sample sizes) | Page 7 | continuous_nbins_5 | n=80 (small)

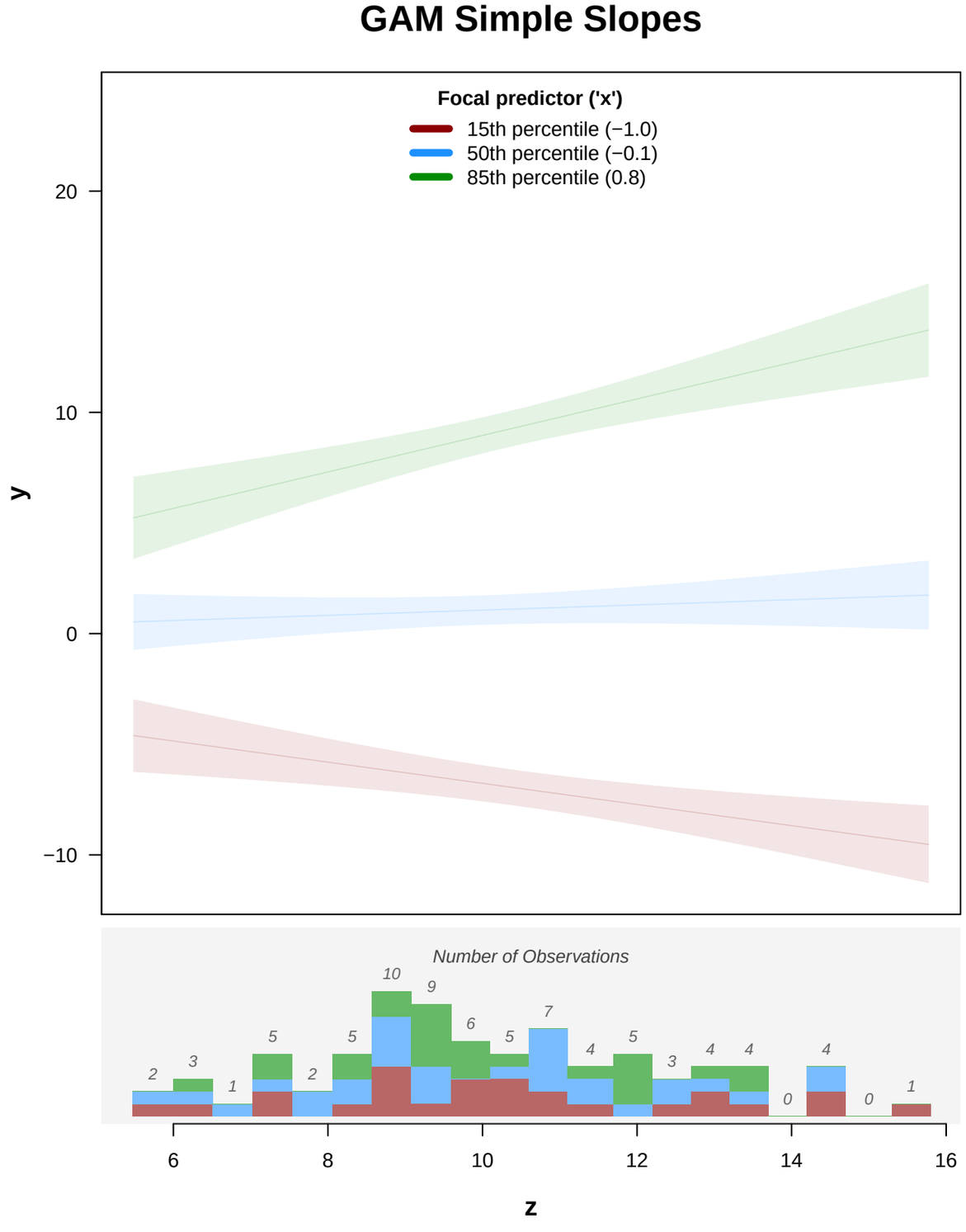
Continuous x & z; n.bin.continuous=5 (coarser frequency bins). Check fewer, wider stacked bars and readable count labels.

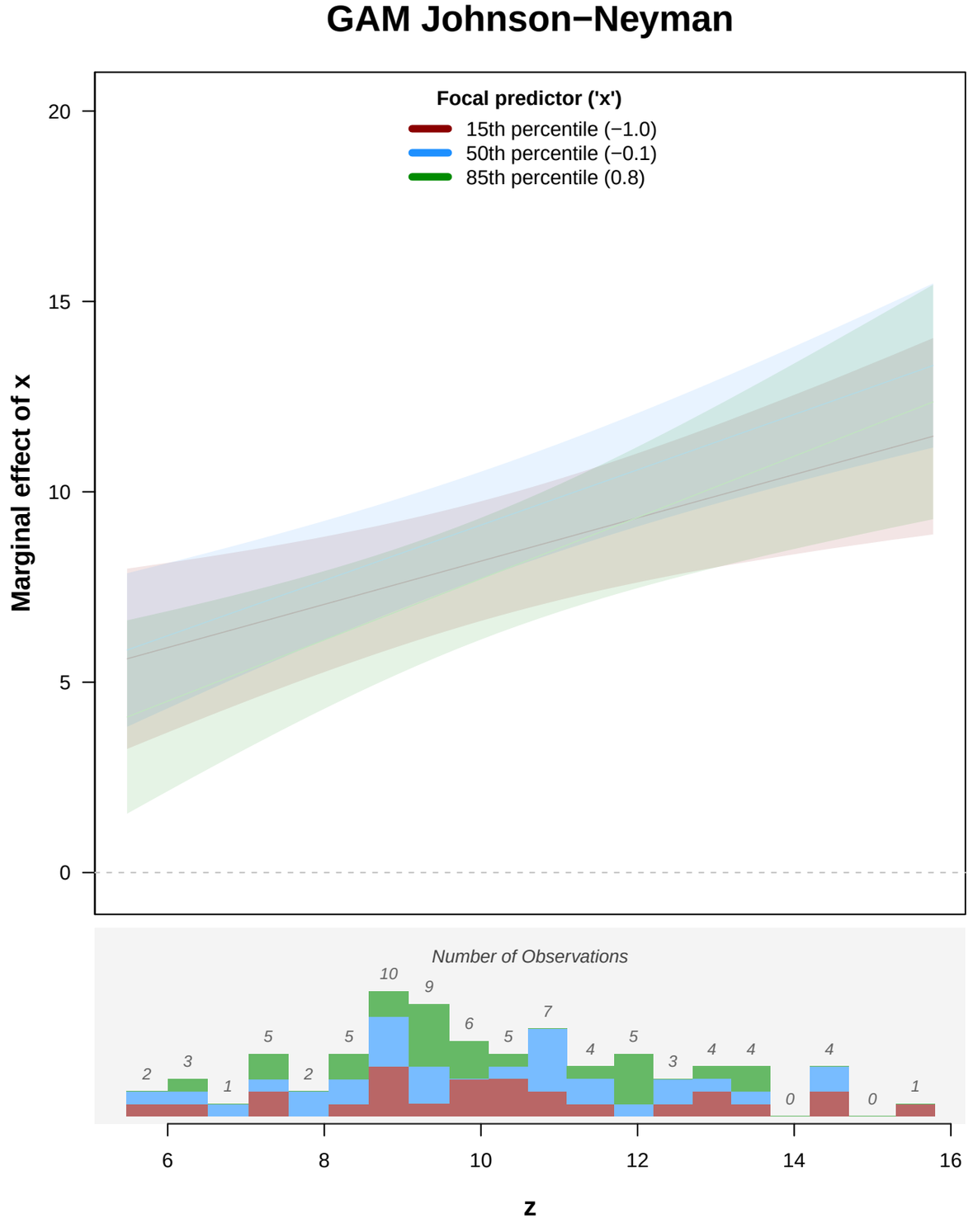

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.3)
50th percentile (−0.4)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
26
21
20
9
4
6
8
10
12
14
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.3)
50th percentile (−0.4)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
26
21
20
9
4
6
8
10
12
14
z

Part A (sample sizes) | Page 8 | continuous_nbins_20 | n=80 (small)

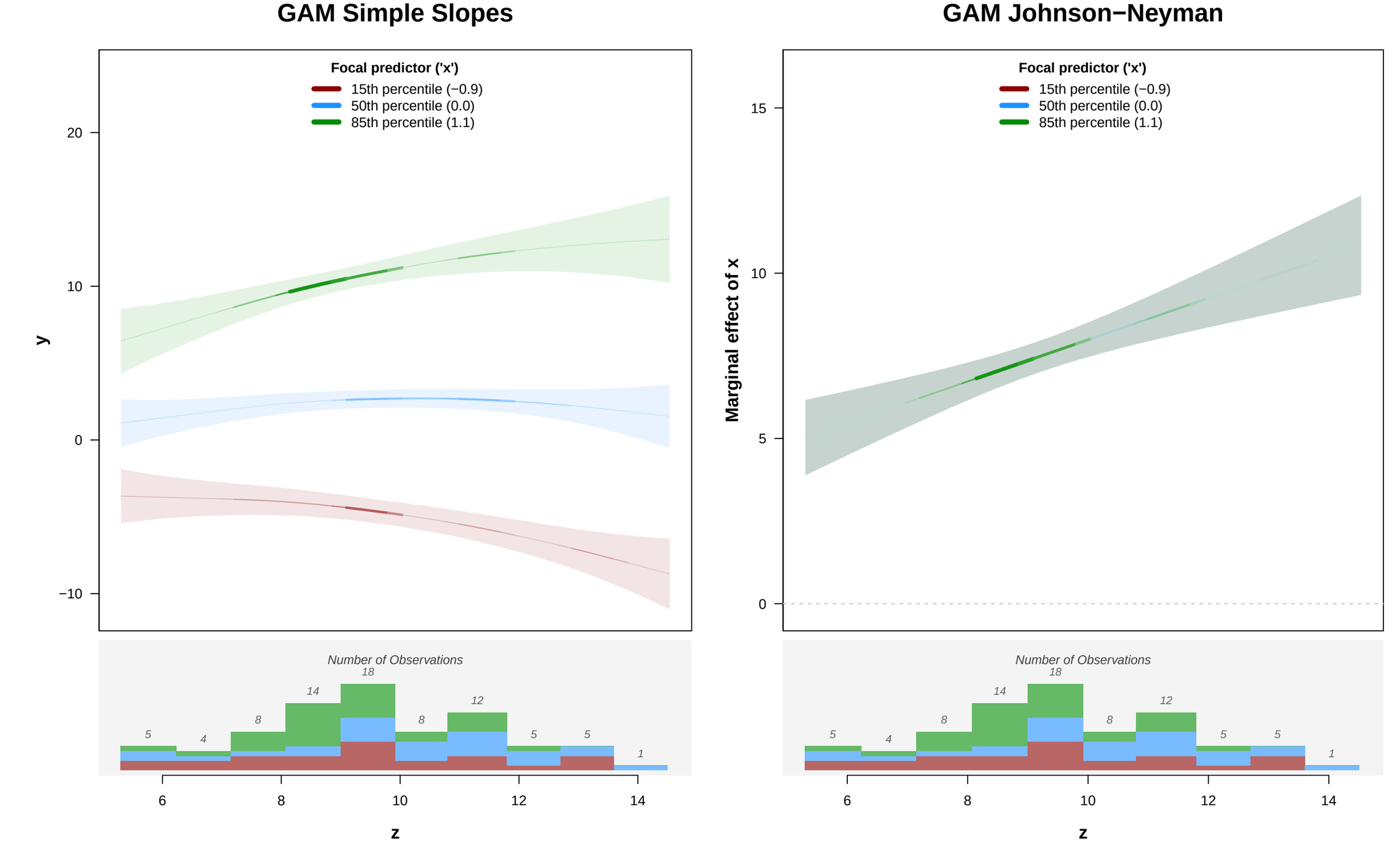
Continuous x & z; n.bin.continuous=20 (finer frequency bins). Check denser bars still readable at small and big n.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.1)
85th percentile (0.8)
y
20
10
0
−10
Number of Observations
2 3 1 5 2 5 10 9 6 5 7 4 5 3 4 4 0 4 0 1
6 8 10 12 14 16
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.1)
85th percentile (0.8)
Marginal effect of x
20
15
10
5
0
Number of Observations
2 3 1 5 2 5 10 9 6 5 7 4 5 3 4 4 0 4 0 1
6 8 10 12 14 16
z

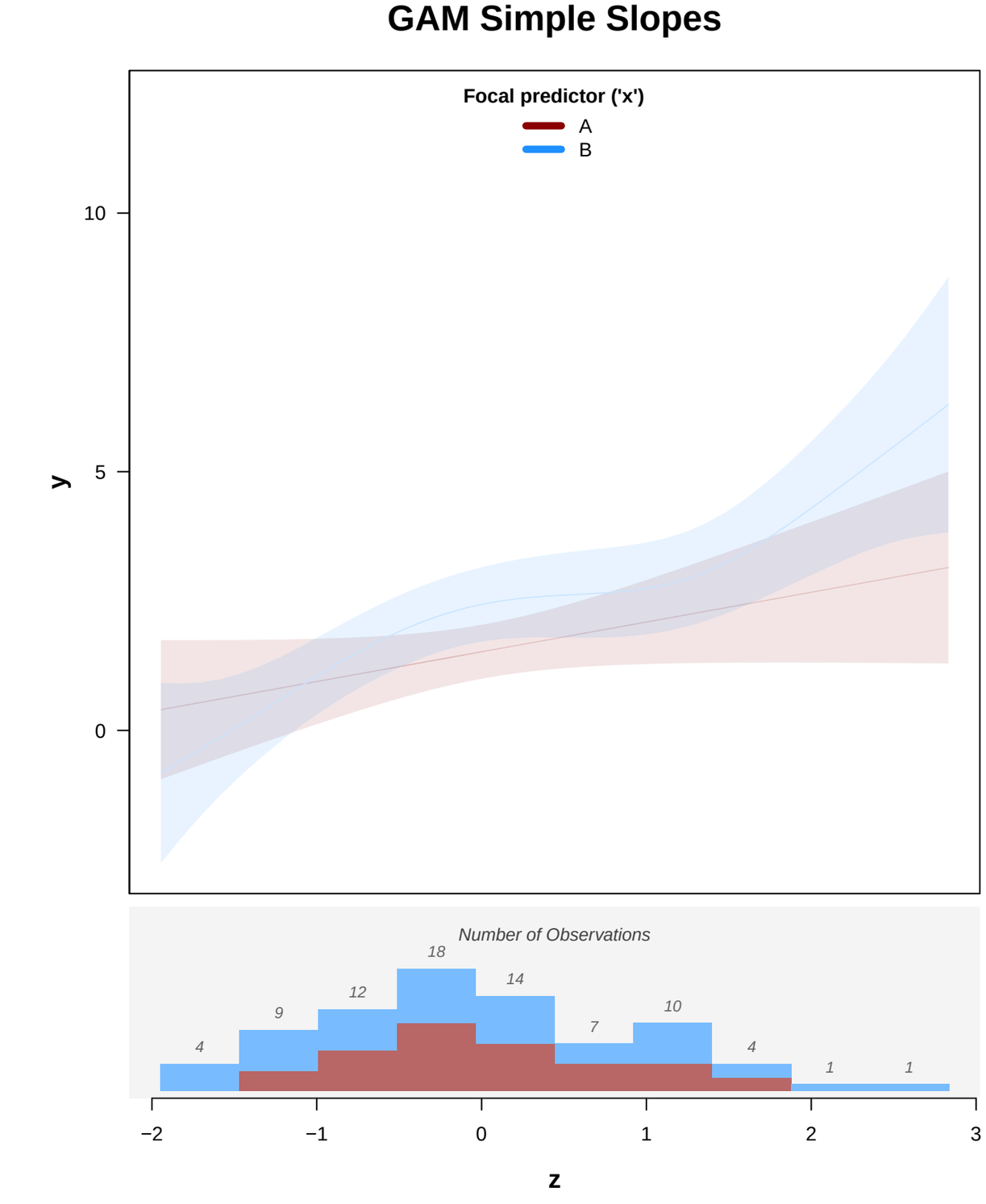
Part A (sample sizes) | Page 9 | continuous_nmax_10 | n=80 (small)

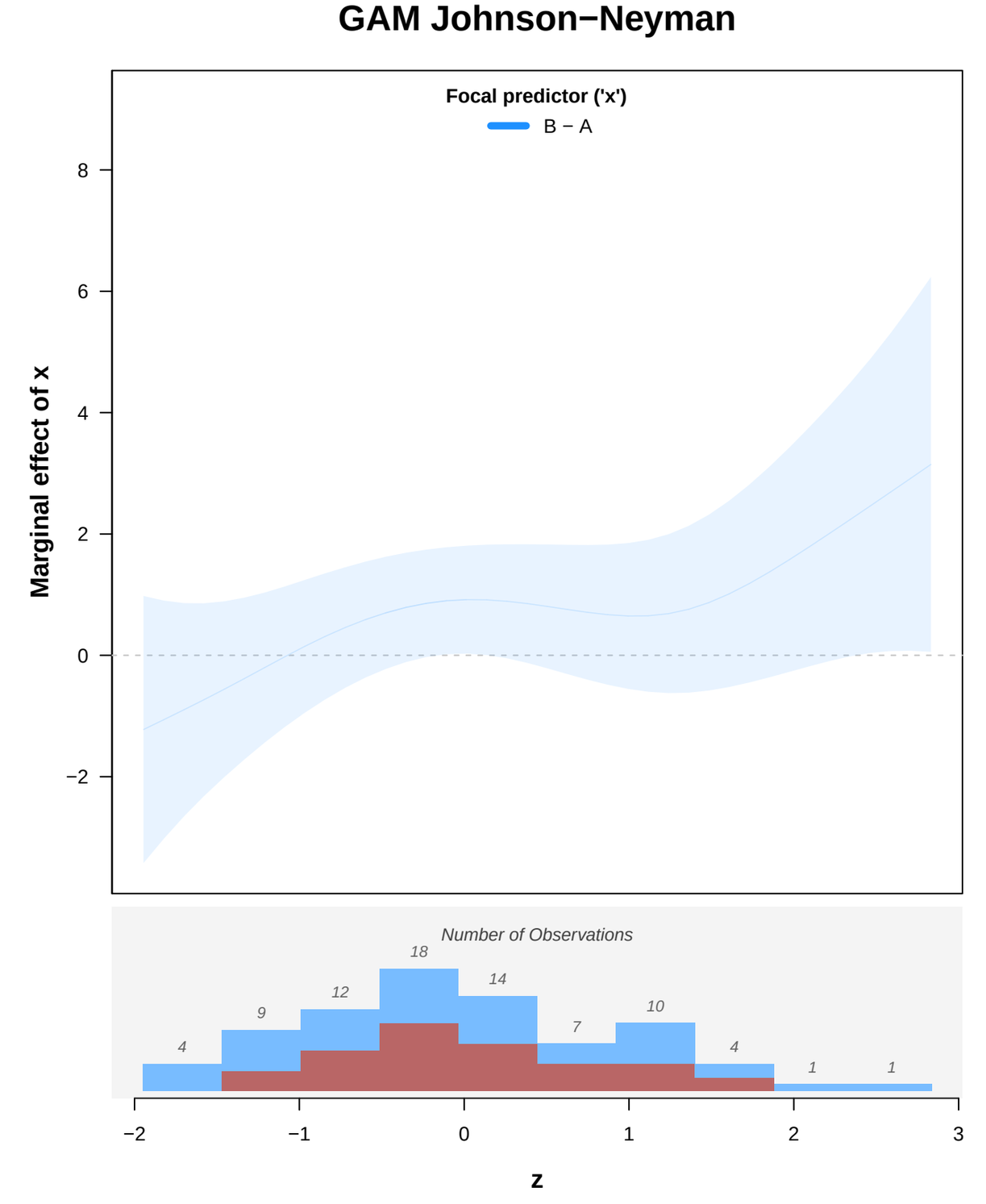
Continuous x & z; n.max=10 so line darkness/width saturates early. Check that sparse bins stay light and dense bins look fully saturated.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.0)
85th percentile (1.1)
y
20
10
0
−10
Number of Observations
5
4
8
14
18
8
12
5
5
1
6
8
10
12
14
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.0)
85th percentile (1.1)
Marginal effect of x
15
10
5
0
Number of Observations
5
4
8
14
18
8
12
5
5
1
6
8
10
12
14
z

Part A (sample sizes) | Page 10 | binary_x_hist | n=80 (small)

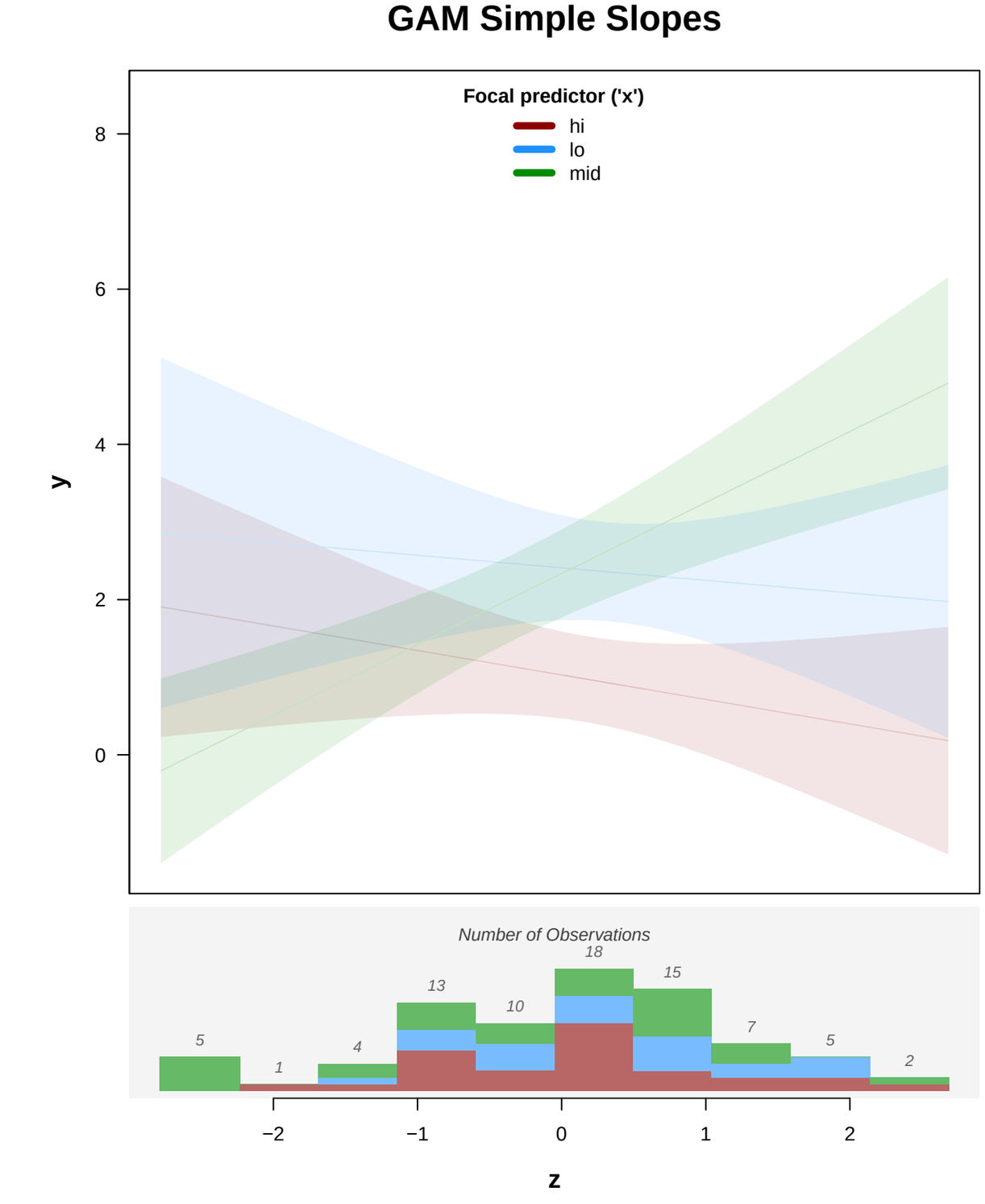
Binary/categorical focal x (2 levels) with continuous z; two colored curves; frequency panels ON.

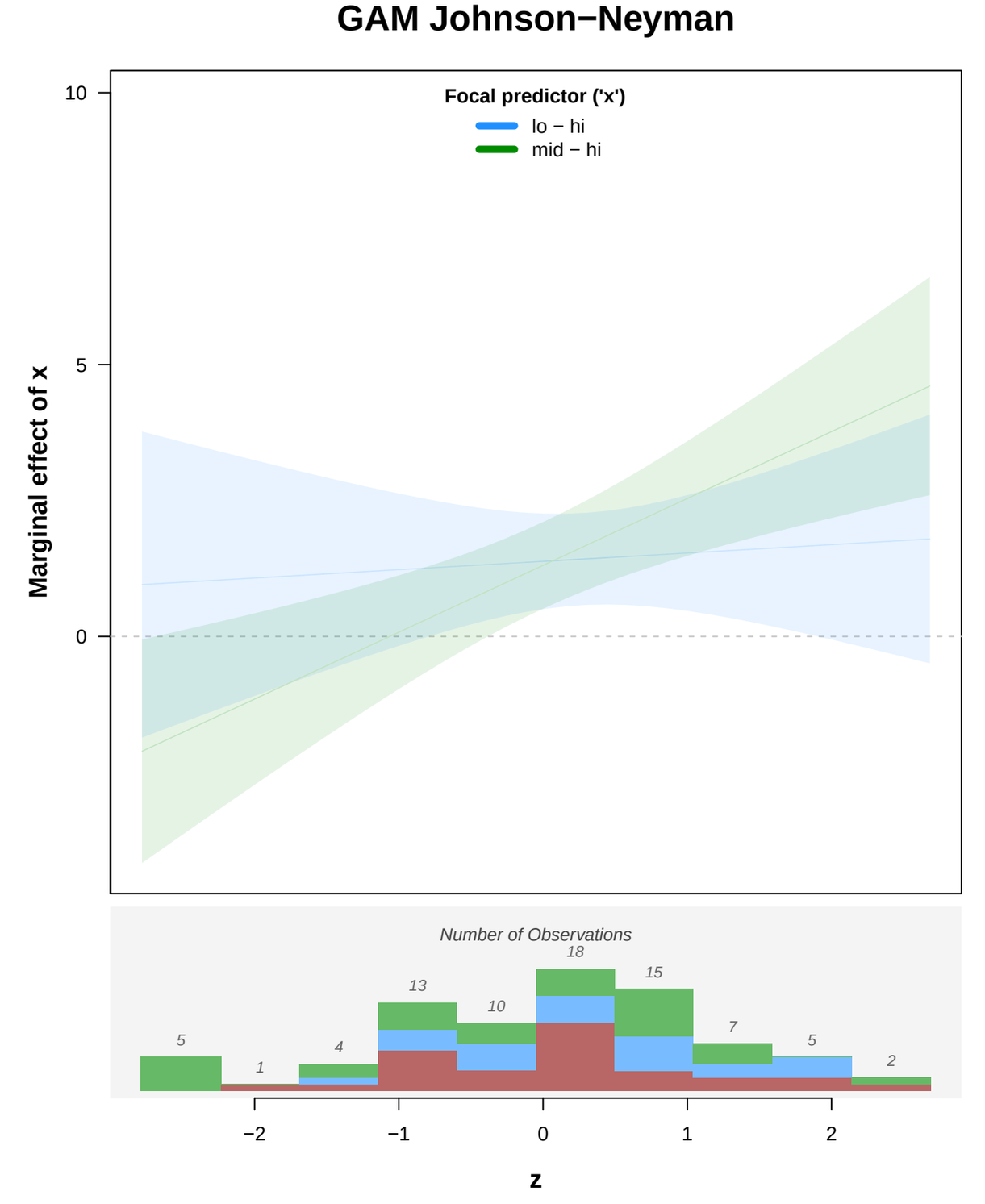

GAM Simple Slopes
Focal predictor ('x')
A
B
y
10
5
0
Number of Observations
18
14
12
10
9
7
4
4
1
1
−2
−1
0
1
2
3
z


GAM Johnson−Neyman
Focal predictor ('x')
B − A
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
18
14
12
10
9
7
4
4
1
1
−2
−1
0
1
2
3
z

Part A (sample sizes) | Page 11 | ternary_x_hist | n=80 (small)

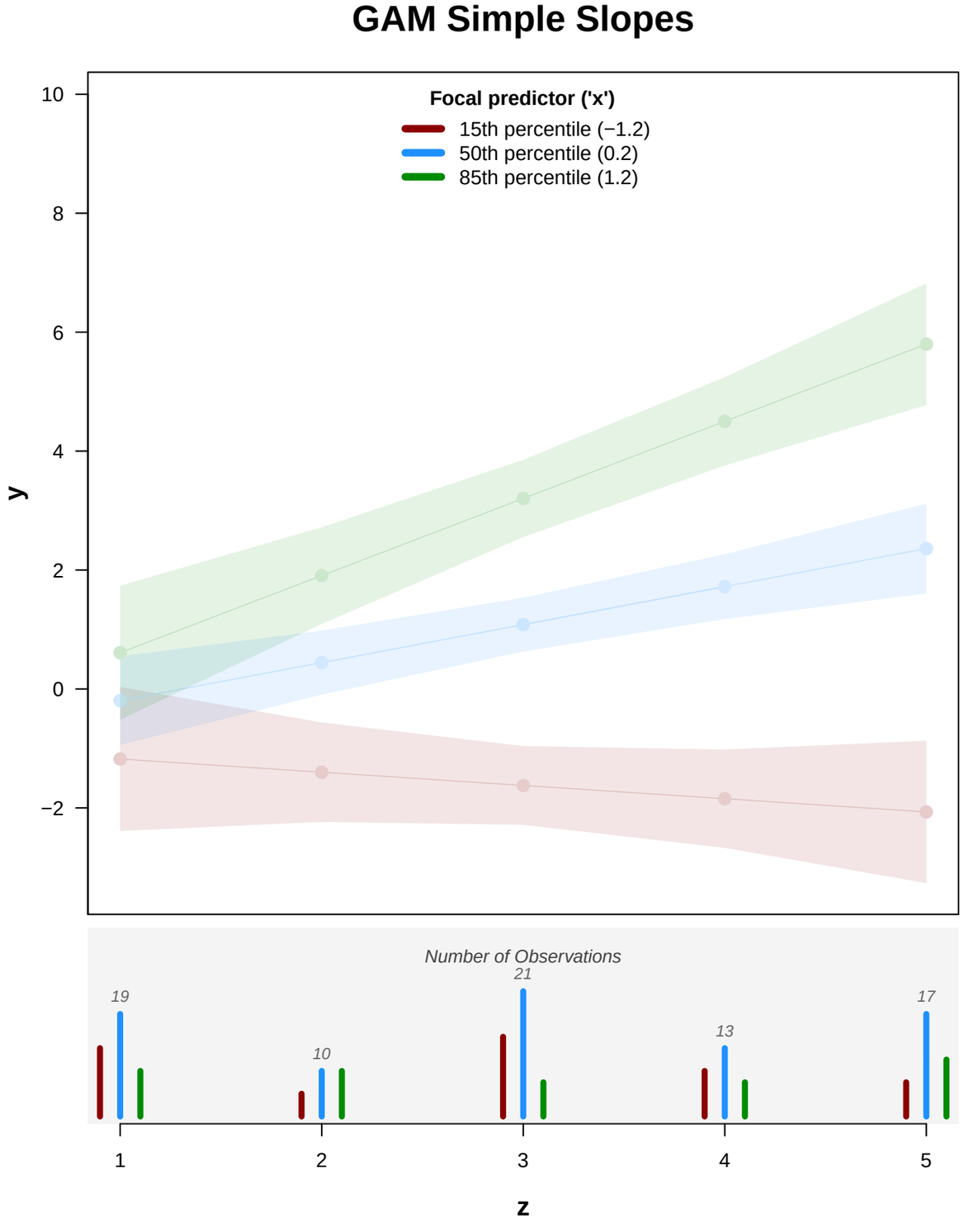
3−level categorical focal x with continuous z; three curves; frequency panels ON.

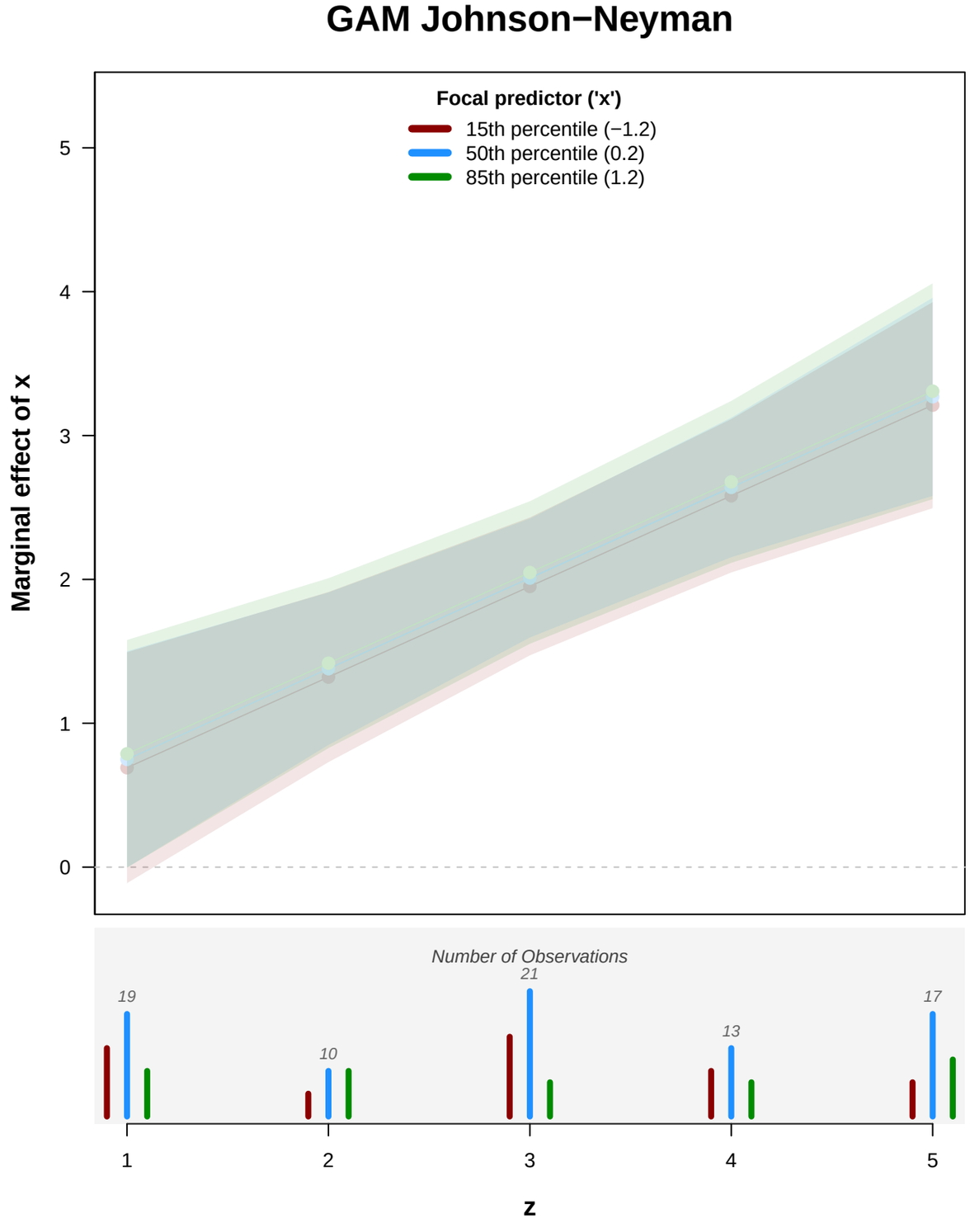

GAM Simple Slopes
Focal predictor ('x')
hi
lo
mid
y
8
6
4
2
0
Number of Observations
5
1
4
13
10
18
15
7
5
2
−2
−1
0
1
2
z


GAM Johnson−Neyman
Focal predictor ('x')
lo − hi
mid − hi
Marginal effect of x
10
5
0
Number of Observations
5
1
4
13
10
18
15
7
5
2
−2
−1
0
1
2
z

Part A (sample sizes) | Page 12 | discrete_z_hist | n=80 (small)

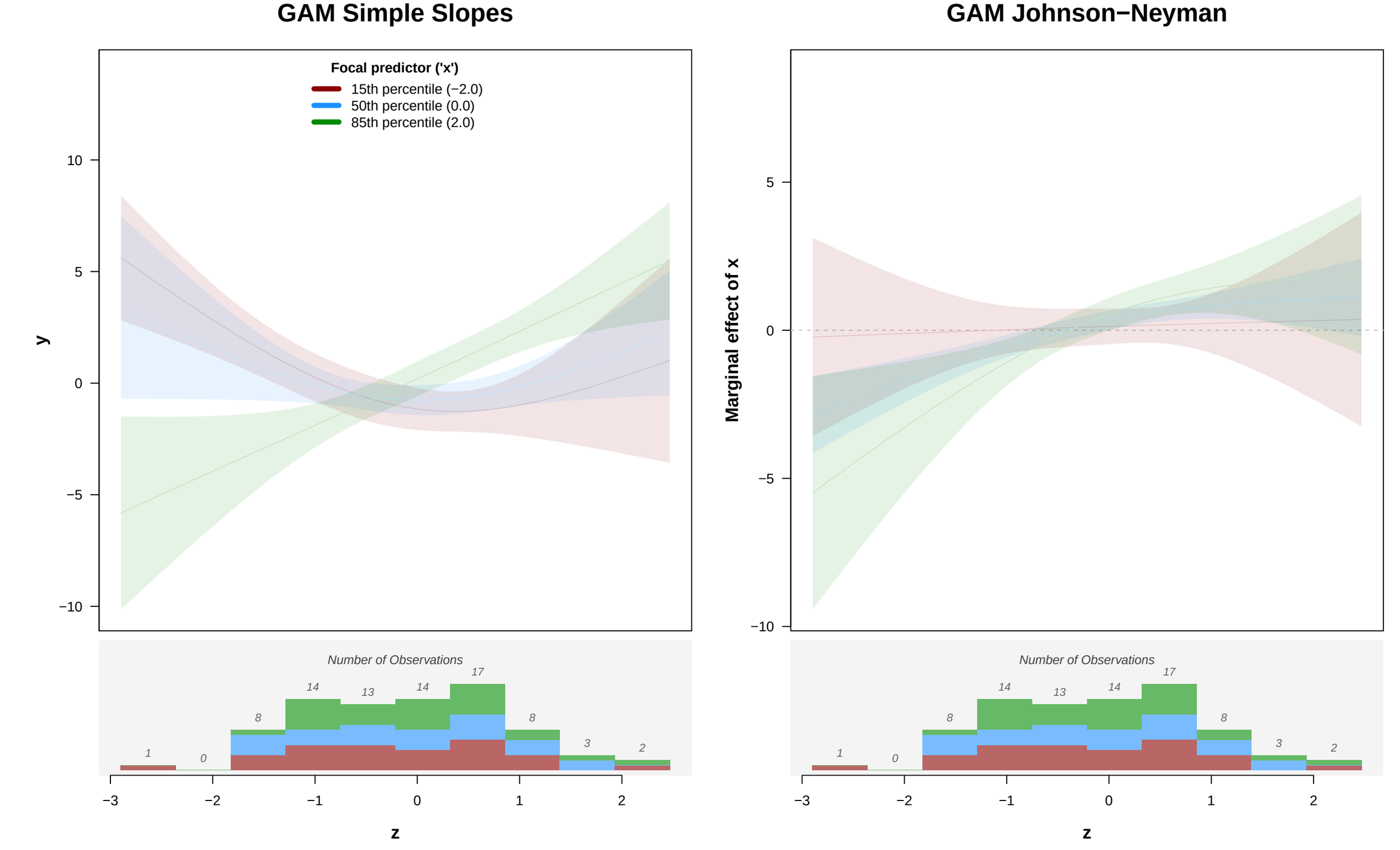
Continuous x with discrete moderator z (5 values); k=3 so GAM can fit; discrete frequency segments.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.2)
50th percentile (0.2)
85th percentile (1.2)
y
10
8
6
4
2
0
−2
Number of Observations
19
10
21
13
17
1
2
3
4
5
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.2)
50th percentile (0.2)
85th percentile (1.2)
Marginal effect of x
5
4
3
2
1
0
Number of Observations
19
10
21
13
17
1
2
3
4
5
z

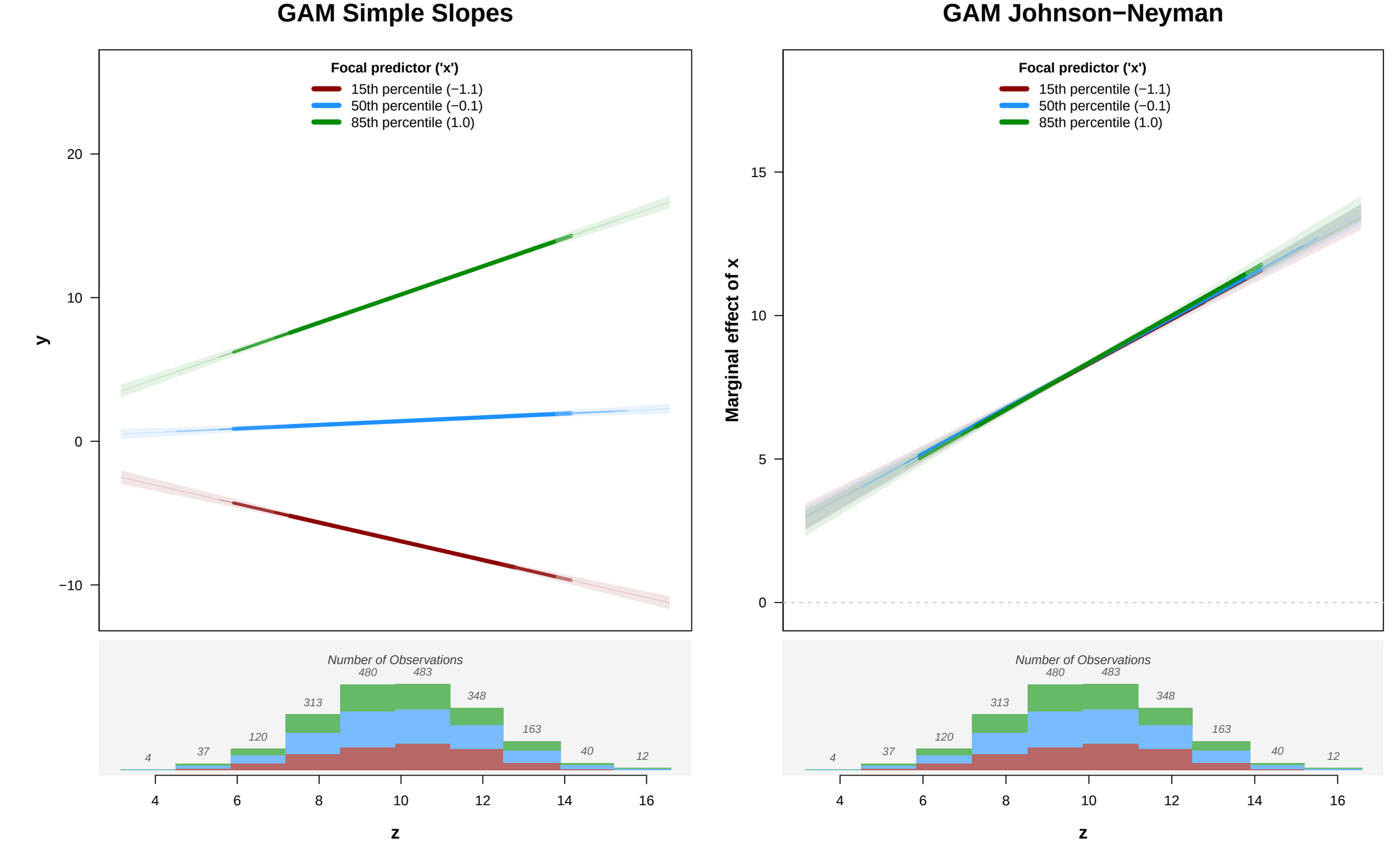
Part A (sample sizes) | Page 13 | discrete_x_hist | n=80 (small)

Discrete focal x (5 values) with continuous z; k=3; spotlight curves with continuous−z frequency bins.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−2.0)
50th percentile (0.0)
85th percentile (2.0)
y
10
5
0
−5
−10
Number of Observations
1
0
8
14
13
14
17
8
3
2
−3
−2
−1
0
1
2
z
GAM Johnson−Neyman
Marginal effect of x
5
0
−5
−10
Number of Observations
1
0
8
14
13
14
17
8
3
2
−3
−2
−1
0
1
2
z

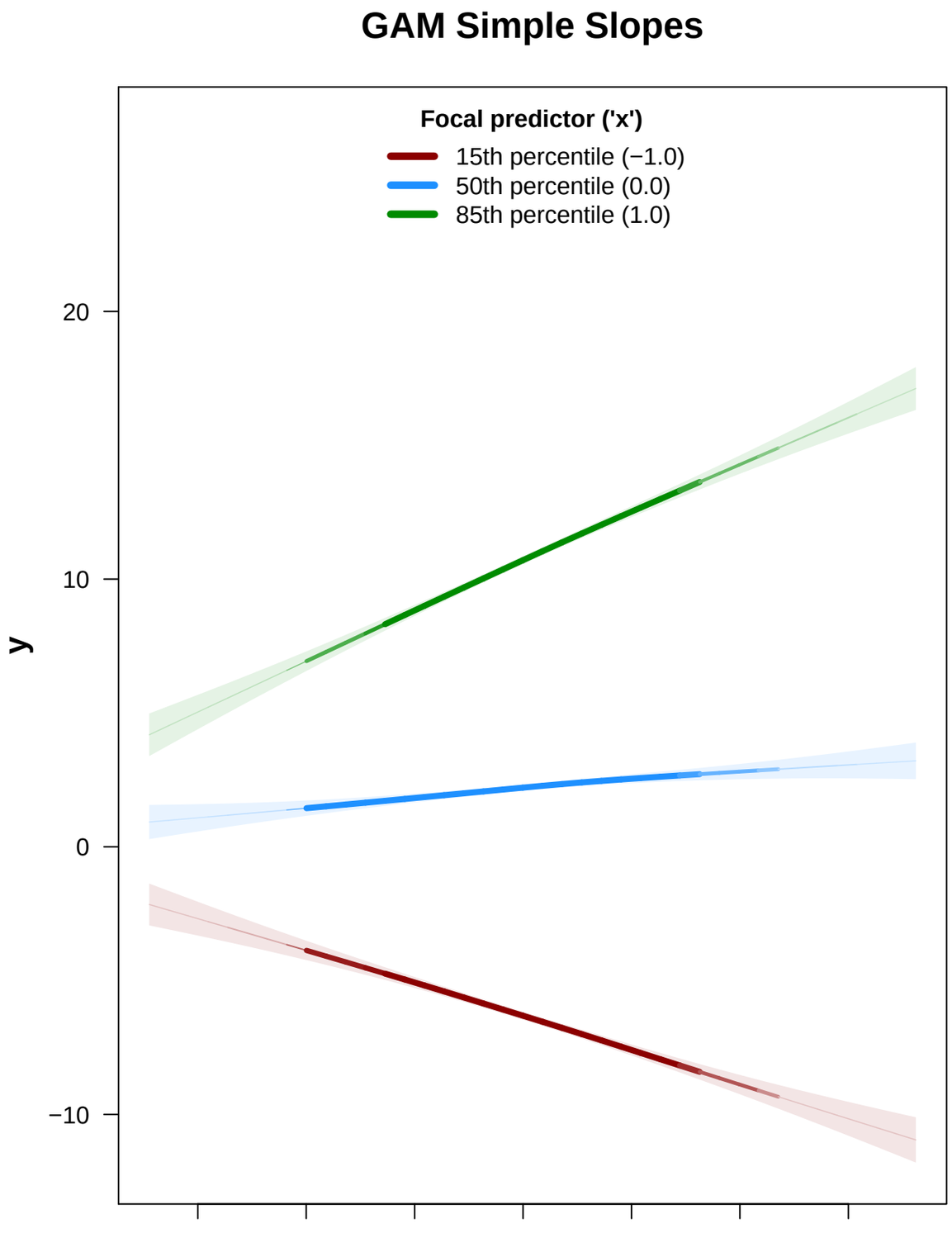
Part A (sample sizes) | Page 14 | continuous_both_hist | n=2000 (big)

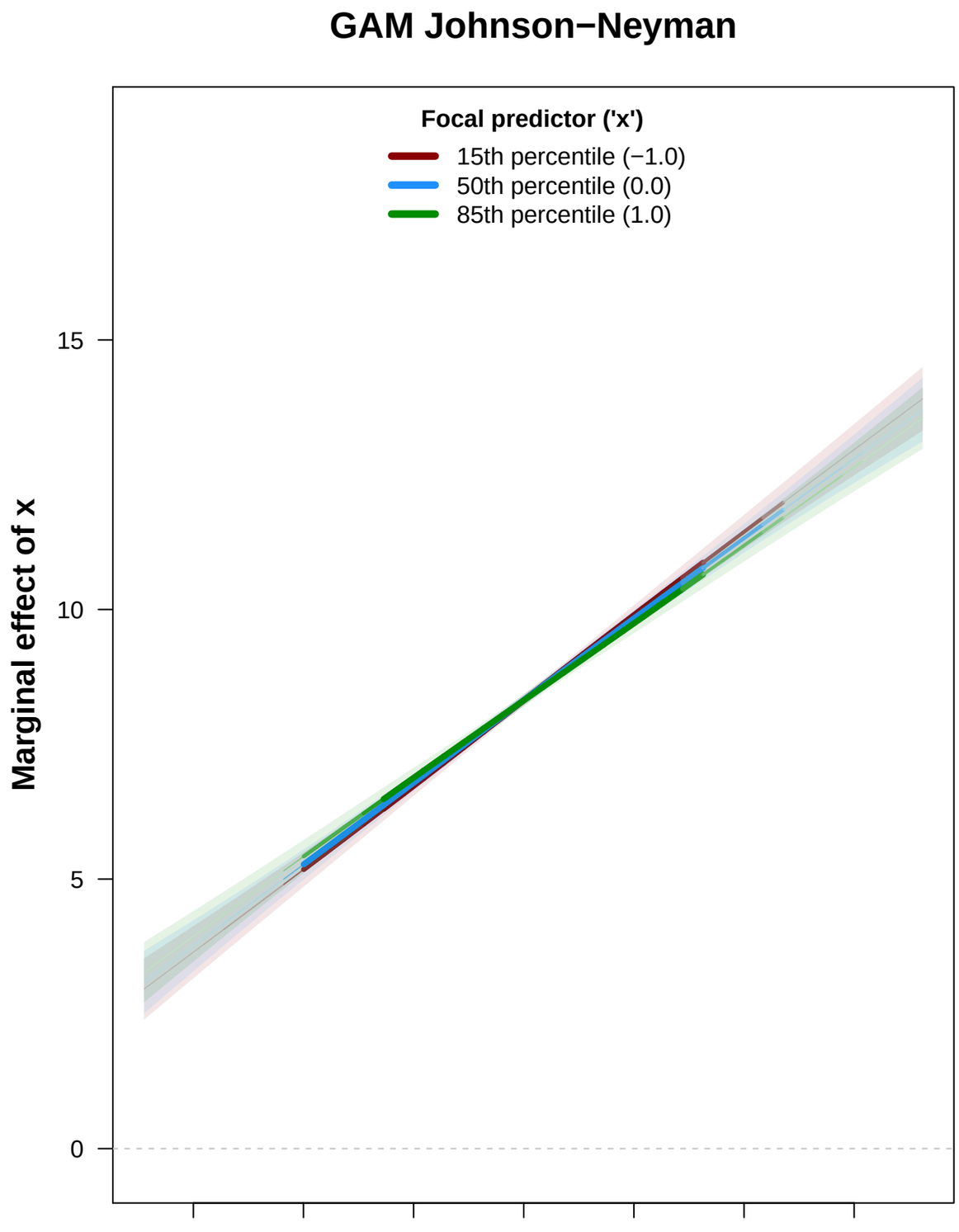
Continuous x & z; default GAM; draw='both'; frequency panels ON. Check stacked hist under both curves, count labels inside gray band, no collisions.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.1)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
4
37
120
313
480
483
348
163
40
12
4
6
8
10
12
14
16
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.1)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
4
37
120
313
480
483
348
163
40
12
4
6
8
10
12
14
16
z

Part A (sample sizes) | Page 15 | continuous_both_nohist | n=2000 (big)

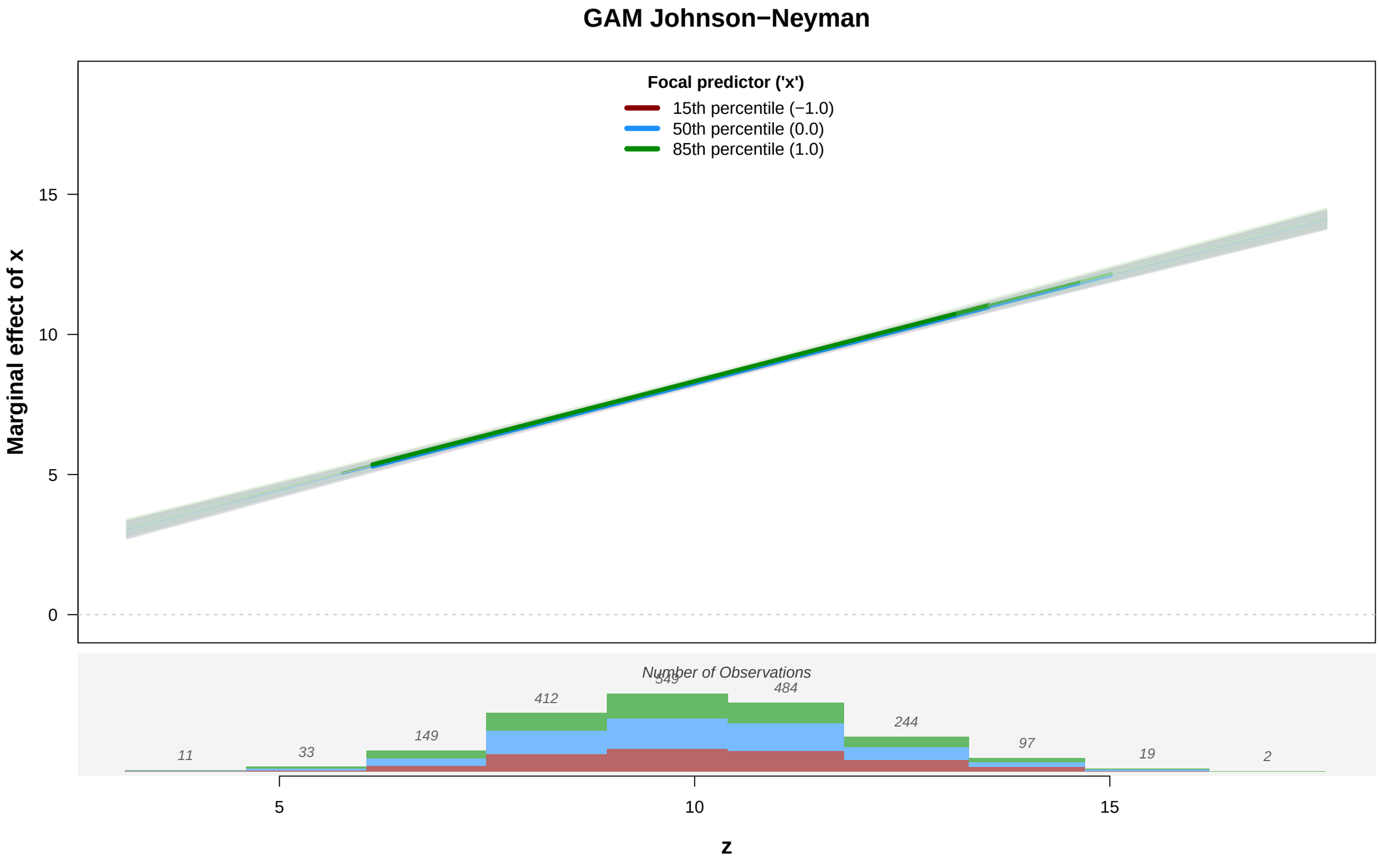
Continuous x & z; default GAM; draw='both'; frequency panels OFF. Baseline layout without the bottom frequency row.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
y
20
10
0
−10


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
Marginal effect of x
15
10
5
0

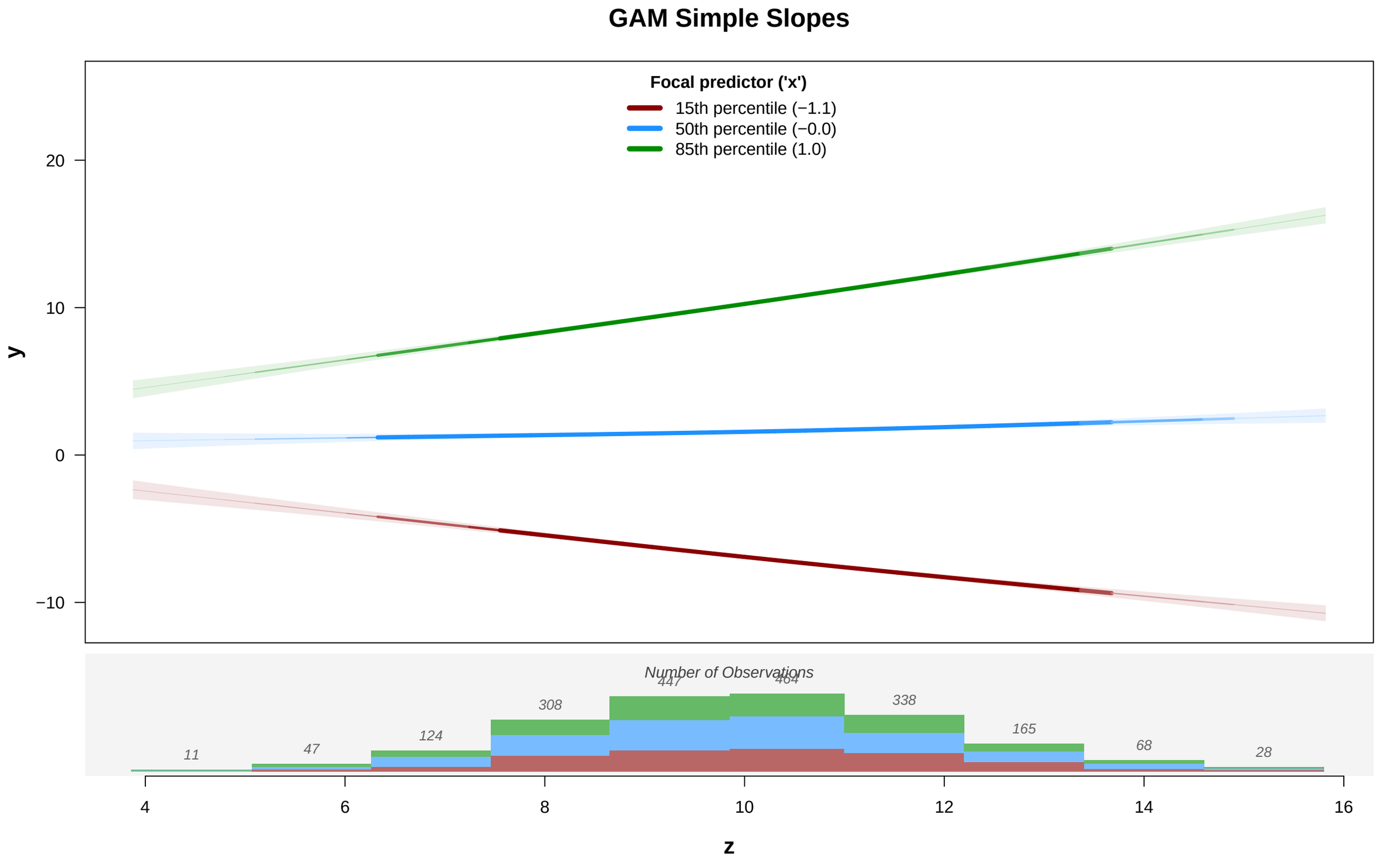
Part A (sample sizes) | Page 16 | continuous_jn_hist | n=2000 (big)

Continuous x & z; single Johnson−Neyman panel with frequency subplot underneath.


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
11
33
149
412
549
484
244
97
19
2
5
10
15
z

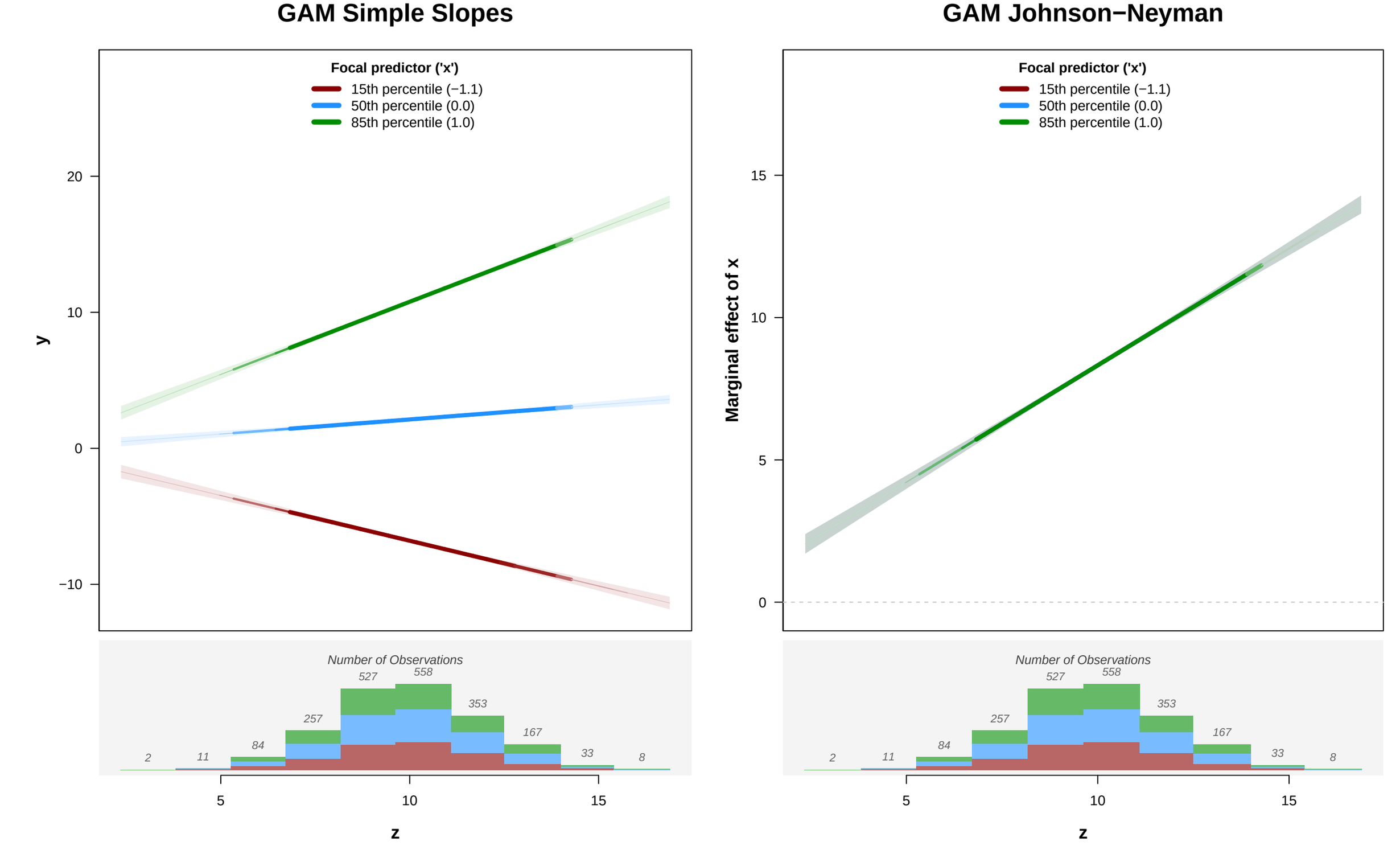
Part A (sample sizes) | Page 17 | continuous_slopes_hist | n=2000 (big)

Continuous x & z; single simple−slopes panel with frequency subplot underneath.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.0)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
11
47
124
308
447
464
338
165
68
28
4
6
8
10
12
14
16
z

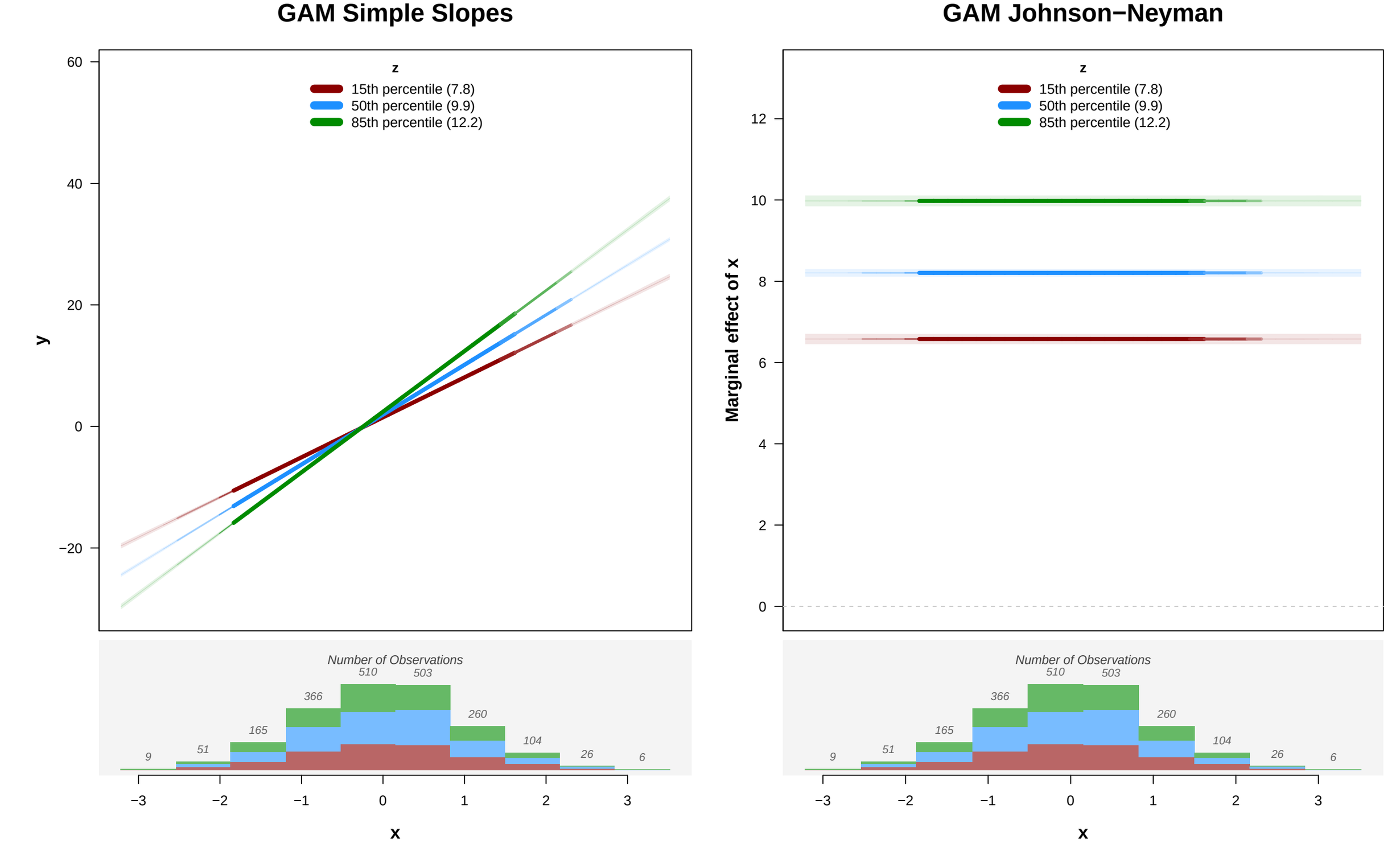
Part A (sample sizes) | Page 18 | continuous_linear_hist | n=2000 (big)

Continuous x & z; model='linear' (lm2) instead of GAM; both panels with frequency subplots.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (0.0)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
2
11
84
257
527
558
353
167
33
8
5
10
15
z
GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (0.0)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
2
11
84
257
527
558
353
167
33
8
5
10
15
z

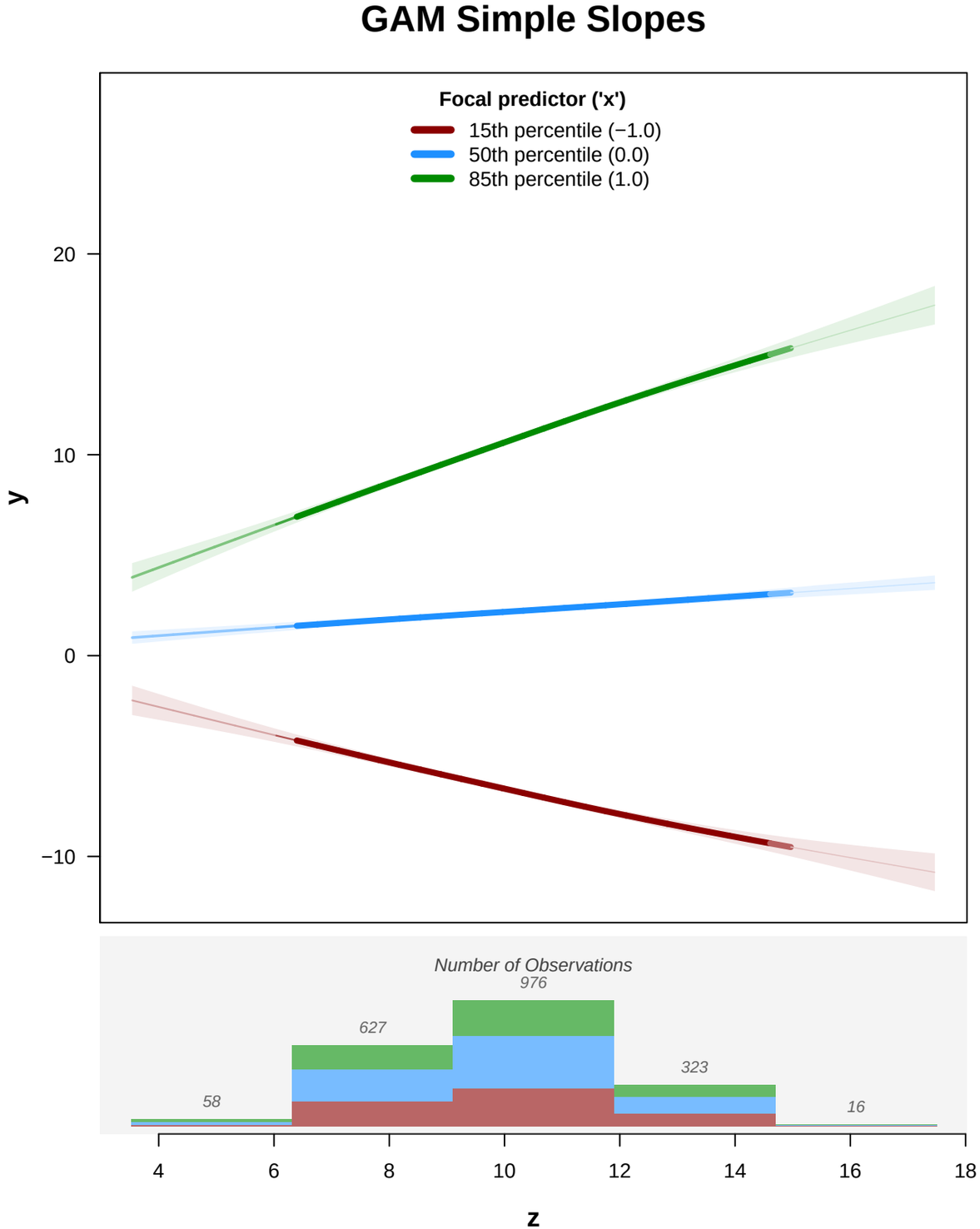
Part A (sample sizes) | Page 19 | continuous_x_on_axis_hist | n=2000 (big)

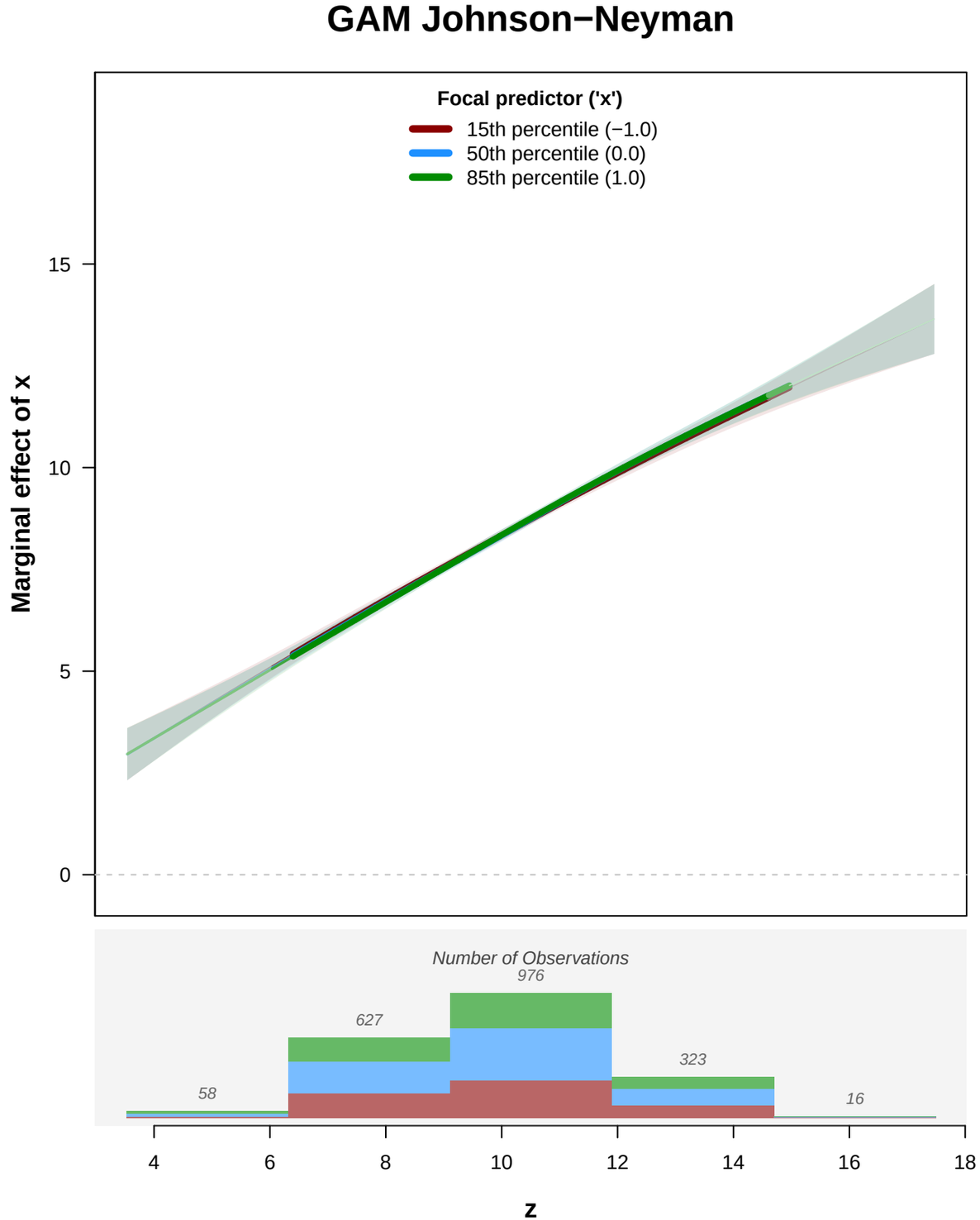
Continuous x & z; moderator.on.x.axis=FALSE so focal x is on the x−axis; frequency panels still aligned to that axis.


GAM Simple Slopes
z
15th percentile (7.8)
50th percentile (9.9)
85th percentile (12.2)
y
60
40
20
0
−20
Number of Observations
9
51
165
366
510
503
260
104
26
6
−3
−2
−1
0
1
2
3
x
GAM Johnson−Neyman
Marginal effect of x
12
10
8
6
4
2
0

Part A (sample sizes) | Page 20 | continuous_nbins_5 | n=2000 (big)

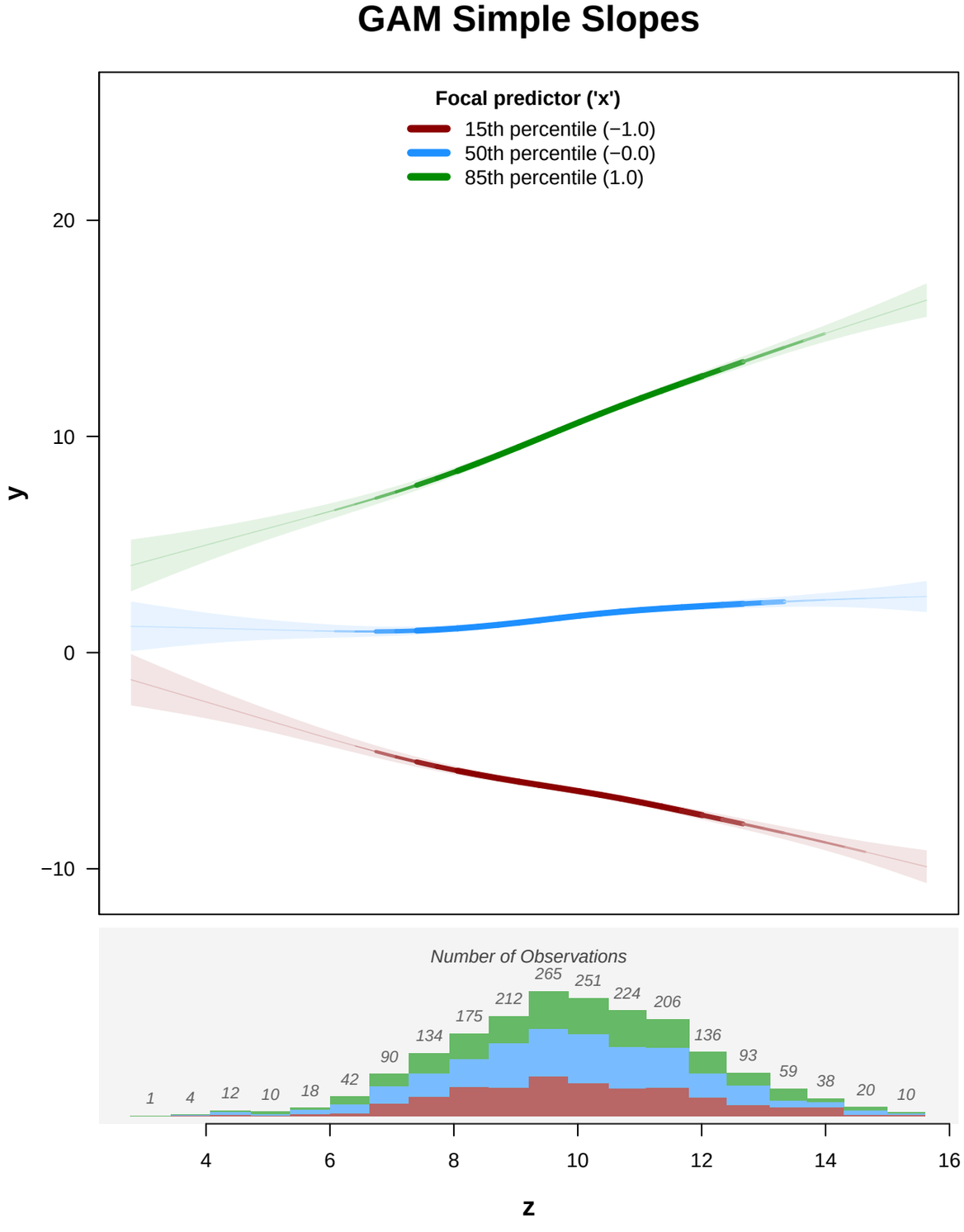
Continuous x & z; n.bin.continuous=5 (coarser frequency bins). Check fewer, wider stacked bars and readable count labels.

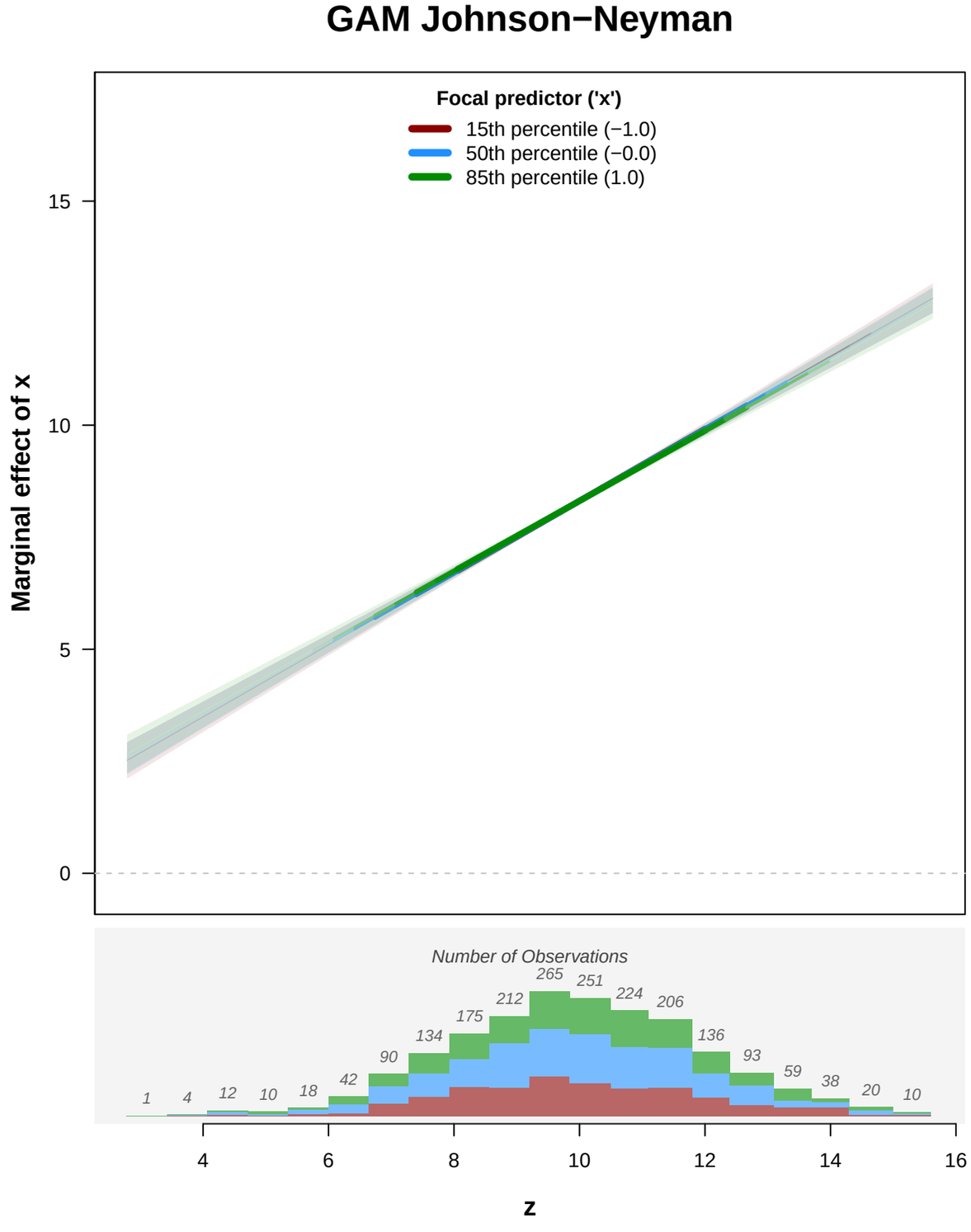

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
976
627
323
58
16
4
6
8
10
12
14
16
18
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
976
627
323
58
16
4
6
8
10
12
14
16
18
z

Part A (sample sizes) | Page 21 | continuous_nbins_20 | n=2000 (big)

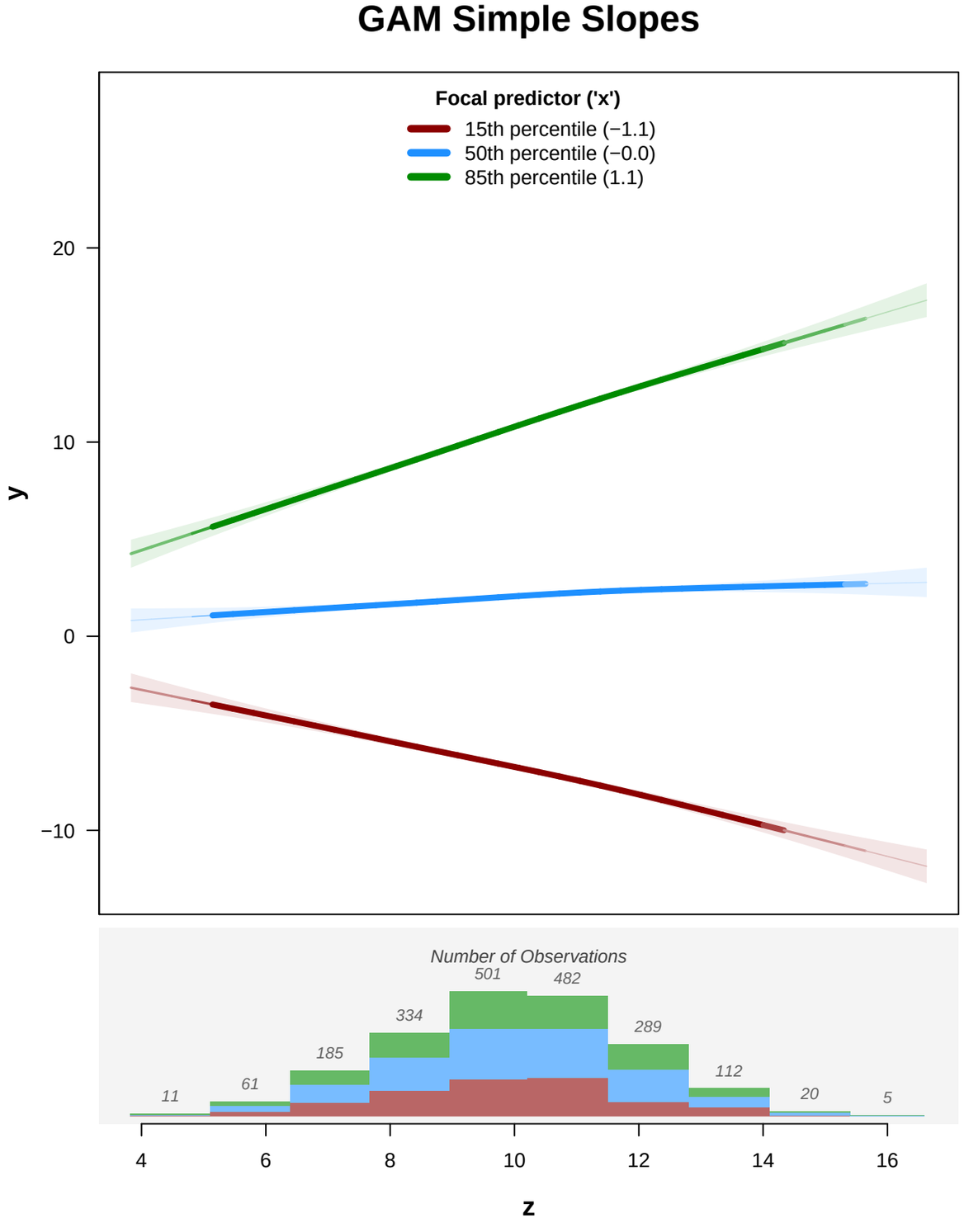
Continuous x & z; n.bin.continuous=20 (finer frequency bins). Check denser bars still readable at small and big n.

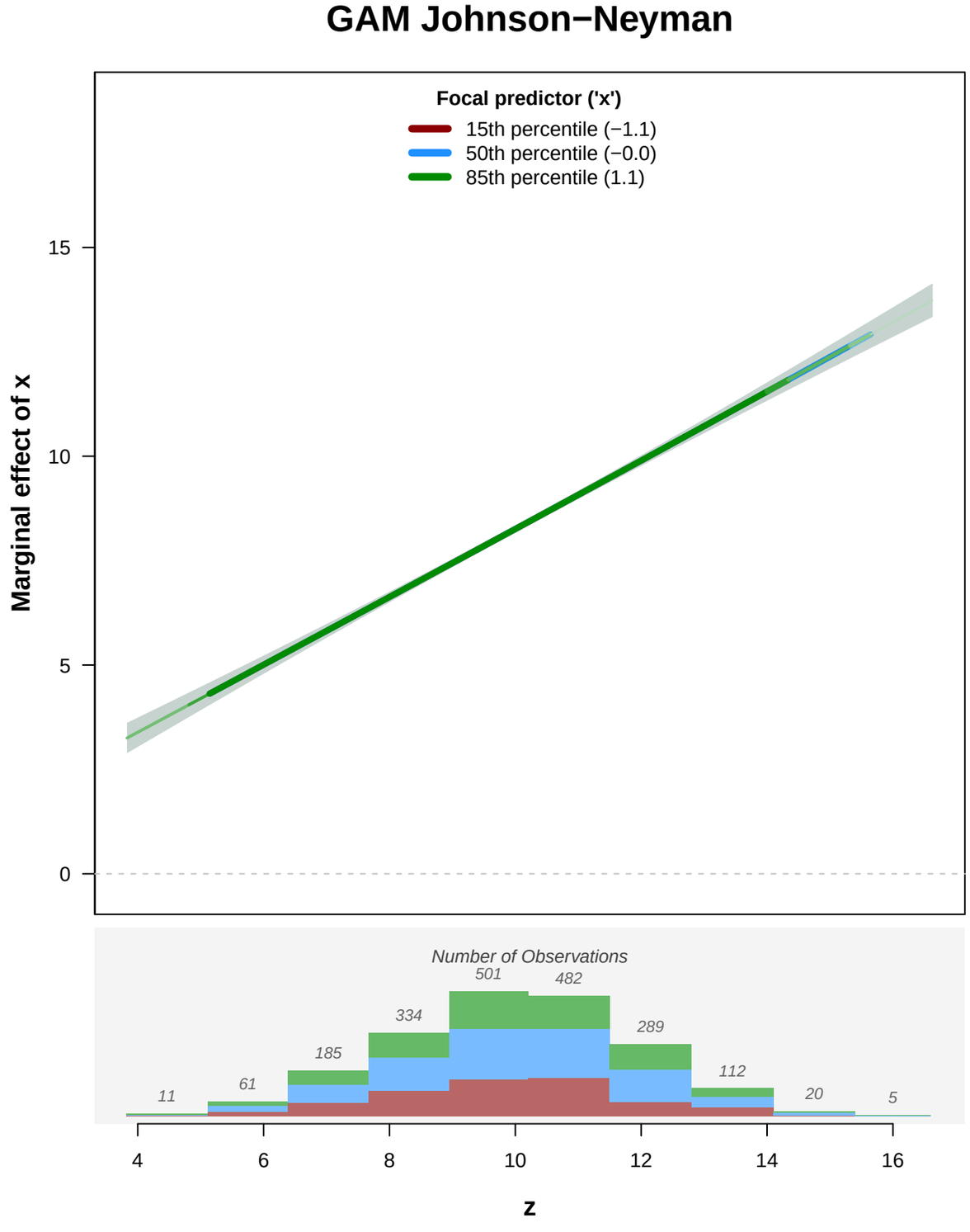

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
y
20
10
0
−10
Number of Observations
1
4
12
10
18
42
90
134
175
212
265
251
224
206
136
93
59
38
20
10
4
6
8
10
12
14
16
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
Marginal effect of x
15
10
5
0
Number of Observations
1
4
12
10
18
42
90
134
175
212
265
251
224
206
136
93
59
38
20
10
4
6
8
10
12
14
16
z

Part A (sample sizes) | Page 22 | continuous_nmax_10 | n=2000 (big)

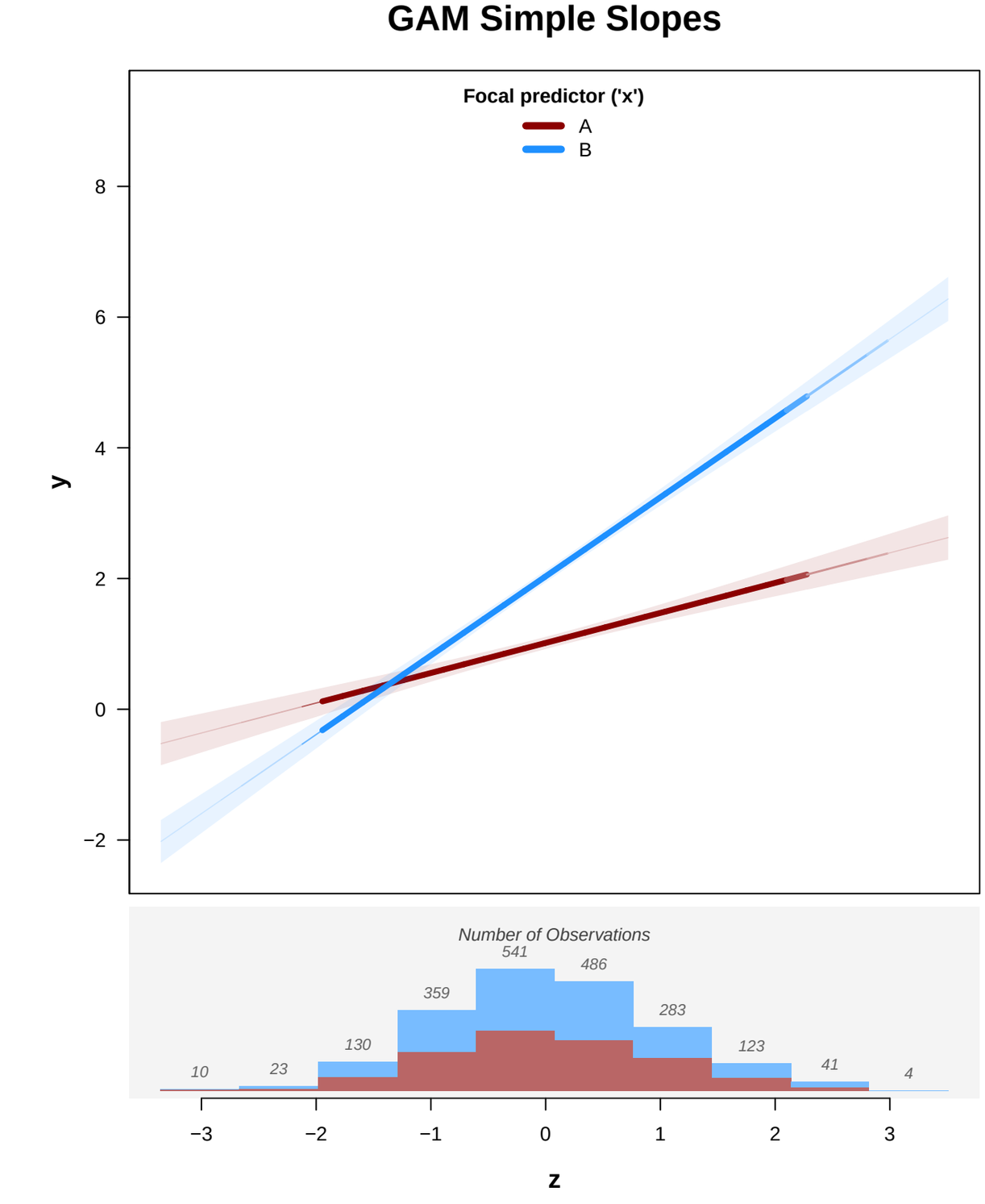
Continuous x & z; n.max=10 so line darkness/width saturates early. Check that sparse bins stay light and dense bins look fully saturated.

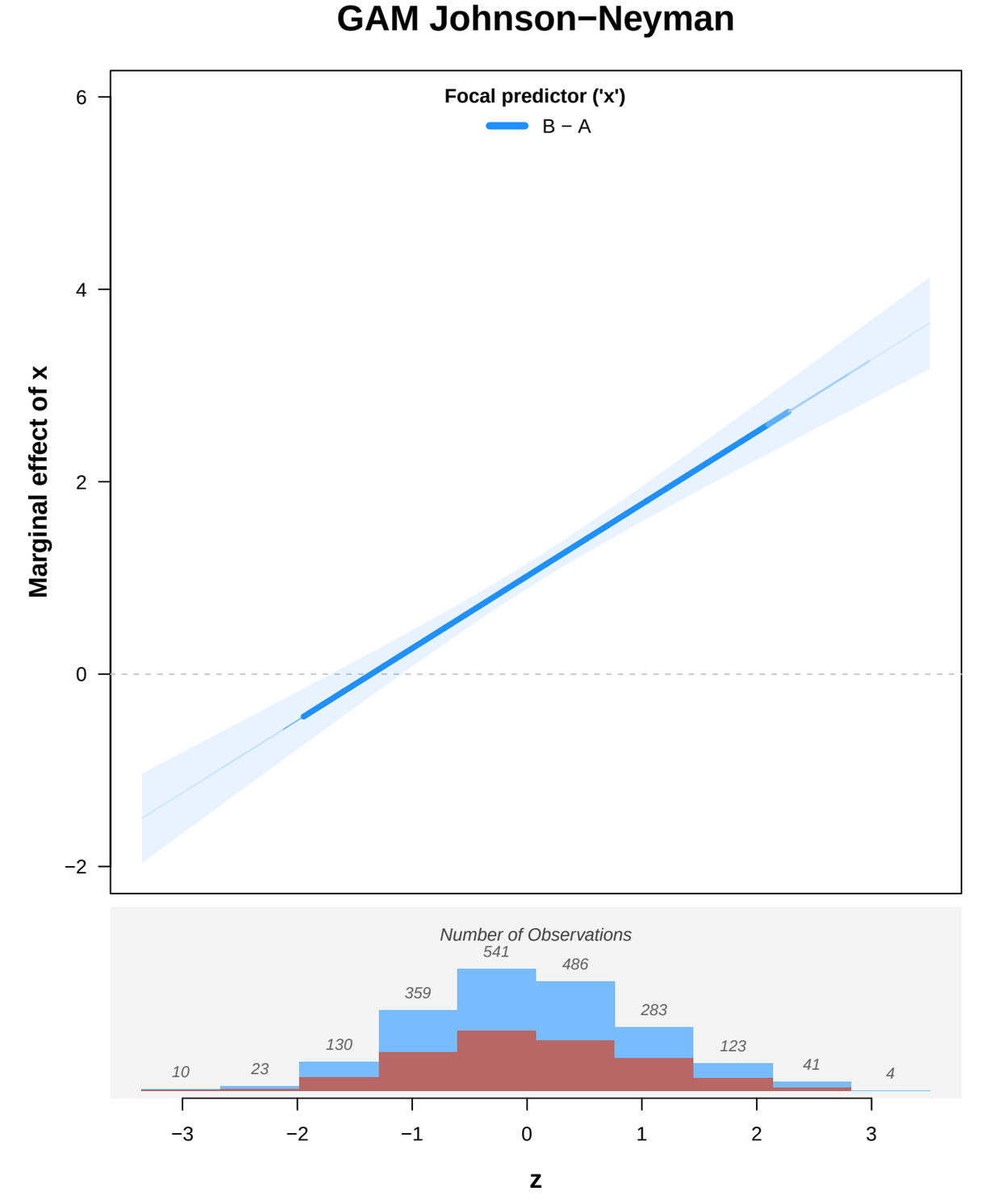

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.0)
85th percentile (1.1)
y
20
10
0
−10
Number of Observations
11
61
185
334
501
482
289
112
20
5
4
6
8
10
12
14
16
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.0)
85th percentile (1.1)
Marginal effect of x
15
10
5
0
Number of Observations
11
61
185
334
501
482
289
112
20
5
4
6
8
10
12
14
16
z

Part A (sample sizes) | Page 23 | binary_x_hist | n=2000 (big)

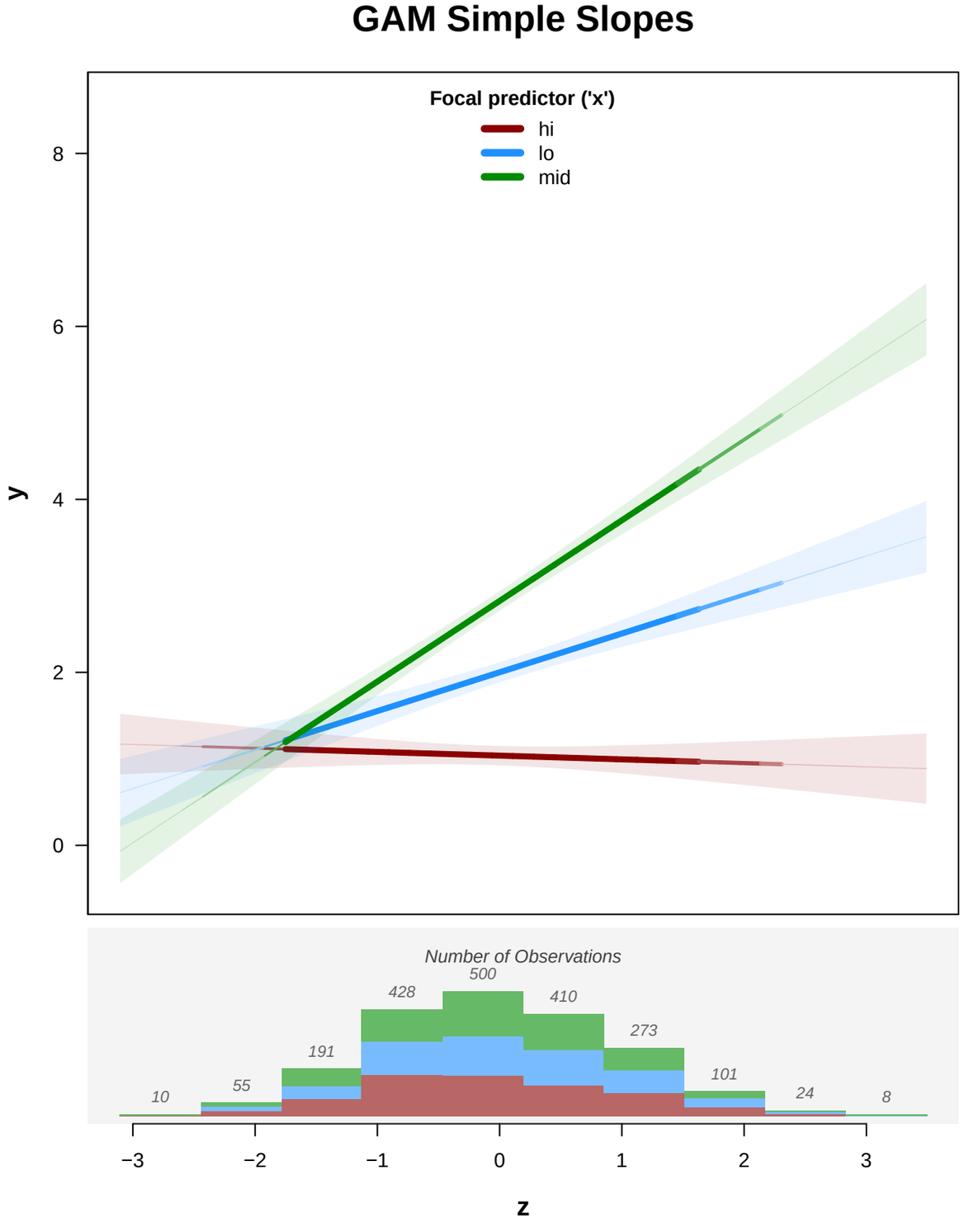
Binary/categorical focal x (2 levels) with continuous z; two colored curves; frequency panels ON.

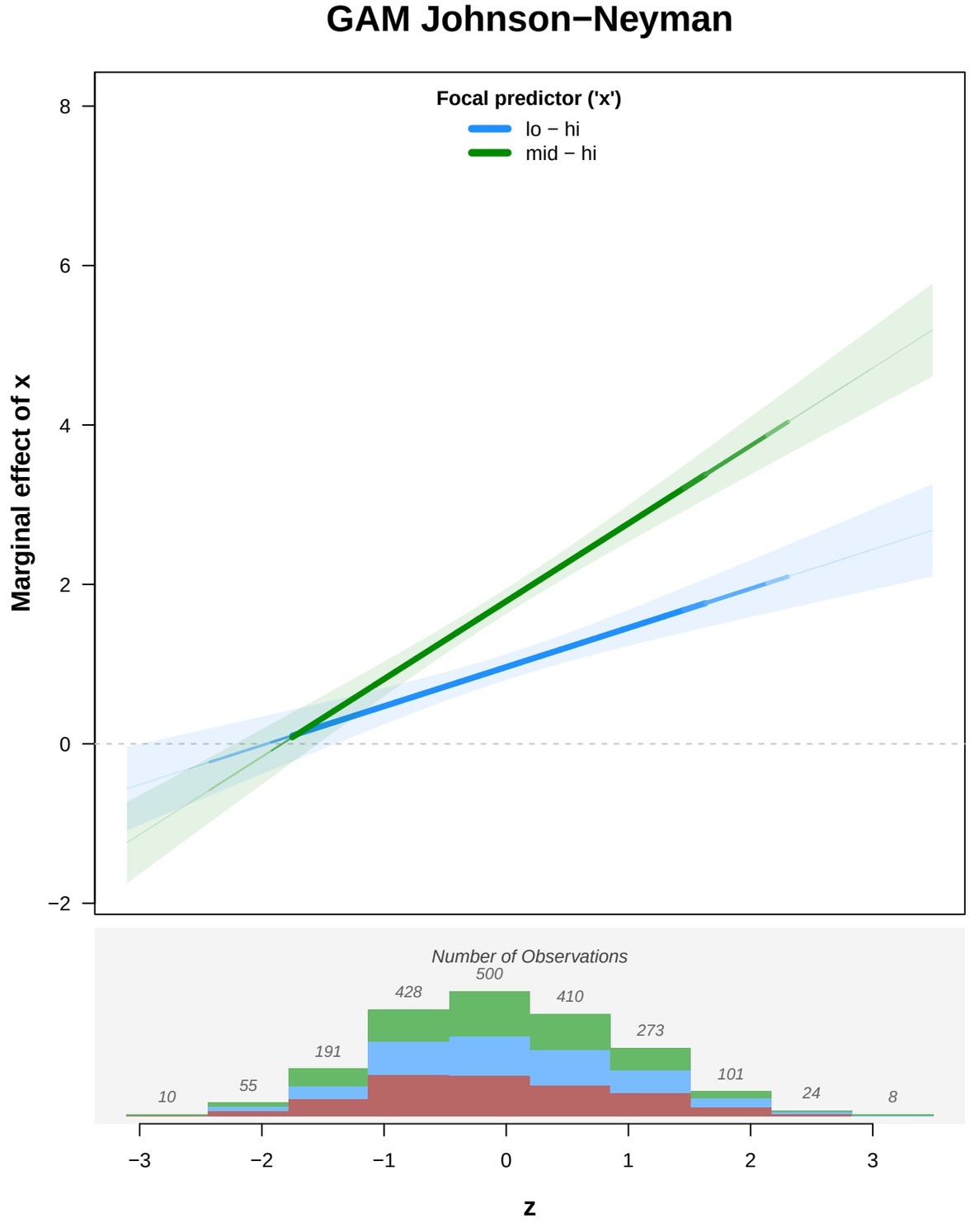

GAM Simple Slopes
Focal predictor ('x')
A
B
y
8
6
4
2
0
−2
Number of Observations
10
23
130
359
541
486
283
123
41
4
−3
−2
−1
0
1
2
3
z


GAM Johnson−Neyman
Focal predictor ('x')
B − A
Marginal effect of x
6
4
2
0
−2
Number of Observations
10
23
130
359
541
486
283
123
41
4
−3
−2
−1
0
1
2
3
z

Part A (sample sizes) | Page 24 | ternary_x_hist | n=2000 (big)

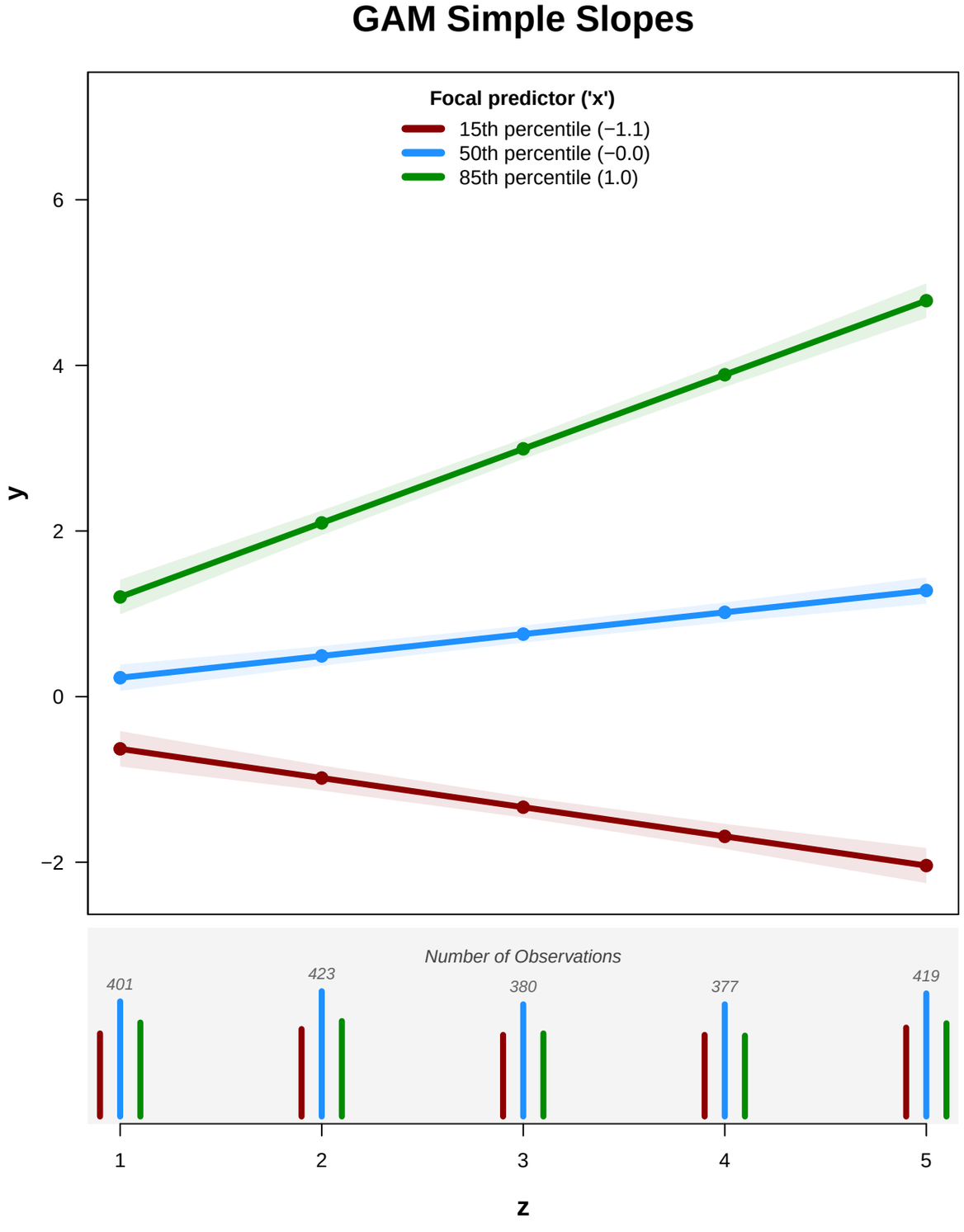
3−level categorical focal x with continuous z; three curves; frequency panels ON.

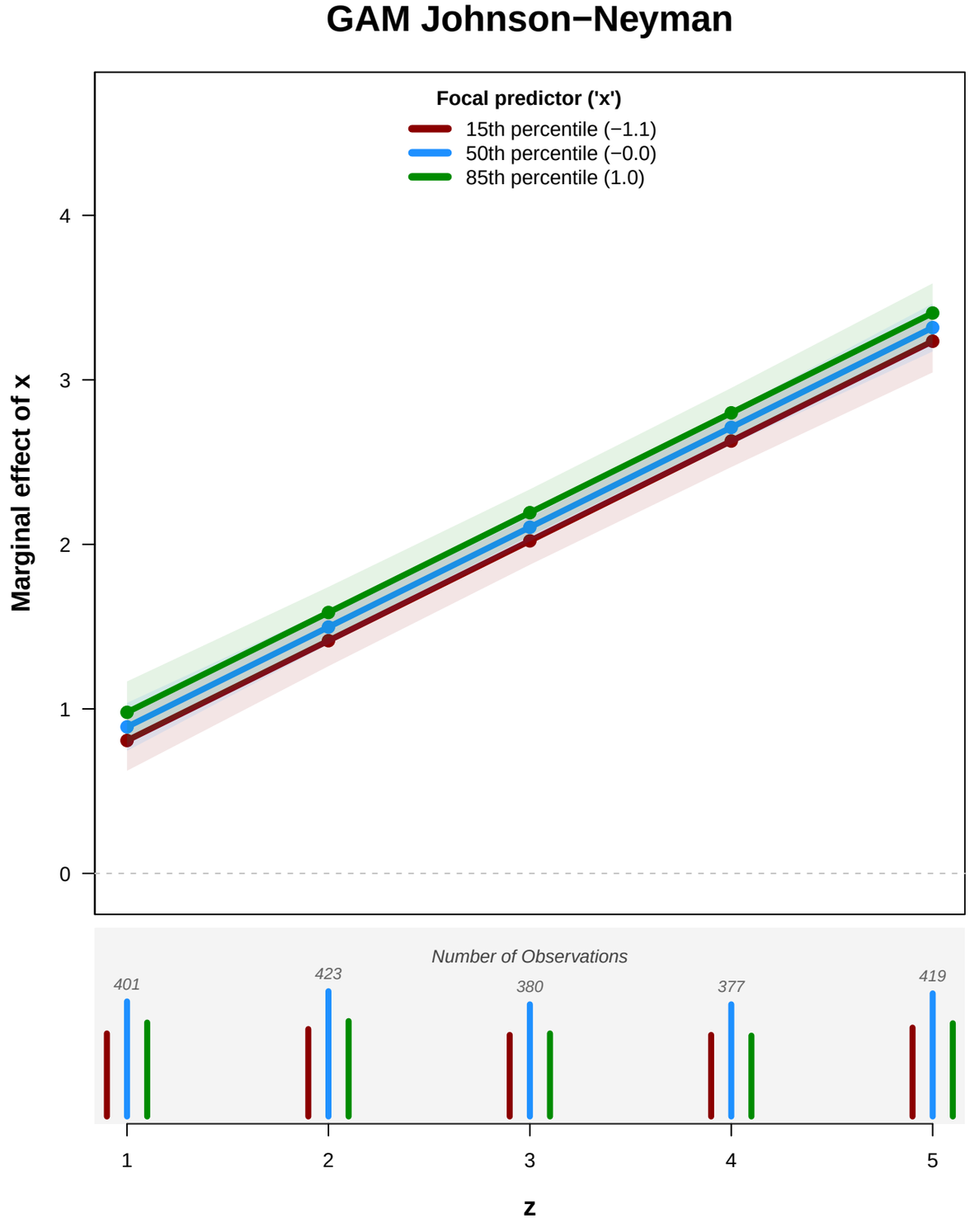

GAM Simple Slopes
Focal predictor ('x')
hi
lo
mid
y
8
6
4
2
0
Number of Observations
10
55
191
428
500
410
273
101
24
8
−3
−2
−1
0
1
2
3
z


GAM Johnson−Neyman
Focal predictor ('x')
lo − hi
mid − hi
Marginal effect of x
8
6
4
2
0
−2
Number of Observations
10
55
191
428
500
410
273
101
24
8
−3
−2
−1
0
1
2
3
z

Part A (sample sizes) | Page 25 | discrete_z_hist | n=2000 (big)

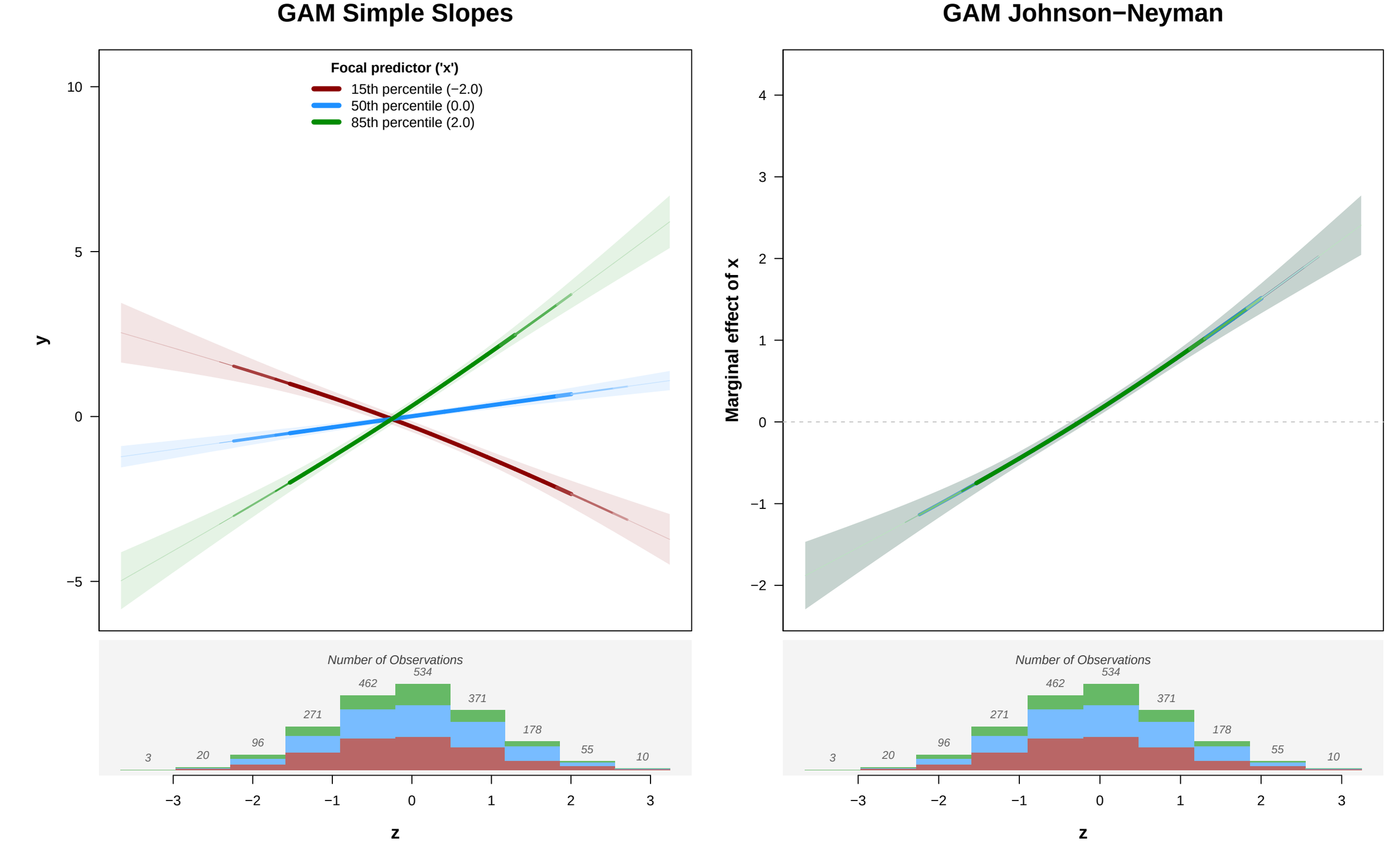
Continuous x with discrete moderator z (5 values); k=3 so GAM can fit; discrete frequency segments.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.0)
85th percentile (1.0)
y
6
4
2
0
−2
Number of Observations
401
423
380
377
419
1
2
3
4
5
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.1)
50th percentile (−0.0)
85th percentile (1.0)
Marginal effect of x
4
3
2
1
0
Number of Observations
401
423
380
377
419
1
2
3
4
5
z

Part A (sample sizes) | Page 26 | discrete_x_hist | n=2000 (big)

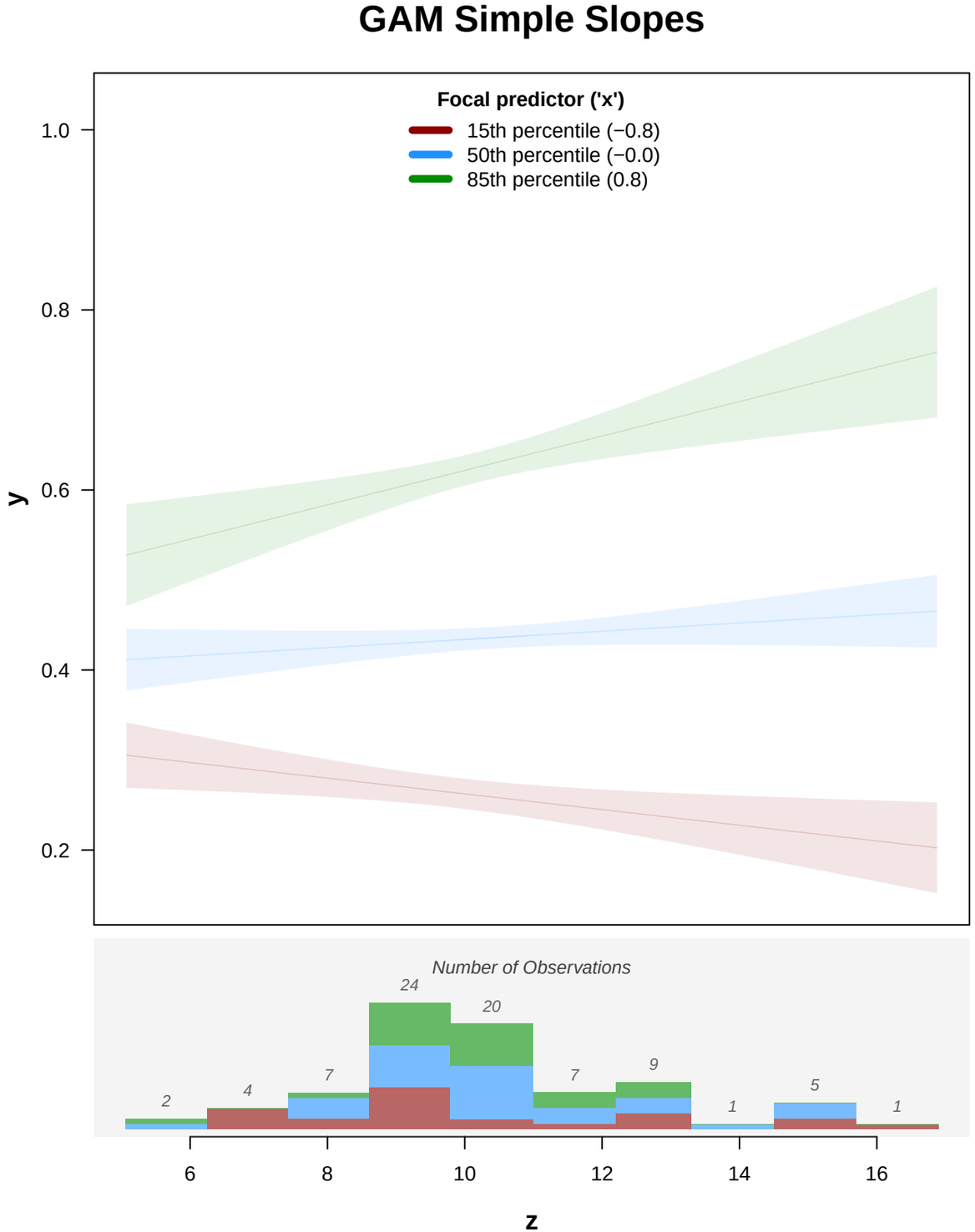
Discrete focal x (5 values) with continuous z; k=3; spotlight curves with continuous−z frequency bins.

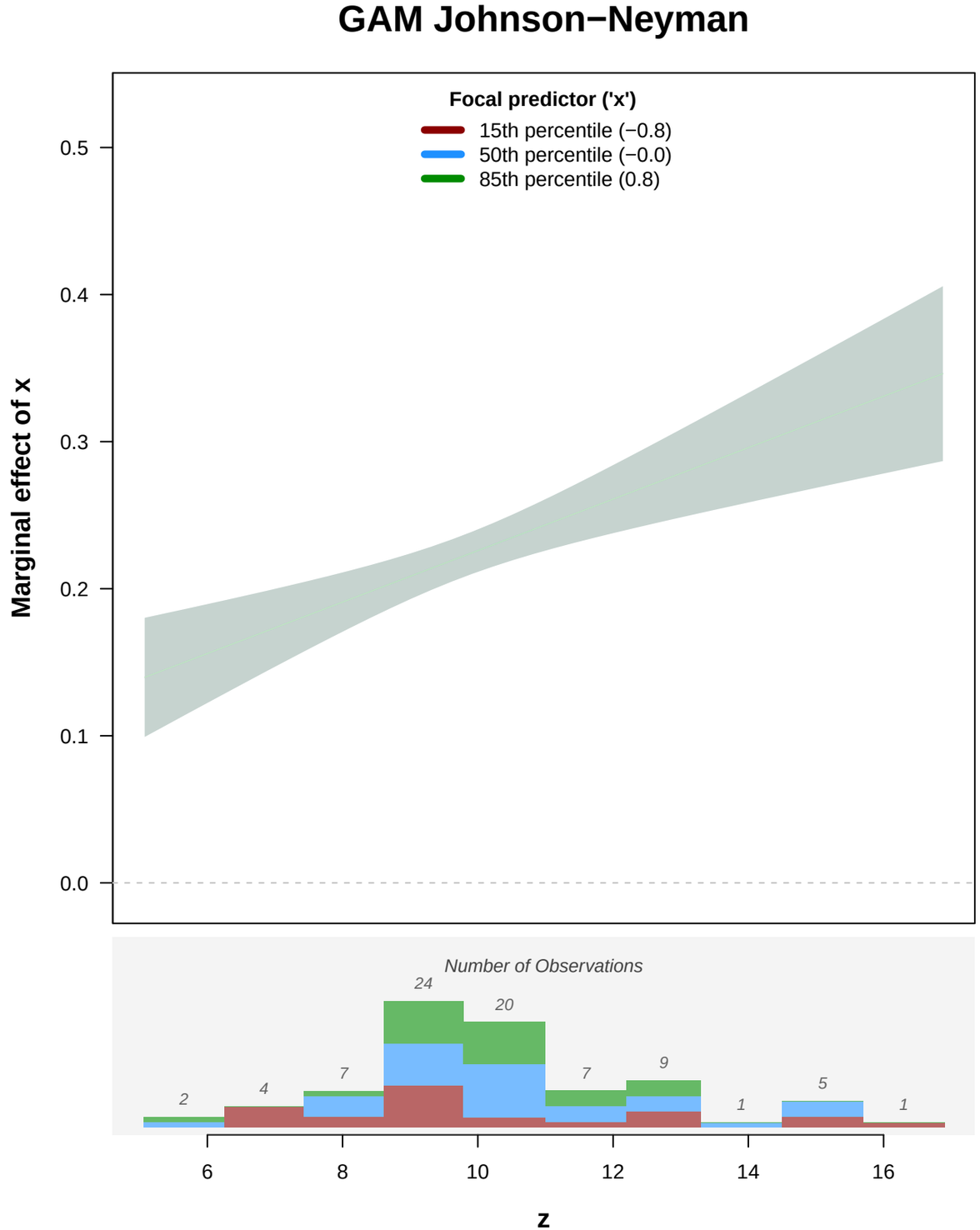

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−2.0)
50th percentile (0.0)
85th percentile (2.0)
y
10
5
0
−5
Number of Observations
3
20
96
271
462
534
371
178
55
10
−3
−2
−1
0
1
2
3
z
GAM Johnson−Neyman
Marginal effect of x
4
3
2
1
0
−1
−2
Number of Observations
3
20
96
271
462
534
371
178
55
10
−3
−2
−1
0
1
2
3
z

Part B (y−values) | Page 27 | y_0_1 | y in [0, 1] | n=80 (small)

DV scaled to [0, 1]. Check y−tick decimals and ylab spacing (ticks like 0.0, 0.2, …).

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

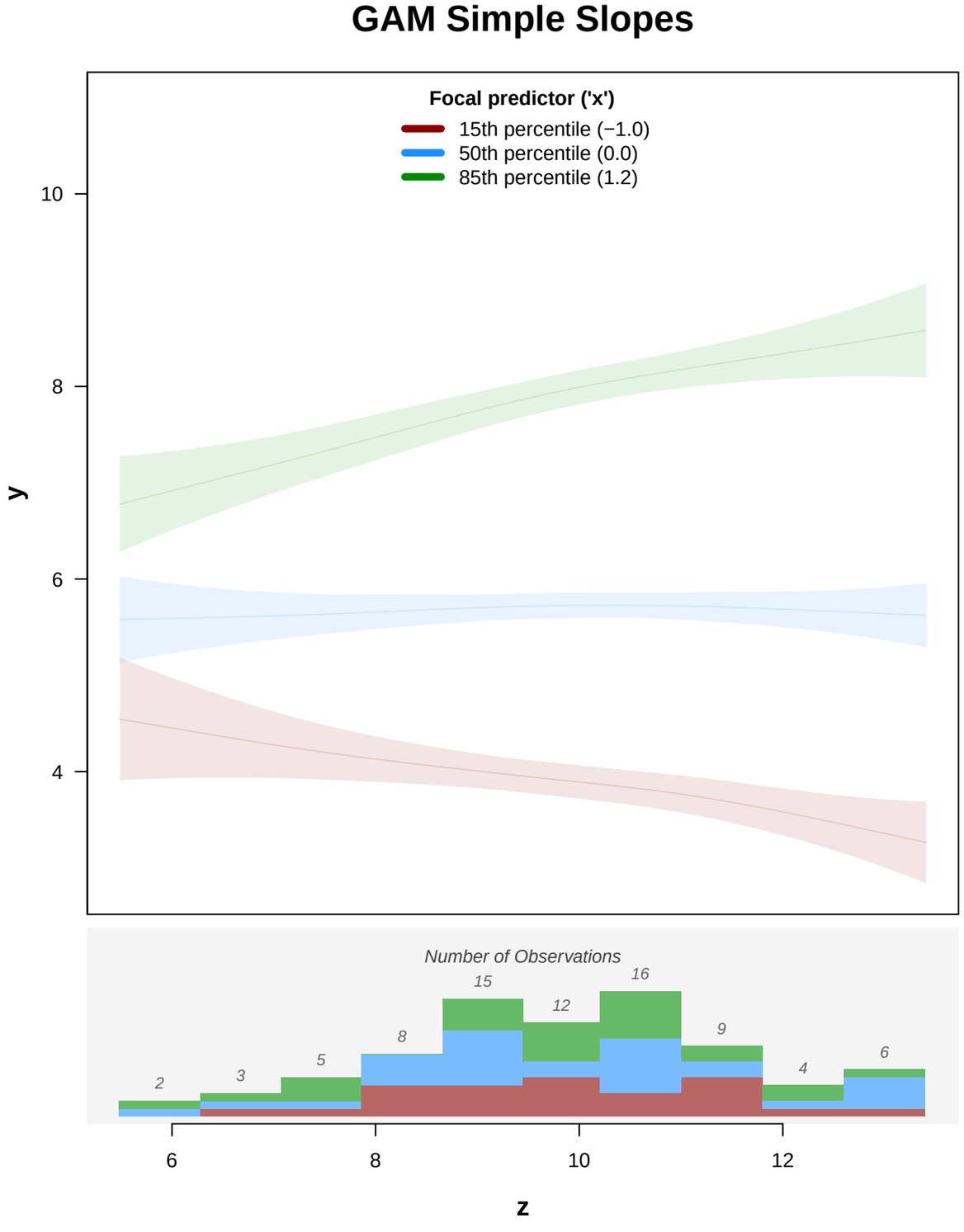

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−0.8)
50th percentile (−0.0)
85th percentile (0.8)
y
1.0
0.8
0.6
0.4
0.2
Number of Observations
2
4
7
24
20
7
9
1
5
1
6
8
10
12
14
16
z

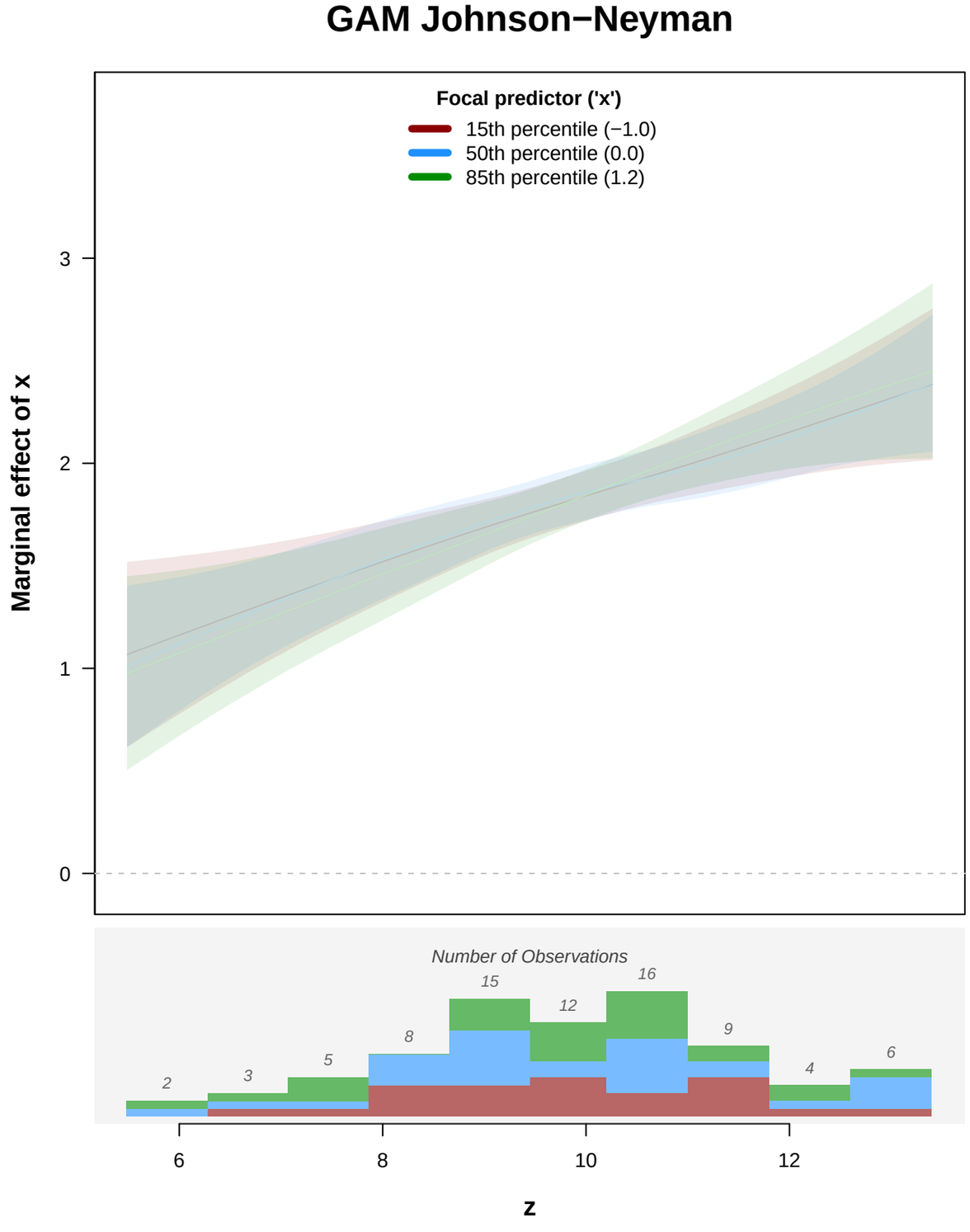

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−0.8)
50th percentile (−0.0)
85th percentile (0.8)
Marginal effect of x
0.5
0.4
0.3
0.2
0.1
0.0
Number of Observations
2
4
7
24
20
7
9
1
5
1
6
8
10
12
14
16
z

Part B (y−values) | Page 28 | y_1_10 | y in [1, 10] | n=80 (small)

DV scaled to [1, 10]. Check short integer/decimal ticks and ylab clearance.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

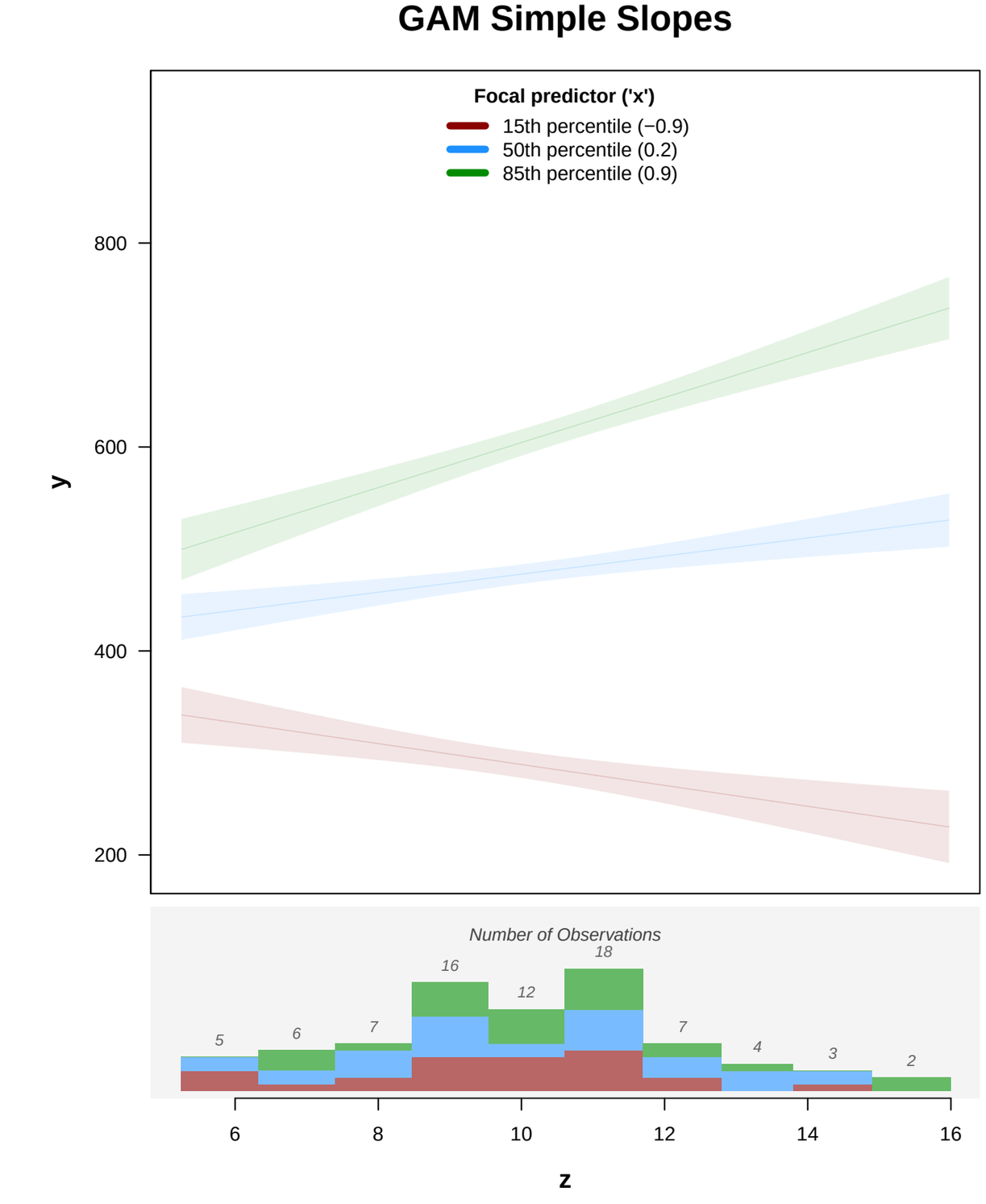

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.2)
y
10
8
6
4
Number of Observations
2
3
5
8
15
12
16
9
4
6
6
8
10
12
z

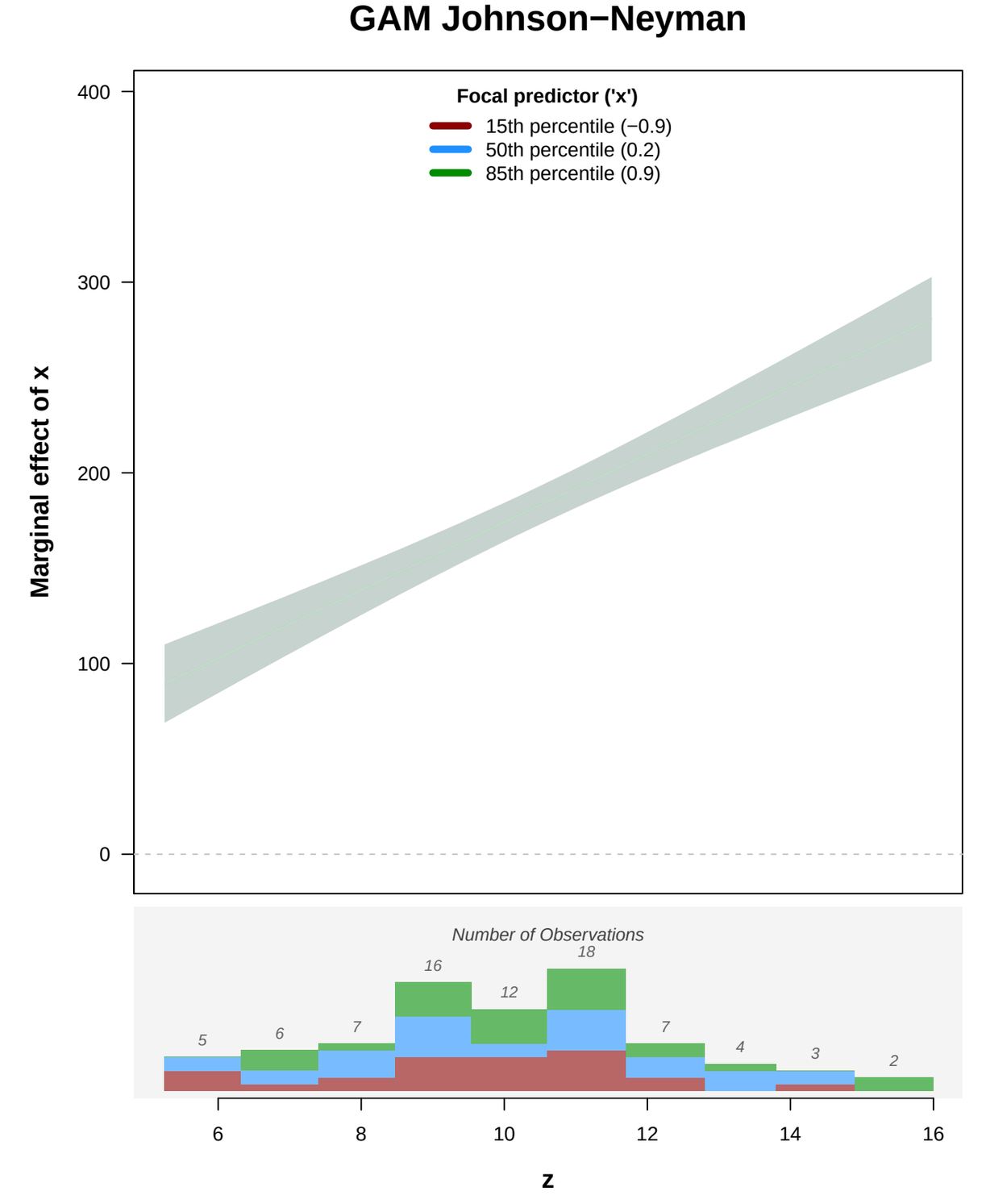

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.2)
Marginal effect of x
3
2
1
0
Number of Observations
2
3
5
8
15
12
16
9
4
6
6
8
10
12
z

Part B (y−values) | Page 29 | y_1_1000 | y in [1, 1000] | n=80 (small)

DV scaled to [1, 1000]. Check medium−width tick labels (e.g. 200, 400) vs ylab.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

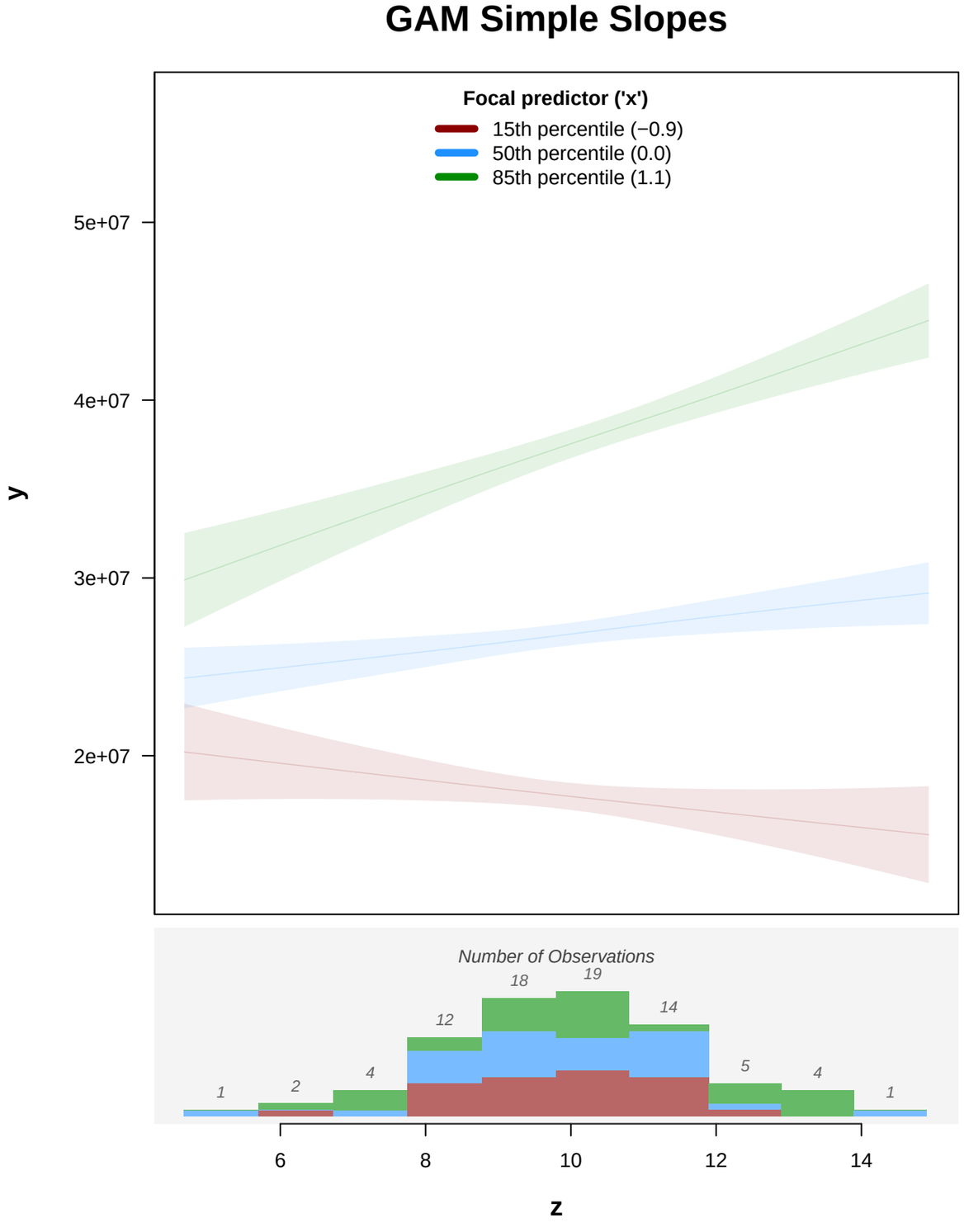

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.2)
85th percentile (0.9)
y
800
600
400
200
Number of Observations
5
6
7
16
12
18
7
4
3
2
6
8
10
12
14
16
z

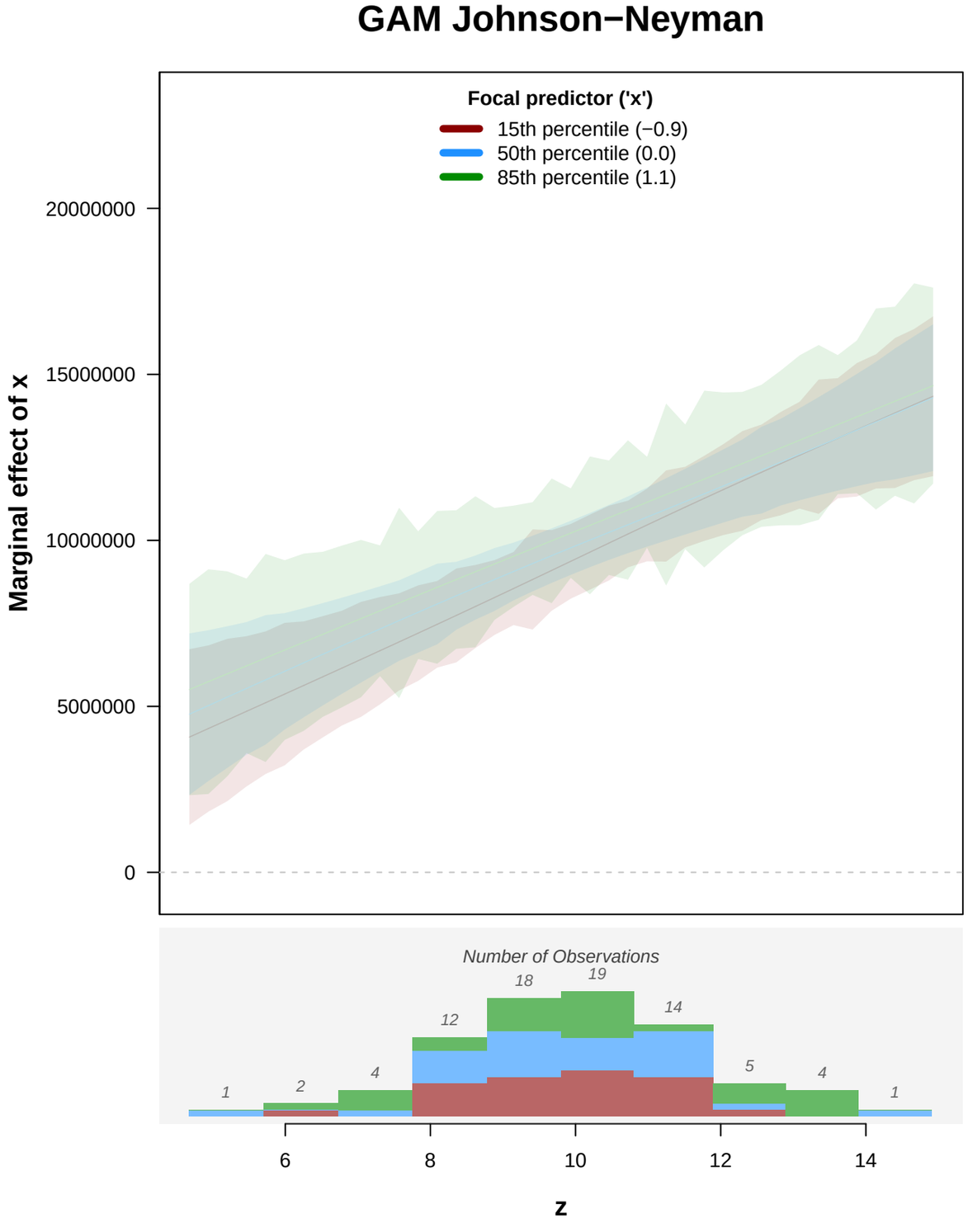

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.2)
85th percentile (0.9)
Marginal effect of x
400
300
200
100
0
Number of Observations
5
6
7
16
12
18
7
4
3
2
6
8
10
12
14
16
z

Part B (y−values) | Page 30 | y_1_5e7 | y in [1, 50000000] | n=80 (small)

DV scaled to [1, 5e7]. Check wide tick labels (scientific or large integers) vs ylab left padding.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

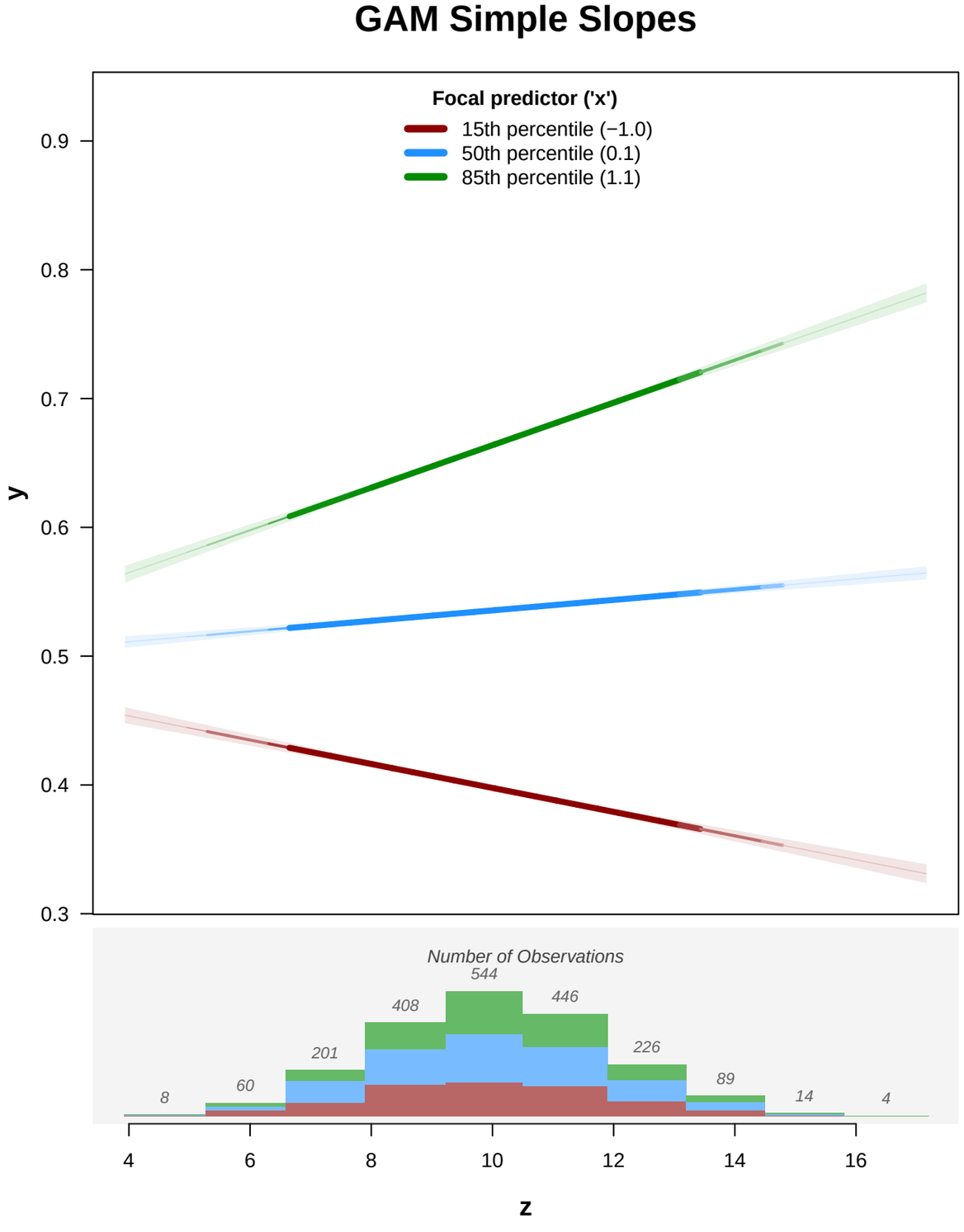

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.0)
85th percentile (1.1)
y
5e+07
4e+07
3e+07
2e+07
Number of Observations
1
2
4
12
18
19
14
5
4
1
6
8
10
12
14
z

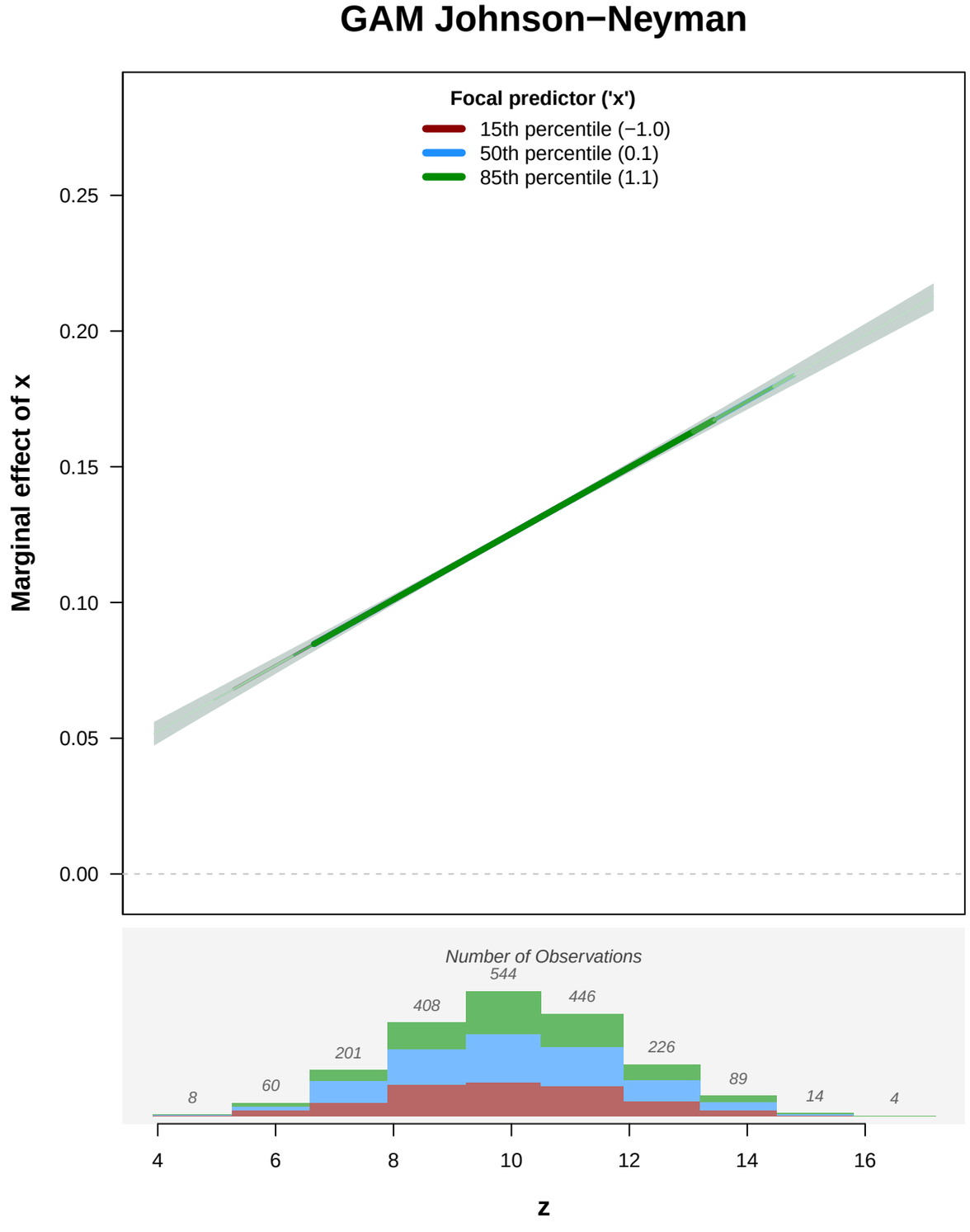

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−0.9)
50th percentile (0.0)
85th percentile (1.1)
Marginal effect of x
20000000
15000000
10000000
5000000
0
Number of Observations
1
2
4
12
18
19
14
5
4
1
6
8
10
12
14
z

Part B (y−values) | Page 31 | y_0_1 | y in [0, 1] | n=2000 (big)

DV scaled to [0, 1]. Check y−tick decimals and ylab spacing (ticks like 0.0, 0.2, …).

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

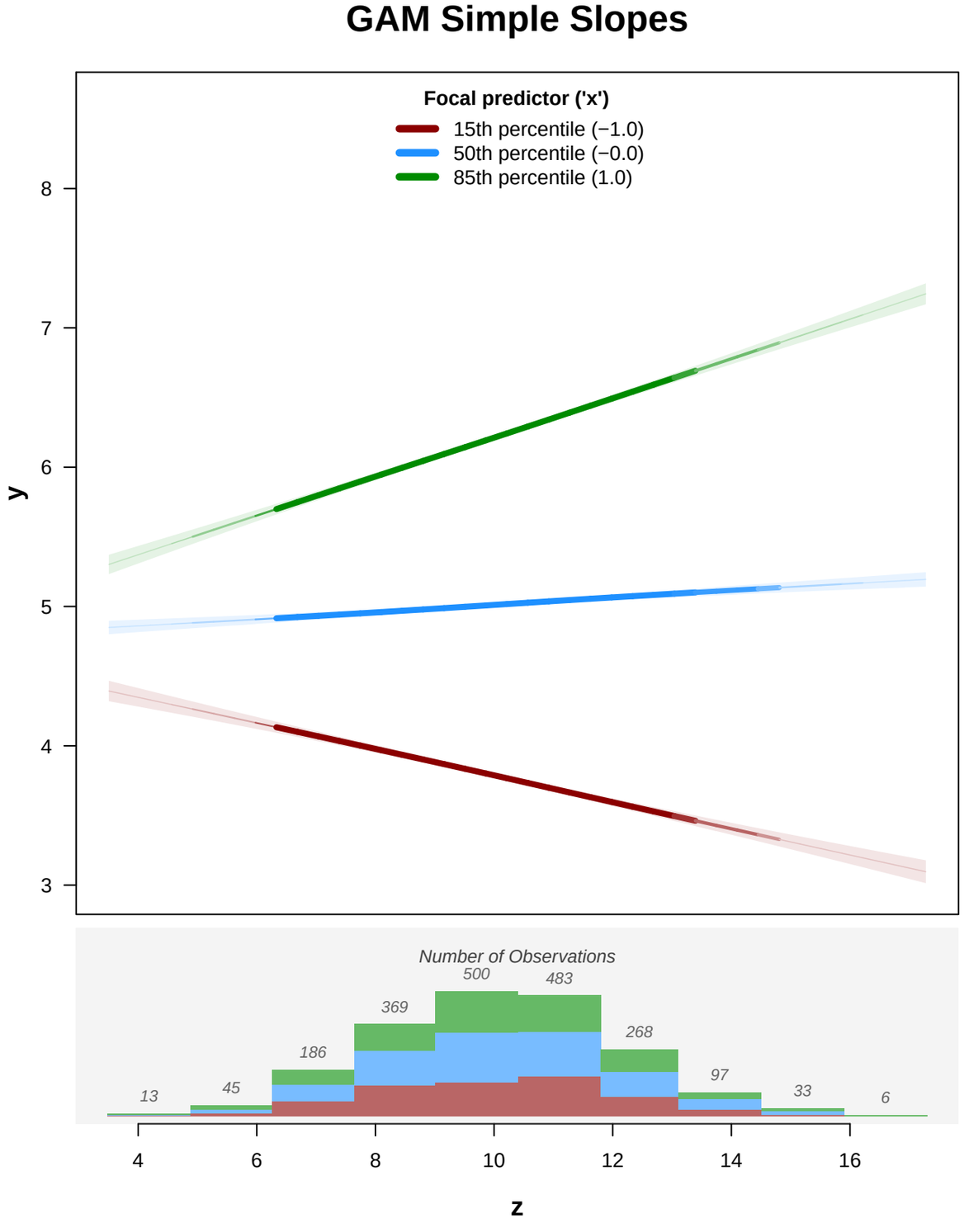

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.1)
85th percentile (1.1)
y
0.9
0.8
0.7
0.6
0.5
0.4
0.3
Number of Observations
8
60
201
408
544
446
226
89
14
4
4
6
8
10
12
14
16
z

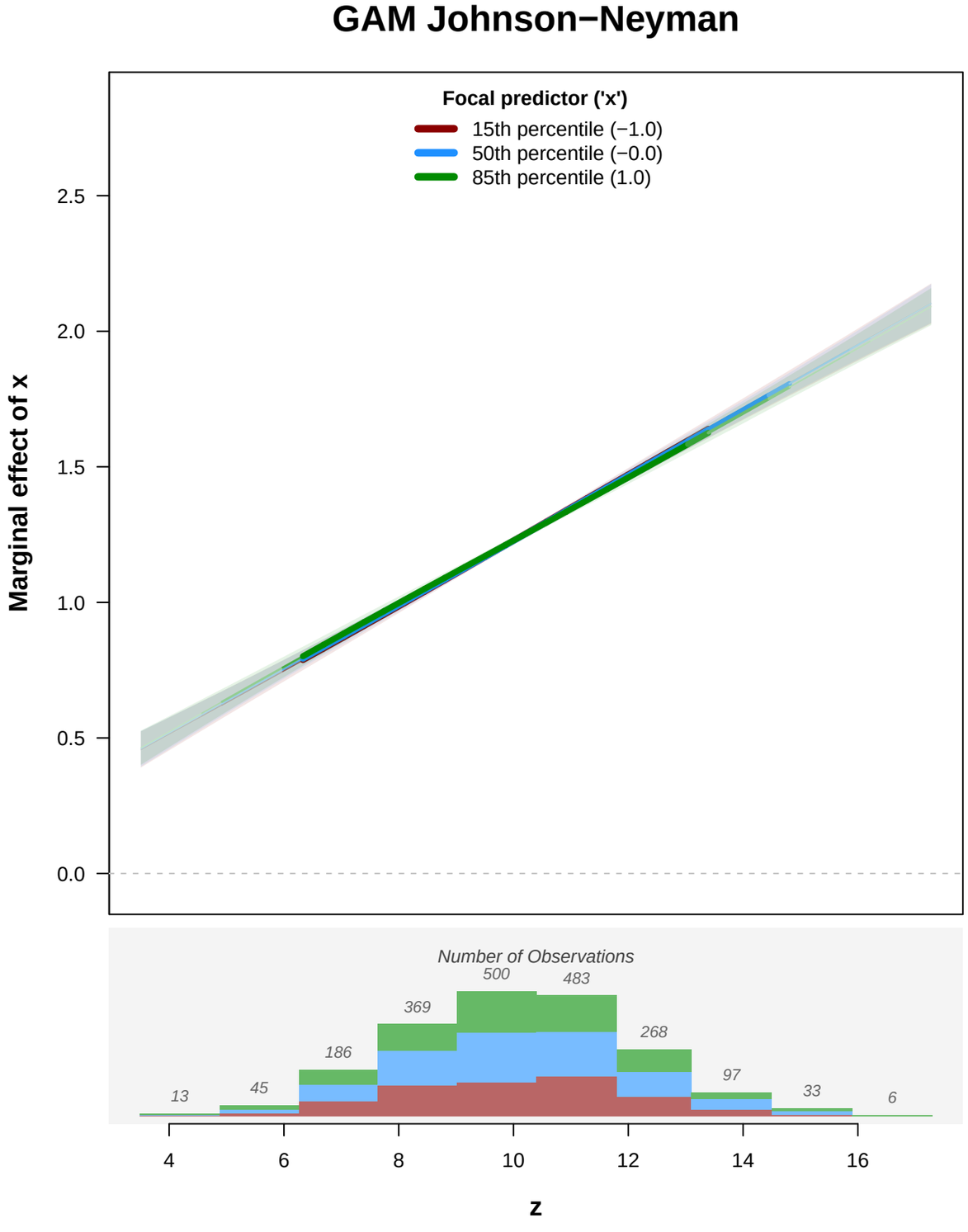

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.1)
85th percentile (1.1)
Marginal effect of x
0.25
0.20
0.15
0.10
0.05
0.00
Number of Observations
8
60
201
408
544
446
226
89
14
4
4
6
8
10
12
14
16
z

Part B (y−values) | Page 32 | y_1_10 | y in [1, 10] | n=2000 (big)

DV scaled to [1, 10]. Check short integer/decimal ticks and ylab clearance.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

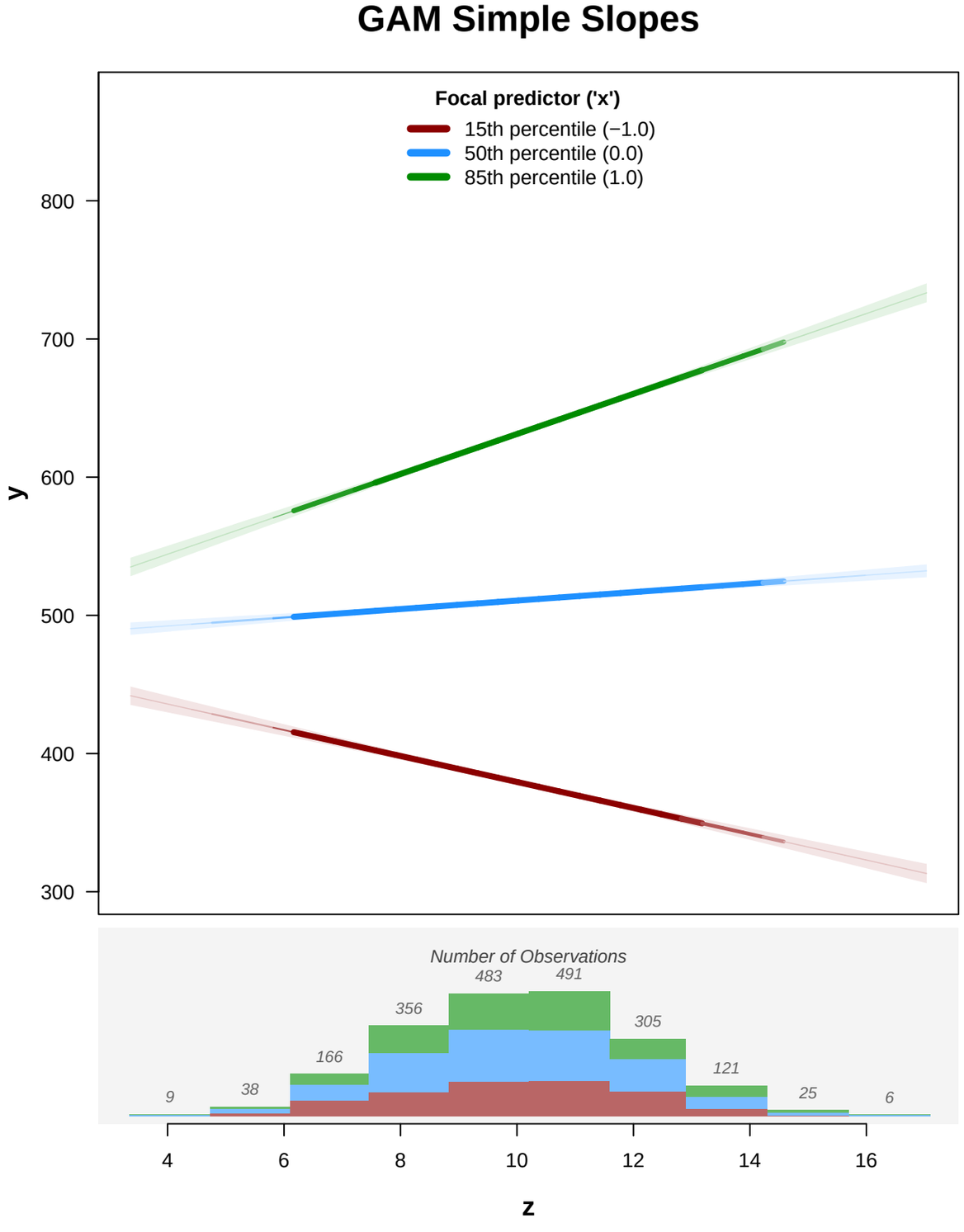

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
y
8
7
6
5
4
3
Number of Observations
13
45
186
369
500
483
268
97
33
6
4
6
8
10
12
14
16
z

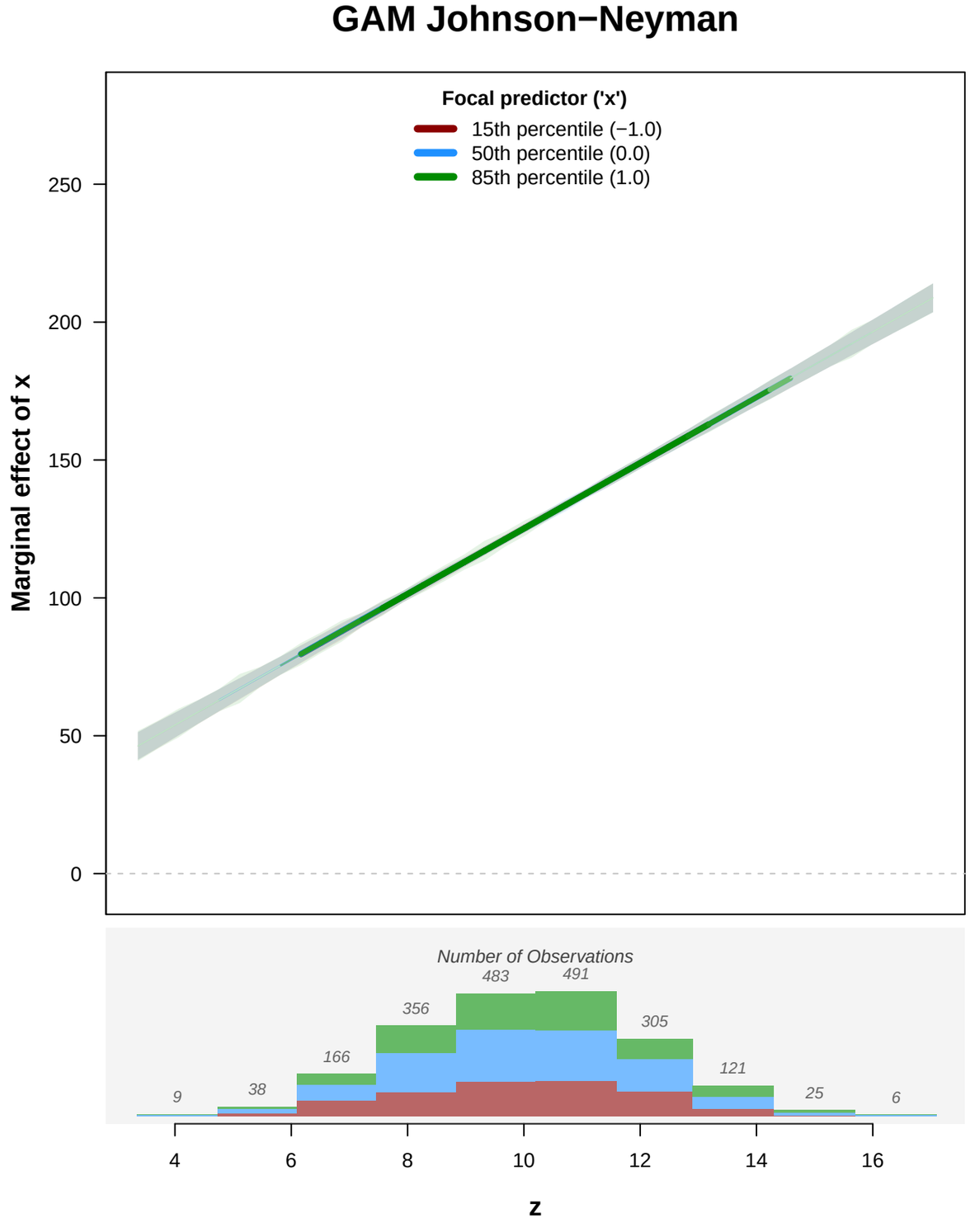

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
Marginal effect of x
2.5
2.0
1.5
1.0
0.5
0.0
Number of Observations
13
45
186
369
500
483
268
97
33
6
4
6
8
10
12
14
16
z

Part B (y−values) | Page 33 | y_1_1000 | y in [1, 1000] | n=2000 (big)

DV scaled to [1, 1000]. Check medium−width tick labels (e.g. 200, 400) vs ylab.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.

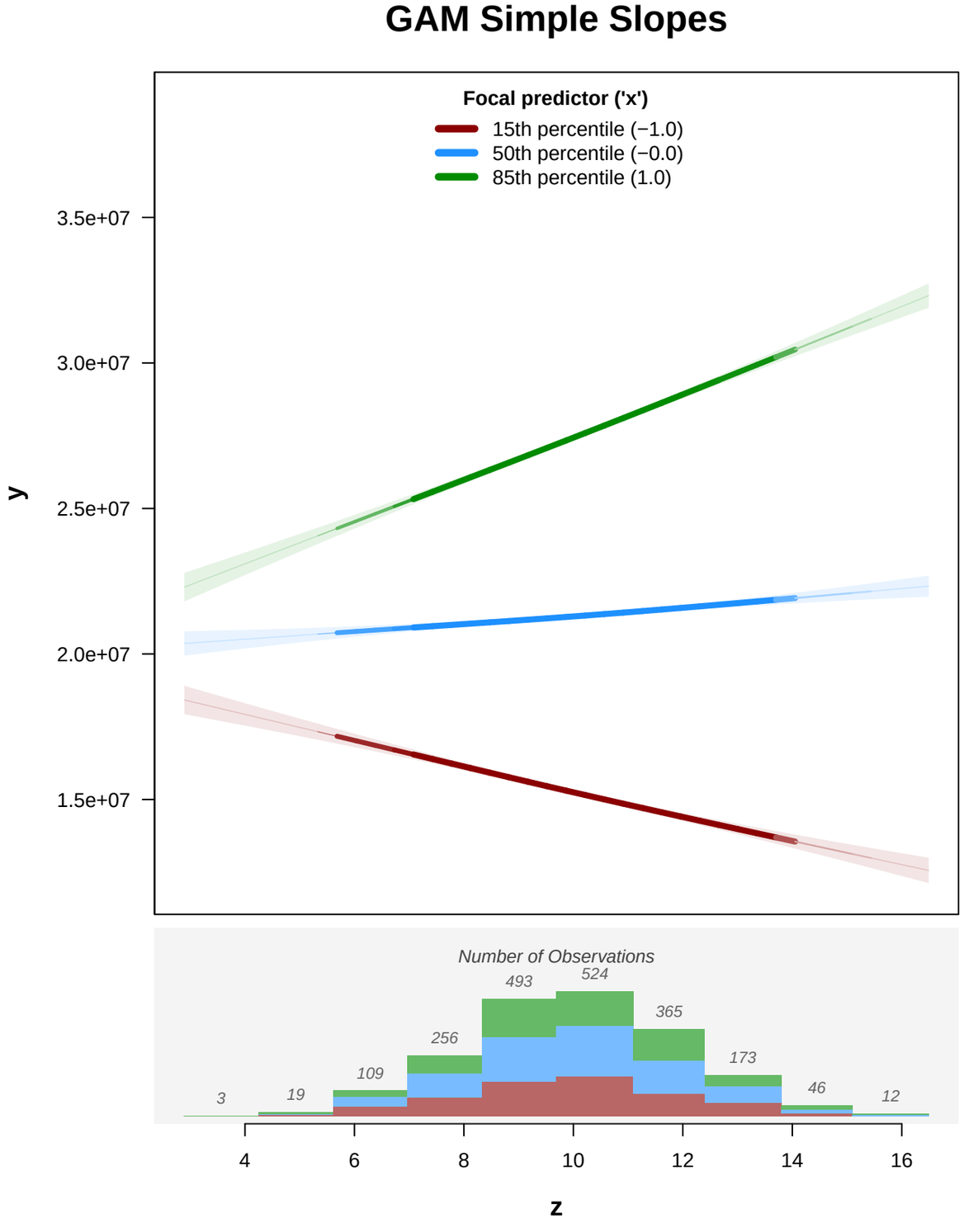

GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
y
800
700
600
500
400
300
Number of Observations
9
38
166
356
483
491
305
121
25
6
4
6
8
10
12
14
16
z

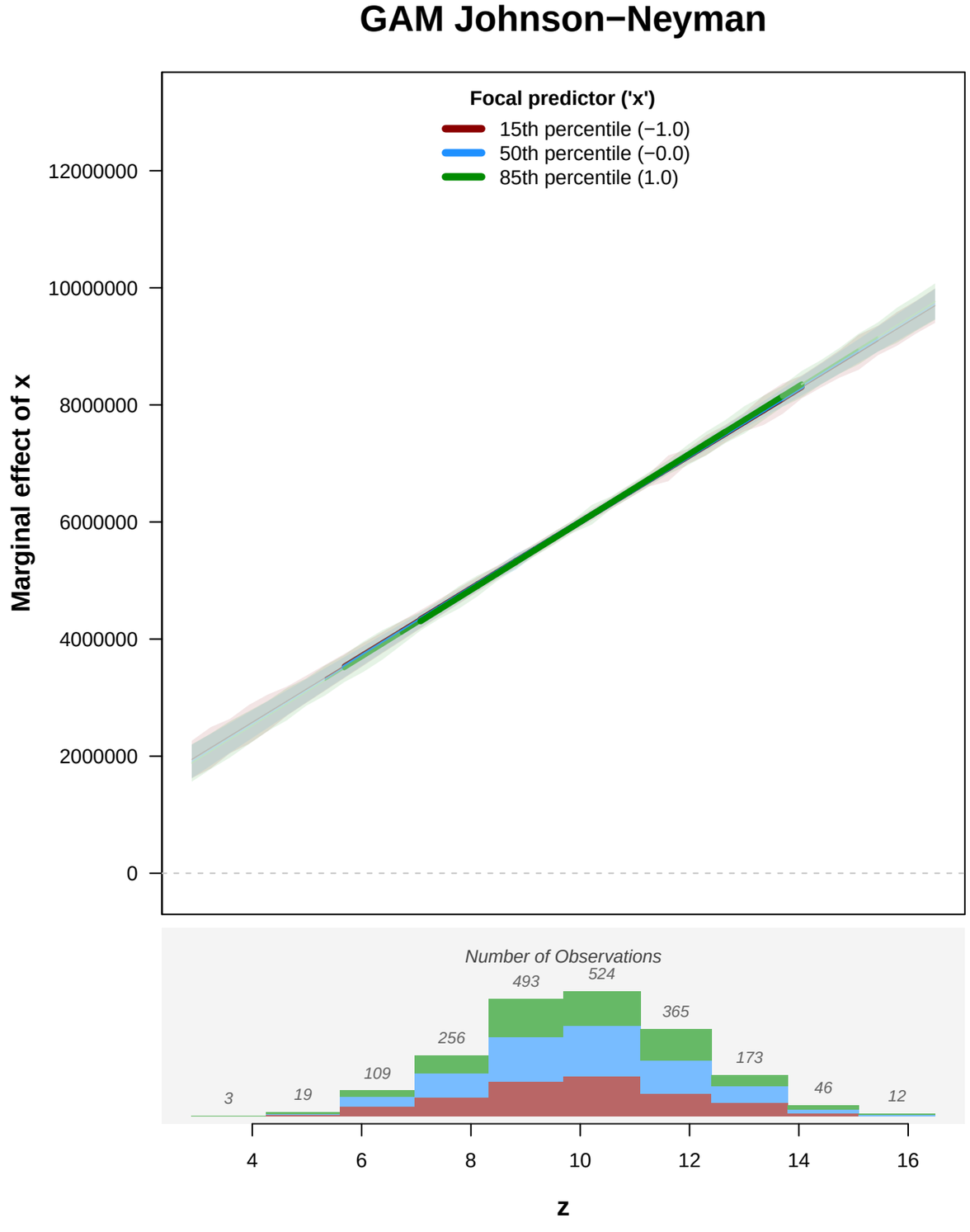

GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (0.0)
85th percentile (1.0)
Marginal effect of x
250
200
150
100
50
0
Number of Observations
9
38
166
356
483
491
305
121
25
6
4
6
8
10
12
14
16
z

Part B (y−values) | Page 34 | y_1_5e7 | y in [1, 50000000] | n=2000 (big)

DV scaled to [1, 5e7]. Check wide tick labels (scientific or large integers) vs ylab left padding.

Same continuous x*z GAM, draw='both', histogram=TRUE. Focus: y−axis tick width and ylab placement.


GAM Simple Slopes
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
y
3.5e+07
3.0e+07
2.5e+07
2.0e+07
1.5e+07
Number of Observations
3
19
109
256
493
524
365
173
46
12
4
6
8
10
12
14
16
z


GAM Johnson−Neyman
Focal predictor ('x')
15th percentile (−1.0)
50th percentile (−0.0)
85th percentile (1.0)
Marginal effect of x
12000000
10000000
8000000
6000000
4000000
2000000
0
Number of Observations
3
19
109
256
493
524
365
173
46
12
4
6
8
10
12
14
16
z